Final Project

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RNA-seq count and meta data from TCGA-BRCA cohort

Outline -

1. Data Preparation

2. Gene selection and summary statistics

2a. ERBB2 Gene Selection and Summary Statistics

2b. Covariate Summary Statistics

3. Data visualization

3a. Histogram — distribution of ERBB2 expression

3b. Scatterplot — relationship between ERBB2 and CDK4

3c. Violin + box plot — ERBB2 expression across PAM50 subtypes

3d. Cumulative distribution function — additional new plot type

3e. Heatmap — expression patterns across the 10 selected genes with PAM50 annotation

1. Data Preparation

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(plotly)# load packages
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##     last_plot
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following object is masked from 'package:graphics':
##
##     layout
```

```r
# Import data
counts <- read.csv("counts.csv", row.names = 1, check.names = FALSE)
# Initially did not use check.names = FALSE, which caused problems when combining with metadata
# read.csv() changed the hyphens to periods in the column names of counts

metadata <- read.csv("meta_data.csv", row.names = 1)

## Counts - gene-expression count - the numbers represent how many sequencing reads/fragments
# were assigned to the RNA transcripts from those genes
# Higher RNA-seq count generally means more RNA from that gene was detected → higher gene expression
## Meta data - Information describing each sample/patient
```

```r
# Inspect data
dim(counts)
```

```
## [1] 60660  1231
```

```r
dim(metadata)
```

```
## [1] 1231   97
```

```r
# head(counts)
head(metadata)
```

```
##                                     patient           sample shortLetterCode
## TCGA-GM-A2DL-01A-11R-A18M-07 TCGA-GM-A2DL TCGA-GM-A2DL-01A               TP
## TCGA-AC-A2QI-01A-12R-A19W-07 TCGA-AC-A2QI TCGA-AC-A2QI-01A               TP
## TCGA-A8-A06R-01A-11R-A00Z-07 TCGA-A8-A06R TCGA-A8-A06R-01A               TP
## TCGA-EW-A1PD-01A-11R-A144-07 TCGA-EW-A1PD TCGA-EW-A1PD-01A               TP
## TCGA-AO-A12D-01A-11R-A115-07 TCGA-AO-A12D TCGA-AO-A12D-01A               TP
## TCGA-AR-A24N-01A-11R-A169-07 TCGA-AR-A24N TCGA-AR-A24N-01A               TP
##                                        definition sample_submitter_id
## TCGA-GM-A2DL-01A-11R-A18M-07 Primary solid Tumor    TCGA-GM-A2DL-01A
## TCGA-AC-A2QI-01A-12R-A19W-07 Primary solid Tumor    TCGA-AC-A2QI-01A
```

```
## TCGA-A8-A06R-01A-11R-A00Z-07 Primary solid Tumor    TCGA-A8-A06R-01A
## TCGA-EW-A1PD-01A-11R-A144-07 Primary solid Tumor    TCGA-EW-A1PD-01A
## TCGA-AO-A12D-01A-11R-A115-07 Primary solid Tumor    TCGA-AO-A12D-01A
## TCGA-AR-A24N-01A-11R-A169-07 Primary solid Tumor    TCGA-AR-A24N-01A
##                              sample_type_id tumor_descriptor
## TCGA-GM-A2DL-01A-11R-A18M-07              1          Primary
## TCGA-AC-A2QI-01A-12R-A19W-07              1          Primary
## TCGA-A8-A06R-01A-11R-A00Z-07              1          Primary
## TCGA-EW-A1PD-01A-11R-A144-07              1          Primary
## TCGA-AO-A12D-01A-11R-A115-07              1          Primary
## TCGA-AR-A24N-01A-11R-A169-07              1          Primary
##                                                      sample_id   sample_type
## TCGA-GM-A2DL-01A-11R-A18M-07 d6e981ee-1afc-4f84-a3be-e87e0d1460cd Primary Tumor
## TCGA-AC-A2QI-01A-12R-A19W-07 08f10bb1-6dd4-4f36-86a3-1801f9762187 Primary Tumor
## TCGA-A8-A06R-01A-11R-A00Z-07 7fa91f3b-34db-41ba-8447-d745eebac2db Primary Tumor
## TCGA-EW-A1PD-01A-11R-A144-07 9f52ae8b-7fe7-441c-9fb6-c55a9bd1a0d0 Primary Tumor
## TCGA-AO-A12D-01A-11R-A115-07 b6359ede-df10-44b9-a2e2-bbf900eeb241 Primary Tumor
## TCGA-AR-A24N-01A-11R-A169-07 dcc31bf1-9890-4fb1-9677-b5fb77e1fc37 Primary Tumor
##                               composition days_to_collection    state
## TCGA-GM-A2DL-01A-11R-A18M-07 Not Reported               2723 released
## TCGA-AC-A2QI-01A-12R-A19W-07 Not Reported                 91 released
## TCGA-A8-A06R-01A-11R-A00Z-07 Not Reported               1184 released
## TCGA-EW-A1PD-01A-11R-A144-07 Not Reported                112 released
## TCGA-AO-A12D-01A-11R-A115-07 Not Reported               1948 released
## TCGA-AR-A24N-01A-11R-A169-07 Not Reported               2219 released
##                              initial_weight preservation_method
## TCGA-GM-A2DL-01A-11R-A18M-07            130             Unknown
## TCGA-AC-A2QI-01A-12R-A19W-07            140             Unknown
## TCGA-A8-A06R-01A-11R-A00Z-07            220             Unknown
## TCGA-EW-A1PD-01A-11R-A144-07            120                 OCT
## TCGA-AO-A12D-01A-11R-A115-07            750                 OCT
## TCGA-AR-A24N-01A-11R-A169-07            310                 OCT
##                                                 pathology_report_uuid submitter_id
## TCGA-GM-A2DL-01A-11R-A18M-07 8CF4E53E-924F-4DC6-8748-BC8A1F7D5662 TCGA-GM-A2DL
## TCGA-AC-A2QI-01A-12R-A19W-07 5CF85AF1-D227-430C-99CD-C2A524CFFA29 TCGA-AC-A2QI
## TCGA-A8-A06R-01A-11R-A00Z-07 B6B7939C-63AF-4B52-B494-AA09A9C03871 TCGA-A8-A06R
## TCGA-EW-A1PD-01A-11R-A144-07 B6D80CA9-F4DA-4DFB-80BF-08BEE17142BD TCGA-EW-A1PD
## TCGA-AO-A12D-01A-11R-A115-07 9363D53D-BC3B-4AC0-8185-536C6DB3A6AE TCGA-AO-A12D
## TCGA-AR-A24N-01A-11R-A169-07 AAF32BB7-716A-4C0E-AE7E-04F6A5BF5628 TCGA-AR-A24N
##                              oct_embedded specimen_type is_ffpe tissue_type
## TCGA-GM-A2DL-01A-11R-A18M-07        false  Solid Tissue   FALSE       Tumor
## TCGA-AC-A2QI-01A-12R-A19W-07        false  Solid Tissue   FALSE       Tumor
## TCGA-A8-A06R-01A-11R-A00Z-07        false       Unknown   FALSE       Tumor
## TCGA-EW-A1PD-01A-11R-A144-07         true  Solid Tissue   FALSE       Tumor
## TCGA-AO-A12D-01A-11R-A115-07         true       Unknown   FALSE       Tumor
## TCGA-AR-A24N-01A-11R-A169-07         true  Solid Tissue   FALSE       Tumor
##                              synchronous_malignancy ajcc_pathologic_stage
## TCGA-GM-A2DL-01A-11R-A18M-07                     No               Stage I
## TCGA-AC-A2QI-01A-12R-A19W-07                     No            Stage IIIA
## TCGA-A8-A06R-01A-11R-A00Z-07                     No             Stage IIB
## TCGA-EW-A1PD-01A-11R-A144-07                     No             Stage IIA
## TCGA-AO-A12D-01A-11R-A115-07                     No             Stage IIA
## TCGA-AR-A24N-01A-11R-A169-07                     No               Stage I
##                              days_to_diagnosis laterality
```

```
## TCGA-GM-A2DL-01A-11R-A18M-07                0       Left
## TCGA-AC-A2QI-01A-12R-A19W-07                0      Right
## TCGA-A8-A06R-01A-11R-A00Z-07                0      Right
## TCGA-EW-A1PD-01A-11R-A144-07                0      Right
## TCGA-AO-A12D-01A-11R-A115-07                0       Left
## TCGA-AR-A24N-01A-11R-A169-07                0       Left
##
## TCGA-GM-A2DL-01A-11R-A18M-07
## TCGA-AC-A2QI-01A-12R-A19W-07
## TCGA-A8-A06R-01A-11R-A00Z-07
## TCGA-EW-A1PD-01A-11R-A144-07
## TCGA-AO-A12D-01A-11R-A115-07 list("Breast", NULL, NULL, NULL, NULL, "Regional Site");c("First-Line T
## TCGA-AR-A24N-01A-11R-A169-07
##                              tissue_or_organ_of_origin age_at_diagnosis
## TCGA-GM-A2DL-01A-11R-A18M-07              Breast, NOS            18475
## TCGA-AC-A2QI-01A-12R-A19W-07              Breast, NOS            27865
## TCGA-A8-A06R-01A-11R-A00Z-07              Breast, NOS            25477
## TCGA-EW-A1PD-01A-11R-A144-07              Breast, NOS            22536
## TCGA-AO-A12D-01A-11R-A115-07              Breast, NOS            15774
## TCGA-AR-A24N-01A-11R-A169-07              Breast, NOS            19989
##                                primary_diagnosis
## TCGA-GM-A2DL-01A-11R-A18M-07 Infiltrating duct carcinoma, NOS
## TCGA-AC-A2QI-01A-12R-A19W-07          Lobular carcinoma, NOS
## TCGA-A8-A06R-01A-11R-A00Z-07 Infiltrating duct carcinoma, NOS
## TCGA-EW-A1PD-01A-11R-A144-07 Infiltrating duct carcinoma, NOS
## TCGA-AO-A12D-01A-11R-A115-07 Infiltrating duct carcinoma, NOS
## TCGA-AR-A24N-01A-11R-A169-07 Infiltrating duct carcinoma, NOS
##                              metastasis_at_diagnosis prior_malignancy
## TCGA-GM-A2DL-01A-11R-A18M-07          No Metastasis               no
## TCGA-AC-A2QI-01A-12R-A19W-07          No Metastasis               no
## TCGA-A8-A06R-01A-11R-A00Z-07                   <NA>               no
## TCGA-EW-A1PD-01A-11R-A144-07          No Metastasis               no
## TCGA-AO-A12D-01A-11R-A115-07                   <NA>               no
## TCGA-AR-A24N-01A-11R-A169-07                   <NA>               no
##                              year_of_diagnosis prior_treatment
## TCGA-GM-A2DL-01A-11R-A18M-07              2003              No
## TCGA-AC-A2QI-01A-12R-A19W-07              2011              No
## TCGA-A8-A06R-01A-11R-A00Z-07              2007              No
## TCGA-EW-A1PD-01A-11R-A144-07              2010              No
## TCGA-AO-A12D-01A-11R-A115-07              2005              No
## TCGA-AR-A24N-01A-11R-A169-07              2005              No
##                              diagnosis_is_primary_disease method_of_diagnosis
## TCGA-GM-A2DL-01A-11R-A18M-07                         TRUE         Core Biopsy
## TCGA-AC-A2QI-01A-12R-A19W-07                         TRUE         Core Biopsy
## TCGA-A8-A06R-01A-11R-A00Z-07                         TRUE                <NA>
## TCGA-EW-A1PD-01A-11R-A144-07                         TRUE         Core Biopsy
## TCGA-AO-A12D-01A-11R-A115-07                         TRUE         Core Biopsy
## TCGA-AR-A24N-01A-11R-A169-07                         TRUE         Core Biopsy
##                              ajcc_staging_system_edition ajcc_pathologic_t
## TCGA-GM-A2DL-01A-11R-A18M-07                         6th               T1c
## TCGA-AC-A2QI-01A-12R-A19W-07                         7th                T3
## TCGA-A8-A06R-01A-11R-A00Z-07                         5th                T2
## TCGA-EW-A1PD-01A-11R-A144-07                         7th               T1c
## TCGA-AO-A12D-01A-11R-A115-07                         6th               T1c
```

```
## TCGA-AR-A24N-01A-11R-A169-07                         6th                T1
##                               morphology ajcc_pathologic_n ajcc_pathologic_m
## TCGA-GM-A2DL-01A-11R-A18M-07     8500/3           N0 (i-)                M0
## TCGA-AC-A2QI-01A-12R-A19W-07     8520/3               N1a                MX
## TCGA-A8-A06R-01A-11R-A00Z-07     8500/3               N1a                M0
## TCGA-EW-A1PD-01A-11R-A144-07     8500/3               N1a                M0
## TCGA-AO-A12D-01A-11R-A115-07     8500/3               N1a                M0
## TCGA-AR-A24N-01A-11R-A169-07     8500/3                N0                M0
##                               classification_of_tumor
## TCGA-GM-A2DL-01A-11R-A18M-07                 primary
## TCGA-AC-A2QI-01A-12R-A19W-07                 primary
## TCGA-A8-A06R-01A-11R-A00Z-07                 primary
## TCGA-EW-A1PD-01A-11R-A144-07                 primary
## TCGA-AO-A12D-01A-11R-A115-07                 primary
## TCGA-AR-A24N-01A-11R-A169-07                 primary
##                                                                diagnosis_id
## TCGA-GM-A2DL-01A-11R-A18M-07 3185a0c5-83e5-5523-81b8-6973708a75b9
## TCGA-AC-A2QI-01A-12R-A19W-07 0b952469-20d9-5831-b719-8d55fd60d25b
## TCGA-A8-A06R-01A-11R-A00Z-07 2804f895-5d16-5096-9ade-e6852295b87f
## TCGA-EW-A1PD-01A-11R-A144-07 4b25cd81-e384-5a36-9863-4150b90a85dc
## TCGA-AO-A12D-01A-11R-A115-07 9e050418-bc62-5830-86f9-8b5703a7fd44
## TCGA-AR-A24N-01A-11R-A169-07 f9630094-86ba-5a9e-9e86-556b19c2b338
##                               icd_10_code
## TCGA-GM-A2DL-01A-11R-A18M-07       C50.9
## TCGA-AC-A2QI-01A-12R-A19W-07       C50.9
## TCGA-A8-A06R-01A-11R-A00Z-07       C50.9
## TCGA-EW-A1PD-01A-11R-A144-07       C50.9
## TCGA-AO-A12D-01A-11R-A115-07       C50.9
## TCGA-AR-A24N-01A-11R-A169-07       C50.9
##                               site_of_resection_or_biopsy
## TCGA-GM-A2DL-01A-11R-A18M-07                Breast, NOS
## TCGA-AC-A2QI-01A-12R-A19W-07                Breast, NOS
## TCGA-A8-A06R-01A-11R-A00Z-07                Breast, NOS
## TCGA-EW-A1PD-01A-11R-A144-07                Breast, NOS
## TCGA-AO-A12D-01A-11R-A115-07                Breast, NOS
## TCGA-AR-A24N-01A-11R-A169-07                Breast, NOS
##                                                sites_of_involvement
## TCGA-GM-A2DL-01A-11R-A18M-07            Breast, Left Upper Outer
## TCGA-AC-A2QI-01A-12R-A19W-07                         Breast, NOS
## TCGA-A8-A06R-01A-11R-A00Z-07           Breast, Right Upper Outer
## TCGA-EW-A1PD-01A-11R-A144-07                         Breast, NOS
## TCGA-AO-A12D-01A-11R-A115-07 Breast, Left Upper Outer;Breast, NOS
## TCGA-AR-A24N-01A-11R-A169-07            Breast, Left Upper Inner
##                               days_to_last_follow_up follow_ups_disease_response
## TCGA-GM-A2DL-01A-11R-A18M-07                   3519               TF-Tumor Free
## TCGA-AC-A2QI-01A-12R-A19W-07                    588               TF-Tumor Free
## TCGA-A8-A06R-01A-11R-A00Z-07                    547               TF-Tumor Free
## TCGA-EW-A1PD-01A-11R-A144-07                    424               TF-Tumor Free
## TCGA-AO-A12D-01A-11R-A115-07                   2515               TF-Tumor Free
## TCGA-AR-A24N-01A-11R-A169-07                   3035               TF-Tumor Free
##                                      race gender              ethnicity
## TCGA-GM-A2DL-01A-11R-A18M-07        white female not hispanic or latino
## TCGA-AC-A2QI-01A-12R-A19W-07        white female not hispanic or latino
## TCGA-A8-A06R-01A-11R-A00Z-07 not reported female           not reported
## TCGA-EW-A1PD-01A-11R-A144-07        white   male     hispanic or latino
## TCGA-AO-A12D-01A-11R-A115-07        white female not hispanic or latino
## TCGA-AR-A24N-01A-11R-A169-07        white female not hispanic or latino
##                               vital_status age_at_index days_to_birth
## TCGA-GM-A2DL-01A-11R-A18M-07        Alive           50        -18475
## TCGA-AC-A2QI-01A-12R-A19W-07        Alive           76        -27865
## TCGA-A8-A06R-01A-11R-A00Z-07        Alive           69        -25477
```

```
## TCGA-EW-A1PD-01A-11R-A144-07        Alive           61        -22536
## TCGA-AO-A12D-01A-11R-A115-07        Alive           43        -15774
## TCGA-AR-A24N-01A-11R-A169-07        Alive           54        -19989
##                                                  demographic_id
## TCGA-GM-A2DL-01A-11R-A18M-07 b94ef8b8-13a9-57c5-90ec-d7b640476cd4
## TCGA-AC-A2QI-01A-12R-A19W-07 de88ff20-a93a-572e-9060-34f64f281360
## TCGA-A8-A06R-01A-11R-A00Z-07 365d369f-8b9a-5db8-b341-41024f115964
## TCGA-EW-A1PD-01A-11R-A144-07 42627028-1318-501d-864a-1a8645ccab14
## TCGA-AO-A12D-01A-11R-A115-07 9585e3b9-74b5-503d-ac8e-3340e56f6a8d
## TCGA-AR-A24N-01A-11R-A169-07 2ed6a99f-7873-5c07-b14c-f5db4727de41
##                              age_is_obfuscated
## TCGA-GM-A2DL-01A-11R-A18M-07             FALSE
## TCGA-AC-A2QI-01A-12R-A19W-07             FALSE
## TCGA-A8-A06R-01A-11R-A00Z-07             FALSE
## TCGA-EW-A1PD-01A-11R-A144-07             FALSE
## TCGA-AO-A12D-01A-11R-A115-07             FALSE
## TCGA-AR-A24N-01A-11R-A169-07             FALSE
##                              country_of_residence_at_enrollment
## TCGA-GM-A2DL-01A-11R-A18M-07                      United States
## TCGA-AC-A2QI-01A-12R-A19W-07                      United States
## TCGA-A8-A06R-01A-11R-A00Z-07                            Germany
## TCGA-EW-A1PD-01A-11R-A144-07                      United States
## TCGA-AO-A12D-01A-11R-A115-07                               <NA>
## TCGA-AR-A24N-01A-11R-A169-07                      United States
##                              bcr_patient_barcode primary_site project_id
## TCGA-GM-A2DL-01A-11R-A18M-07   TCGA-GM-A2DL-01A       Breast  TCGA-BRCA
## TCGA-AC-A2QI-01A-12R-A19W-07   TCGA-AC-A2QI-01A       Breast  TCGA-BRCA
## TCGA-A8-A06R-01A-11R-A00Z-07   TCGA-A8-A06R-01A       Breast  TCGA-BRCA
## TCGA-EW-A1PD-01A-11R-A144-07   TCGA-EW-A1PD-01A       Breast  TCGA-BRCA
## TCGA-AO-A12D-01A-11R-A115-07   TCGA-AO-A12D-01A       Breast  TCGA-BRCA
## TCGA-AR-A24N-01A-11R-A169-07   TCGA-AR-A24N-01A       Breast  TCGA-BRCA
##
## TCGA-GM-A2DL-01A-11R-A18M-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-AC-A2QI-01A-12R-A19W-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-A8-A06R-01A-11R-A00Z-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-EW-A1PD-01A-11R-A144-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-AO-A12D-01A-11R-A115-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-AR-A24N-01A-11R-A169-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
##                                                  name releasable released
## TCGA-GM-A2DL-01A-11R-A18M-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-AC-A2QI-01A-12R-A19W-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-A8-A06R-01A-11R-A00Z-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-EW-A1PD-01A-11R-A144-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-AO-A12D-01A-11R-A115-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-AR-A24N-01A-11R-A169-07 Breast Invasive Carcinoma       TRUE     TRUE
##                              figo_stage figo_staging_edition_year days_to_death
## TCGA-GM-A2DL-01A-11R-A18M-07       <NA>                       NA            NA
## TCGA-AC-A2QI-01A-12R-A19W-07       <NA>                       NA            NA
## TCGA-A8-A06R-01A-11R-A00Z-07       <NA>                       NA            NA
## TCGA-EW-A1PD-01A-11R-A144-07       <NA>                       NA            NA
## TCGA-AO-A12D-01A-11R-A115-07       <NA>                       NA            NA
## TCGA-AR-A24N-01A-11R-A169-07       <NA>                       NA            NA
##                              tumor_of_origin days_to_sample_procurement
## TCGA-GM-A2DL-01A-11R-A18M-07            <NA>                         NA
```

```
## TCGA-AC-A2QI-01A-12R-A19W-07            <NA>                          NA
## TCGA-A8-A06R-01A-11R-A00Z-07            <NA>                          NA
## TCGA-EW-A1PD-01A-11R-A144-07            <NA>                          NA
## TCGA-AO-A12D-01A-11R-A115-07            <NA>                          NA
## TCGA-AR-A24N-01A-11R-A169-07            <NA>                          NA
##                               last_known_disease_status tumor_grade
## TCGA-GM-A2DL-01A-11R-A18M-07                      <NA>        <NA>
## TCGA-AC-A2QI-01A-12R-A19W-07                      <NA>        <NA>
## TCGA-A8-A06R-01A-11R-A00Z-07                      <NA>        <NA>
## TCGA-EW-A1PD-01A-11R-A144-07                      <NA>        <NA>
## TCGA-AO-A12D-01A-11R-A115-07                      <NA>        <NA>
## TCGA-AR-A24N-01A-11R-A169-07                      <NA>        <NA>
##                               progression_or_recurrence alcohol_history
## TCGA-GM-A2DL-01A-11R-A18M-07                      <NA>            <NA>
## TCGA-AC-A2QI-01A-12R-A19W-07                      <NA>            <NA>
## TCGA-A8-A06R-01A-11R-A00Z-07                      <NA>            <NA>
## TCGA-EW-A1PD-01A-11R-A144-07                      <NA>            <NA>
## TCGA-AO-A12D-01A-11R-A115-07                      <NA>            <NA>
## TCGA-AR-A24N-01A-11R-A169-07                      <NA>            <NA>
##                               exposure_id year_of_birth paper_patient
## TCGA-GM-A2DL-01A-11R-A18M-07        <NA>            NA  TCGA-GM-A2DL
## TCGA-AC-A2QI-01A-12R-A19W-07        <NA>            NA  TCGA-AC-A2QI
## TCGA-A8-A06R-01A-11R-A00Z-07        <NA>            NA  TCGA-A8-A06R
## TCGA-EW-A1PD-01A-11R-A144-07        <NA>            NA          <NA>
## TCGA-AO-A12D-01A-11R-A115-07        <NA>            NA  TCGA-AO-A12D
## TCGA-AR-A24N-01A-11R-A169-07        <NA>            NA  TCGA-AR-A24N
##                               paper_Tumor.Type
## TCGA-GM-A2DL-01A-11R-A18M-07             BRCA
## TCGA-AC-A2QI-01A-12R-A19W-07             BRCA
## TCGA-A8-A06R-01A-11R-A00Z-07             BRCA
## TCGA-EW-A1PD-01A-11R-A144-07             <NA>
## TCGA-AO-A12D-01A-11R-A115-07             BRCA
## TCGA-AR-A24N-01A-11R-A169-07             BRCA
##                               paper_Included_in_previous_marker_papers
## TCGA-GM-A2DL-01A-11R-A18M-07                                      YES
## TCGA-AC-A2QI-01A-12R-A19W-07                                       NO
## TCGA-A8-A06R-01A-11R-A00Z-07                                      YES
## TCGA-EW-A1PD-01A-11R-A144-07                                     <NA>
## TCGA-AO-A12D-01A-11R-A115-07                                      YES
## TCGA-AR-A24N-01A-11R-A169-07                                      YES
##                               paper_vital_status paper_days_to_birth
## TCGA-GM-A2DL-01A-11R-A18M-07              Alive              -18475
## TCGA-AC-A2QI-01A-12R-A19W-07              Alive              -27865
## TCGA-A8-A06R-01A-11R-A00Z-07              Alive              -25477
## TCGA-EW-A1PD-01A-11R-A144-07               <NA>                  NA
## TCGA-AO-A12D-01A-11R-A115-07              Alive              -15774
## TCGA-AR-A24N-01A-11R-A169-07              Alive              -19989
##                               paper_days_to_death paper_days_to_last_followup
## TCGA-GM-A2DL-01A-11R-A18M-07                  NA                        3519
## TCGA-AC-A2QI-01A-12R-A19W-07                  NA                         588
## TCGA-A8-A06R-01A-11R-A00Z-07                  NA                         547
## TCGA-EW-A1PD-01A-11R-A144-07                  NA                          NA
## TCGA-AO-A12D-01A-11R-A115-07                  NA                        2515
## TCGA-AR-A24N-01A-11R-A169-07                  NA                        3035
```

```
##                                paper_age_at_initial_pathologic_diagnosis
## TCGA-GM-A2DL-01A-11R-A18M-07                                        50
## TCGA-AC-A2QI-01A-12R-A19W-07                                        76
## TCGA-A8-A06R-01A-11R-A00Z-07                                        69
## TCGA-EW-A1PD-01A-11R-A144-07                                        NA
## TCGA-AO-A12D-01A-11R-A115-07                                        43
## TCGA-AR-A24N-01A-11R-A169-07                                        54
##                                paper_pathologic_stage paper_Tumor_Grade
## TCGA-GM-A2DL-01A-11R-A18M-07               Stage_I               NA
## TCGA-AC-A2QI-01A-12R-A19W-07              Stage_III              NA
## TCGA-A8-A06R-01A-11R-A00Z-07               Stage_II              NA
## TCGA-EW-A1PD-01A-11R-A144-07                   <NA>              NA
## TCGA-AO-A12D-01A-11R-A115-07               Stage_II              NA
## TCGA-AR-A24N-01A-11R-A169-07                Stage_I              NA
##                                paper_BRCA_Pathology paper_BRCA_Subtype_PAM50
## TCGA-GM-A2DL-01A-11R-A18M-07                  IDC                      LumA
## TCGA-AC-A2QI-01A-12R-A19W-07                  ILC                      LumA
## TCGA-A8-A06R-01A-11R-A00Z-07                  IDC                      LumB
## TCGA-EW-A1PD-01A-11R-A144-07                 <NA>                      <NA>
## TCGA-AO-A12D-01A-11R-A115-07                 <NA>                      Her2
## TCGA-AR-A24N-01A-11R-A169-07                  IDC                      LumB
##                                paper_MSI_status paper_HPV_Status
## TCGA-GM-A2DL-01A-11R-A18M-07               NA               NA
## TCGA-AC-A2QI-01A-12R-A19W-07               NA               NA
## TCGA-A8-A06R-01A-11R-A00Z-07               NA               NA
## TCGA-EW-A1PD-01A-11R-A144-07               NA               NA
## TCGA-AO-A12D-01A-11R-A115-07               NA               NA
## TCGA-AR-A24N-01A-11R-A169-07               NA               NA
##                                paper_tobacco_smoking_history paper_CNV.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                            NA                 C3
## TCGA-AC-A2QI-01A-12R-A19W-07                            NA                 C1
## TCGA-A8-A06R-01A-11R-A00Z-07                            NA                 C6
## TCGA-EW-A1PD-01A-11R-A144-07                            NA               <NA>
## TCGA-AO-A12D-01A-11R-A115-07                            NA                 C6
## TCGA-AR-A24N-01A-11R-A169-07                            NA                 C1
##                                paper_Mutation.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                      C1
## TCGA-AC-A2QI-01A-12R-A19W-07                      C4
## TCGA-A8-A06R-01A-11R-A00Z-07                      C4
## TCGA-EW-A1PD-01A-11R-A144-07                    <NA>
## TCGA-AO-A12D-01A-11R-A115-07                      C9
## TCGA-AR-A24N-01A-11R-A169-07                      C1
##                                paper_DNA.Methylation.Clusters paper_mRNA.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                             C2                 C1
## TCGA-AC-A2QI-01A-12R-A19W-07                             C1                 C2
## TCGA-A8-A06R-01A-11R-A00Z-07                             C2                 C1
## TCGA-EW-A1PD-01A-11R-A144-07                           <NA>               <NA>
## TCGA-AO-A12D-01A-11R-A115-07                             C4                 C2
## TCGA-AR-A24N-01A-11R-A169-07                             C2                 C1
##                                paper_miRNA.Clusters paper_lncRNA.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                   C2                   C1
## TCGA-AC-A2QI-01A-12R-A19W-07                   C2                 <NA>
## TCGA-A8-A06R-01A-11R-A00Z-07                   C3                   C3
## TCGA-EW-A1PD-01A-11R-A144-07                 <NA>                 <NA>
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```
## TCGA-AO-A12D-01A-11R-A115-07                  C3                    C1
## TCGA-AR-A24N-01A-11R-A169-07                  C3                    C2
##                              paper_Protein.Clusters paper_PARADIGM.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                     C2                      C5
## TCGA-AC-A2QI-01A-12R-A19W-07                     C1                      C6
## TCGA-A8-A06R-01A-11R-A00Z-07                     C1                      C4
## TCGA-EW-A1PD-01A-11R-A144-07                   <NA>                    <NA>
## TCGA-AO-A12D-01A-11R-A115-07                     C1                      C4
## TCGA-AR-A24N-01A-11R-A169-07                     C1                      C5
##                              paper_Pan.Gyn.Clusters
## TCGA-GM-A2DL-01A-11R-A18M-07                     C3
## TCGA-AC-A2QI-01A-12R-A19W-07                     C1
## TCGA-A8-A06R-01A-11R-A00Z-07                     C1
## TCGA-EW-A1PD-01A-11R-A144-07                   <NA>
## TCGA-AO-A12D-01A-11R-A115-07                     C3
## TCGA-AR-A24N-01A-11R-A169-07                     C1
```

```
colnames(counts)
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##    [1] "TCGA-GM-A2DL-01A-11R-A18M-07" "TCGA-AC-A2QI-01A-12R-A19W-07"
##    [3] "TCGA-A8-A06R-01A-11R-A00Z-07" "TCGA-EW-A1PD-01A-11R-A144-07"
##    [5] "TCGA-AO-A12D-01A-11R-A115-07" "TCGA-AR-A24N-01A-11R-A169-07"
##    [7] "TCGA-AR-A24U-01A-11R-A169-07" "TCGA-D8-A1JU-01A-11R-A13Q-07"
##    [9] "TCGA-A8-A0AD-01A-11R-A056-07" "TCGA-E2-A3DX-01A-21R-A213-07"
##   [11] "TCGA-BH-A18J-11A-31R-A12D-07" "TCGA-A2-A0YL-01A-21R-A109-07"
##   [13] "TCGA-C8-A12P-01A-11R-A115-07" "TCGA-AO-A0JA-01A-11R-A056-07"
##   [15] "TCGA-EW-A1OV-01A-11R-A144-07" "TCGA-5L-AAT0-01A-12R-A41B-07"
##   [17] "TCGA-BH-A18J-01A-11R-A12D-07" "TCGA-D8-A27V-01A-12R-A17B-07"
##   [19] "TCGA-A7-A26I-01B-06R-A22O-07" "TCGA-A8-A096-01A-11R-A00Z-07"
##   [21] "TCGA-AR-A256-01A-11R-A169-07" "TCGA-A7-A426-01A-22R-A24H-07"
##   [23] "TCGA-E2-A1BC-11A-32R-A12P-07" "TCGA-E2-A1LH-11A-22R-A14D-07"
##   [25] "TCGA-E2-A1LH-01A-11R-A14D-07" "TCGA-E9-A1N8-01A-11R-A144-07"
##   [27] "TCGA-BH-A0HK-11A-11R-A089-07" "TCGA-EW-A2FR-01A-11R-A21T-07"
##   [29] "TCGA-E2-A154-01A-11R-A115-07" "TCGA-A2-A0T4-01A-31R-A084-07"
##   [31] "TCGA-E9-A1NH-01A-11R-A14D-07" "TCGA-AR-A2LO-01A-31R-A18M-07"
##   [33] "TCGA-Z7-A8R6-01A-11R-A41B-07" "TCGA-AC-A3YJ-01A-11R-A22U-07"
##   [35] "TCGA-A2-A04U-01A-11R-A115-07" "TCGA-A7-A26I-01A-11R-A169-07"
##   [37] "TCGA-A8-A09A-01A-11R-A00Z-07" "TCGA-AO-A126-01A-11R-A10J-07"
##   [39] "TCGA-EW-A1J6-01A-11R-A13Q-07" "TCGA-LQ-A4E4-01A-11R-A266-07"
##   [41] "TCGA-E2-A1IL-01A-11R-A14D-07" "TCGA-D8-A1XW-01A-11R-A14M-07"
##   [43] "TCGA-A8-A07C-01A-11R-A034-07" "TCGA-EW-A2FV-01A-11R-A17B-07"
##   [45] "TCGA-E2-A1AZ-01A-11R-A12P-07" "TCGA-AC-A2FM-11B-32R-A19W-07"
##   [47] "TCGA-BH-A1FJ-01A-11R-A13Q-07" "TCGA-EW-A1J1-01A-11R-A13Q-07"
##   [49] "TCGA-A7-A0DB-11A-33R-A089-07" "TCGA-A7-A0DB-01A-11R-A277-07"
##   [51] "TCGA-A7-A56D-01A-11R-A27Q-07" "TCGA-AC-A3OD-01A-11R-A21T-07"
##   [53] "TCGA-E2-A109-01A-11R-A10J-07" "TCGA-BH-A0DK-11A-13R-A089-07"
##   [55] "TCGA-E2-A15O-01A-11R-A115-07" "TCGA-BH-A0DK-01A-21R-A056-07"
##   [57] "TCGA-B6-A0WW-01A-11R-A109-07" "TCGA-E2-A2P5-01A-11R-A19W-07"
##   [59] "TCGA-E9-A3Q9-01A-11R-A21T-07" "TCGA-E2-A158-01A-11R-A12D-07"
##   [61] "TCGA-A1-A0SD-01A-11R-A115-07" "TCGA-E9-A5FL-01A-11R-A27Q-07"
##   [63] "TCGA-BH-A1FU-11A-23R-A14D-07" "TCGA-AR-A1AT-01A-11R-A12P-07"
##   [65] "TCGA-A1-A0SE-01A-11R-A084-07" "TCGA-AN-A0XN-01A-21R-A109-07"
##   [67] "TCGA-A2-A0YG-01A-21R-A109-07" "TCGA-E9-A1RF-01A-11R-A157-07"
##   [69] "TCGA-B6-A0RQ-01A-11R-A115-07" "TCGA-A7-A6VY-01A-12R-A33J-07"
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##   [71] "TCGA-A7-A13D-01B-04R-A277-07" "TCGA-BH-A203-11A-42R-A169-07"
##   [73] "TCGA-GM-A2DD-01A-11R-A180-07" "TCGA-E9-A2JT-01A-22R-A18M-07"
##   [75] "TCGA-EW-A1OW-01A-21R-A144-07" "TCGA-AN-A03X-01A-21R-A00Z-07"
##   [77] "TCGA-EW-A1PB-01A-11R-A144-07" "TCGA-AC-A2FO-01A-11R-A180-07"
##   [79] "TCGA-BH-A18M-01A-11R-A12D-07" "TCGA-A8-A090-01A-11R-A00Z-07"
##   [81] "TCGA-AC-A2FM-01A-11R-A19W-07" "TCGA-BH-A1FJ-11B-42R-A13Q-07"
##   [83] "TCGA-A8-A099-01A-11R-A00Z-07" "TCGA-AO-A12B-01A-11R-A10J-07"
##   [85] "TCGA-E2-A15F-01A-11R-A115-07" "TCGA-AO-A0JF-01A-11R-A056-07"
##   [87] "TCGA-E2-A576-01A-11R-A31O-07" "TCGA-AN-A0AR-01A-11R-A00Z-07"
##   [89] "TCGA-AC-A3OD-01B-06R-A22O-07" "TCGA-LL-A5YN-01A-11R-A28M-07"
##   [91] "TCGA-E9-A1RO-01A-22R-A16F-07" "TCGA-AN-A03Y-01A-21R-A00Z-07"
##   [93] "TCGA-A7-A4SC-01A-12R-A266-07" "TCGA-OL-A5D6-01A-21R-A27Q-07"
##   [95] "TCGA-EW-A1IZ-01A-11R-A13Q-07" "TCGA-BH-A0EB-01A-11R-A034-07"
##   [97] "TCGA-C8-A3M8-01A-11R-A213-07" "TCGA-AO-A129-01A-21R-A10J-07"
##   [99] "TCGA-D8-A27P-01A-11R-A16F-07" "TCGA-E2-A152-01A-11R-A12D-07"
##  [101] "TCGA-BH-A1FU-01A-11R-A14D-07" "TCGA-E9-A1NI-01A-11R-A14D-07"
##  [103] "TCGA-D8-A27M-01A-11R-A16F-07" "TCGA-A2-A0YF-01A-21R-A109-07"
##  [105] "TCGA-E9-A1RG-01A-11R-A14D-07" "TCGA-B6-A0X4-01A-11R-A109-07"
##  [107] "TCGA-AO-A0J4-01A-11R-A034-07" "TCGA-A7-A425-01A-11R-A24H-07"
##  [109] "TCGA-A8-A08X-01A-21R-A00Z-07" "TCGA-OL-A66N-01A-12R-A31O-07"
##  [111] "TCGA-B6-A1KF-01A-11R-A13Q-07" "TCGA-AC-A2FE-01A-11R-A19W-07"
##  [113] "TCGA-A2-A0CM-01A-31R-A034-07" "TCGA-GM-A5PX-01A-12R-A28M-07"
##  [115] "TCGA-AR-A1AR-01A-31R-A137-07" "TCGA-AO-A0J6-01A-11R-A034-07"
##  [117] "TCGA-E2-A156-01A-11R-A12D-07" "TCGA-D8-A73W-01A-22R-A352-07"
##  [119] "TCGA-E2-A107-01A-11R-A10J-07" "TCGA-E2-A158-11A-22R-A12D-07"
##  [121] "TCGA-AC-A23G-01A-11R-A213-07" "TCGA-A2-A1G4-01A-11R-A13Q-07"
##  [123] "TCGA-AO-A1KO-01A-31R-A13Q-07" "TCGA-A2-A3XS-01A-11R-A22U-07"
##  [125] "TCGA-B6-A0RH-01A-21R-A115-07" "TCGA-D8-A1JH-01A-11R-A13Q-07"
##  [127] "TCGA-AO-A03P-01A-11R-A00Z-07" "TCGA-OL-A5DA-01A-11R-A27Q-07"
##  [129] "TCGA-A8-A07B-01A-11R-A00Z-07" "TCGA-A2-A25A-01A-12R-A16F-07"
##  [131] "TCGA-A2-A04R-01A-41R-A109-07" "TCGA-A2-A04T-01A-21R-A034-07"
##  [133] "TCGA-AR-A1AX-01A-11R-A12P-07" "TCGA-C8-A26V-01A-11R-A16F-07"
##  [135] "TCGA-BH-A18R-11A-42R-A12D-07" "TCGA-AN-A0XR-01A-11R-A109-07"
##  [137] "TCGA-BH-A0HQ-01A-11R-A034-07" "TCGA-LL-A5YM-01A-11R-A28M-07"
##  [139] "TCGA-D8-A1X5-01A-11R-A14D-07" "TCGA-A7-A0DB-01A-11R-A00Z-07"
##  [141] "TCGA-EW-A1P6-01A-11R-A144-07" "TCGA-A7-A0DB-01C-02R-A277-07"
##  [143] "TCGA-LL-A73Y-01A-11R-A33J-07" "TCGA-E2-A1L8-01A-11R-A13Q-07"
##  [145] "TCGA-AR-A1AM-01A-41R-A22K-07" "TCGA-E2-A1LG-01A-21R-A14M-07"
##  [147] "TCGA-E2-A14W-01A-11R-A12D-07" "TCGA-C8-A27B-01A-11R-A169-07"
##  [149] "TCGA-EW-A1P1-01A-31R-A14D-07" "TCGA-AN-A0FN-01A-11R-A034-07"
##  [151] "TCGA-BH-A0AW-01A-11R-A056-07" "TCGA-AR-A0TX-01A-11R-A084-07"
##  [153] "TCGA-E9-A3QA-01A-61R-A22K-07" "TCGA-A2-A0SU-01A-11R-A084-07"
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##  [157] "TCGA-E9-A1RF-11A-32R-A157-07" "TCGA-BH-A0EE-01A-11R-A034-07"
##  [159] "TCGA-BH-A0H6-01A-21R-A056-07" "TCGA-OL-A66L-01A-12R-A31O-07"
##  [161] "TCGA-A7-A13D-01A-13R-A12P-07" "TCGA-A7-A13D-01A-13R-A277-07"
##  [163] "TCGA-E9-A5FK-01A-11R-A27Q-07" "TCGA-D8-A1JL-01A-11R-A13Q-07"
##  [165] "TCGA-BH-A203-01A-12R-A169-07" "TCGA-EW-A1PE-01A-11R-A144-07"
##  [167] "TCGA-AR-A0TP-01A-11R-A084-07" "TCGA-EW-A6SA-01A-21R-A32P-07"
##  [169] "TCGA-BH-A18M-11A-33R-A12D-07" "TCGA-BH-A18R-01A-11R-A12D-07"
##  [171] "TCGA-BH-A0BP-01A-11R-A115-07" "TCGA-E9-A1ND-01A-11R-A144-07"
##  [173] "TCGA-E9-A1ND-11A-43R-A144-07" "TCGA-E2-A1IH-01A-11R-A13Q-07"
##  [175] "TCGA-S3-A6ZH-01A-22R-A32P-07" "TCGA-BH-A0C1-01B-11R-A12D-07"
##  [177] "TCGA-A2-A4RX-01A-11R-A266-07" "TCGA-AC-A5EH-01A-11R-A28M-07"
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##  [179] "TCGA-AR-A0TR-01A-11R-A084-07" "TCGA-E2-A106-01A-11R-A10J-07"
##  [181] "TCGA-AQ-A1H2-01A-11R-A13Q-07" "TCGA-A2-A0CX-01A-21R-A00Z-07"
##  [183] "TCGA-AR-A0TZ-01A-12R-A084-07" "TCGA-A2-A25F-01A-11R-A169-07"
##  [185] "TCGA-E2-A15D-01A-11R-A115-07" "TCGA-E2-A14X-01A-11R-A115-07"
##  [187] "TCGA-D8-A1Y1-01A-21R-A14M-07" "TCGA-A8-A092-01A-11R-A00Z-07"
##  [189] "TCGA-E2-A573-01A-11R-A29R-07" "TCGA-EW-A1IY-01A-11R-A13Q-07"
##  [191] "TCGA-C8-A133-01A-32R-A12D-07" "TCGA-BH-A1EV-11A-24R-A137-07"
##  [193] "TCGA-B6-A2IU-01A-32R-A18M-07" "TCGA-D8-A147-01A-11R-A115-07"
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## [1129] "TCGA-AO-A0JB-01A-11R-A32Y-07" "TCGA-D8-A1JM-01A-11R-A13Q-07"
## [1131] "TCGA-AC-A2B8-01A-11R-A17B-07" "TCGA-B6-A0I8-01A-11R-A034-07"
## [1133] "TCGA-AN-A0AT-01A-11R-A034-07" "TCGA-A2-A0YC-01A-11R-A109-07"
## [1135] "TCGA-BH-A0H3-01A-11R-A12P-07" "TCGA-BH-A204-11A-53R-A157-07"
## [1137] "TCGA-A8-A09Q-01A-11R-A00Z-07" "TCGA-A2-A0T6-01A-11R-A084-07"
## [1139] "TCGA-A2-A0D1-01A-11R-A034-07" "TCGA-B6-A0WZ-01A-11R-A109-07"
## [1141] "TCGA-B6-A0RE-01A-11R-A056-07" "TCGA-A7-A13F-01A-11R-A12P-07"
## [1143] "TCGA-D8-A1XG-01A-11R-A14D-07" "TCGA-A2-A0YD-01A-11R-A109-07"
## [1145] "TCGA-E2-A14U-01A-11R-A22K-07" "TCGA-A8-A076-01A-21R-A00Z-07"
## [1147] "TCGA-BH-A0B7-11A-34R-A115-07" "TCGA-BH-A18U-01A-21R-A12D-07"
## [1149] "TCGA-A8-A06Z-01A-11R-A00Z-07" "TCGA-BH-A0BT-11A-21R-A12P-07"
```

```
## [1151] "TCGA-BH-A0H7-01A-13R-A056-07" "TCGA-BH-A0E0-01A-11R-A056-07"
## [1153] "TCGA-A7-A5ZW-01A-12R-A29R-07" "TCGA-E9-A244-01A-11R-A169-07"
## [1155] "TCGA-D8-A1JD-01A-11R-A13Q-07" "TCGA-BH-A18K-11A-13R-A12D-07"
## [1157] "TCGA-E9-A248-01A-11R-A169-07" "TCGA-A2-A0T1-01A-21R-A084-07"
## [1159] "TCGA-BH-A0AZ-11A-22R-A12P-07" "TCGA-AN-A0FX-01A-11R-A034-07"
## [1161] "TCGA-BH-A1FH-01A-12R-A13Q-07" "TCGA-A7-A3IZ-01A-11R-A213-07"
## [1163] "TCGA-D8-A13Y-01A-11R-A115-07" "TCGA-A2-A0SX-01A-12R-A084-07"
## [1165] "TCGA-B6-A0IN-01A-11R-A034-07" "TCGA-A2-A3Y0-01A-11R-A239-07"
## [1167] "TCGA-BH-A0B3-01A-11R-A056-07" "TCGA-AN-A0XO-01A-11R-A109-07"
## [1169] "TCGA-E2-A15K-11A-13R-A12P-07" "TCGA-BH-A0HF-01A-11R-A056-07"
## [1171] "TCGA-GM-A2D9-01A-11R-A18M-07" "TCGA-A2-A0CZ-01A-11R-A034-07"
## [1173] "TCGA-D8-A1JE-01A-11R-A13Q-07" "TCGA-D8-A1XT-01A-11R-A14M-07"
## [1175] "TCGA-EW-A2FS-01A-11R-A17B-07" "TCGA-A7-A0CJ-01A-21R-A00Z-07"
## [1177] "TCGA-BH-A1FG-01A-11R-A13Q-07" "TCGA-E2-A159-01A-11R-A115-07"
## [1179] "TCGA-AC-A62Y-01A-11R-A29R-07" "TCGA-A7-A13G-01A-11R-A13Q-07"
## [1181] "TCGA-AN-A0XU-01A-11R-A109-07" "TCGA-JL-A3YW-01A-12R-A239-07"
## [1183] "TCGA-BH-A1EY-01A-11R-A13Q-07" "TCGA-BH-A204-01A-11R-A157-07"
## [1185] "TCGA-LL-A5YP-01A-21R-A28M-07" "TCGA-BH-AB28-01A-31R-A41B-07"
## [1187] "TCGA-B6-A0IJ-01A-11R-A034-07" "TCGA-E2-A15E-01A-11R-A12D-07"
## [1189] "TCGA-D8-A1XZ-01A-11R-A14M-07" "TCGA-E2-A15E-06A-11R-A12D-07"
## [1191] "TCGA-A1-A0SQ-01A-21R-A144-07" "TCGA-A7-A13F-11A-42R-A12P-07"
## [1193] "TCGA-GM-A3XN-01A-12R-A22U-07" "TCGA-AC-A7VC-01A-11R-A352-07"
## [1195] "TCGA-S3-AA12-01A-11R-A41B-07" "TCGA-A2-A3KC-01A-11R-A213-07"
## [1197] "TCGA-AN-A0FY-01A-11R-A034-07" "TCGA-BH-A0B7-01A-12R-A115-07"
## [1199] "TCGA-E2-A15J-01A-11R-A12P-07" "TCGA-BH-A1F0-11B-23R-A137-07"
## [1201] "TCGA-AR-A1AJ-01A-21R-A12P-07" "TCGA-A8-A0AB-01A-11R-A034-07"
## [1203] "TCGA-E9-A1RE-01A-11R-A157-07" "TCGA-BH-A18U-11A-23R-A12D-07"
## [1205] "TCGA-BH-A0H7-11A-13R-A089-07" "TCGA-BH-A0BT-01A-11R-A12P-07"
## [1207] "TCGA-OL-A66O-01A-11R-A31O-07" "TCGA-AC-A3YI-01A-21R-A239-07"
## [1209] "TCGA-BH-A0E0-11A-13R-A089-07" "TCGA-E2-A14R-01A-11R-A115-07"
## [1211] "TCGA-LL-A5YL-01A-12R-A29R-07" "TCGA-BH-A18K-01A-11R-A12D-07"
## [1213] "TCGA-A2-A0ST-01A-12R-A084-07" "TCGA-D8-A27F-01A-11R-A16F-07"
## [1215] "TCGA-EW-A1P4-01A-21R-A144-07" "TCGA-B6-A0IB-01A-11R-A034-07"
## [1217] "TCGA-AO-A0JC-01A-11R-A056-07" "TCGA-BH-A1FH-11B-42R-A13Q-07"
## [1219] "TCGA-B6-A0X1-01A-11R-A109-07" "TCGA-E9-A1N3-01A-12R-A157-07"
## [1221] "TCGA-BH-A0B3-11B-21R-A089-07" "TCGA-AN-A0FS-01A-11R-A034-07"
## [1223] "TCGA-E2-A15K-01A-11R-A12P-07" "TCGA-AC-A7VB-01A-11R-A352-07"
## [1225] "TCGA-LD-A7W5-01A-22R-A352-07" "TCGA-B6-A0RL-01A-11R-A084-07"
## [1227] "TCGA-BH-A18P-11A-43R-A12D-07" "TCGA-C8-A8HP-01A-11R-A36F-07"
## [1229] "TCGA-EW-A3U0-01A-11R-A22K-07" "TCGA-D8-A1J9-01A-11R-A13Q-07"
## [1231] "TCGA-V7-A7HQ-01A-11R-A33J-07"
```

```
colnames(metadata)
```

```
##  [1] "patient"
##  [2] "sample"
##  [3] "shortLetterCode"
##  [4] "definition"
##  [5] "sample_submitter_id"
##  [6] "sample_type_id"
##  [7] "tumor_descriptor"
##  [8] "sample_id"
##  [9] "sample_type"
## [10] "composition"
```

```
## [11] "days_to_collection"
## [12] "state"
## [13] "initial_weight"
## [14] "preservation_method"
## [15] "pathology_report_uuid"
## [16] "submitter_id"
## [17] "oct_embedded"
## [18] "specimen_type"
## [19] "is_ffpe"
## [20] "tissue_type"
## [21] "synchronous_malignancy"
## [22] "ajcc_pathologic_stage"
## [23] "days_to_diagnosis"
## [24] "laterality"
## [25] "treatments"
## [26] "tissue_or_organ_of_origin"
## [27] "age_at_diagnosis"
## [28] "primary_diagnosis"
## [29] "metastasis_at_diagnosis"
## [30] "prior_malignancy"
## [31] "year_of_diagnosis"
## [32] "prior_treatment"
## [33] "diagnosis_is_primary_disease"
## [34] "method_of_diagnosis"
## [35] "ajcc_staging_system_edition"
## [36] "ajcc_pathologic_t"
## [37] "morphology"
## [38] "ajcc_pathologic_n"
## [39] "ajcc_pathologic_m"
## [40] "classification_of_tumor"
## [41] "diagnosis_id"
## [42] "icd_10_code"
## [43] "site_of_resection_or_biopsy"
## [44] "sites_of_involvement"
## [45] "days_to_last_follow_up"
## [46] "follow_ups_disease_response"
## [47] "race"
## [48] "gender"
## [49] "ethnicity"
## [50] "vital_status"
## [51] "age_at_index"
## [52] "days_to_birth"
## [53] "demographic_id"
## [54] "age_is_obfuscated"
## [55] "country_of_residence_at_enrollment"
## [56] "bcr_patient_barcode"
## [57] "primary_site"
## [58] "project_id"
## [59] "disease_type"
## [60] "name"
## [61] "releasable"
## [62] "released"
## [63] "figo_stage"
## [64] "figo_staging_edition_year"
```

```
## [65] "days_to_death"
## [66] "tumor_of_origin"
## [67] "days_to_sample_procurement"
## [68] "last_known_disease_status"
## [69] "tumor_grade"
## [70] "progression_or_recurrence"
## [71] "alcohol_history"
## [72] "exposure_id"
## [73] "year_of_birth"
## [74] "paper_patient"
## [75] "paper_Tumor.Type"
## [76] "paper_Included_in_previous_marker_papers"
## [77] "paper_vital_status"
## [78] "paper_days_to_birth"
## [79] "paper_days_to_death"
## [80] "paper_days_to_last_followup"
## [81] "paper_age_at_initial_pathologic_diagnosis"
## [82] "paper_pathologic_stage"
## [83] "paper_Tumor_Grade"
## [84] "paper_BRCA_Pathology"
## [85] "paper_BRCA_Subtype_PAM50"
## [86] "paper_MSI_status"
## [87] "paper_HPV_Status"
## [88] "paper_tobacco_smoking_history"
## [89] "paper_CNV.Clusters"
## [90] "paper_Mutation.Clusters"
## [91] "paper_DNA.Methylation.Clusters"
## [92] "paper_mRNA.Clusters"
## [93] "paper_miRNA.Clusters"
## [94] "paper_lncRNA.Clusters"
## [95] "paper_Protein.Clusters"
## [96] "paper_PARADIGM.Clusters"
## [97] "paper_Pan.Gyn.Clusters"
```

2. Gene Selection and Summary Statistics

2a. ERBB2 Gene Selection and Summary Statistics

```
# I see a lot of pneumotoxicity from ENHERTU which has a targeted antibody used to treat HER-2+ breast
# The gene ERBB2 codes for HER-2 so I looked up the Ensembl ID for ERBB2 and found it to be ENSG0000014
# But Ensembl IDs have version suffixes ENSG00000141736.XX so had to use grep() to find it

# Select ERBB2 gene
grep("ENSG00000141736", rownames(counts), value = TRUE) # Used ChatGPTs assistance to learn grep()
```

```
## [1] "ENSG00000141736.14"
```

```
ERBB2_data <- counts["ENSG00000141736.14", ] # from the data frame of counts,
# I'm selecting the row with my gene of interest
head(ERBB2_data)
```

```
##                    TCGA-GM-A2DL-01A-11R-A18M-07 TCGA-AC-A2QI-01A-12R-A19W-07
## ENSG00000141736.14                         7613                        41006
##                    TCGA-A8-A06R-01A-11R-A00Z-07 TCGA-EW-A1PD-01A-11R-A144-07
## ENSG00000141736.14                         6858                        52772
##                    TCGA-AO-A12D-01A-11R-A115-07 TCGA-AR-A24N-01A-11R-A169-07
## ENSG00000141736.14                       453696                        24483
##                    TCGA-AR-A24U-01A-11R-A169-07 TCGA-D8-A1JU-01A-11R-A13Q-07
## ENSG00000141736.14                       263346                        24239
##                    TCGA-A8-AOAD-01A-11R-A056-07 TCGA-E2-A3DX-01A-21R-A213-07
## ENSG00000141736.14                        27812                        81863
##                    TCGA-BH-A18J-11A-31R-A12D-07 TCGA-A2-AOYL-01A-21R-A109-07
## ENSG00000141736.14                        14617                        14478
##                    TCGA-C8-A12P-01A-11R-A115-07 TCGA-AO-AOJA-01A-11R-A056-07
## ENSG00000141736.14                       145804                        15243
##                    TCGA-EW-A1OV-01A-11R-A144-07 TCGA-5L-AAT0-01A-12R-A41B-07
## ENSG00000141736.14                        11348                        15966
##                    TCGA-BH-A18J-01A-11R-A12D-07 TCGA-D8-A27V-01A-12R-A17B-07
## ENSG00000141736.14                        19069                        24998
##                    TCGA-A7-A26I-01B-06R-A22O-07 TCGA-A8-AO96-01A-11R-A00Z-07
## ENSG00000141736.14                         5205                        29918
##                    TCGA-AR-A256-01A-11R-A169-07 TCGA-A7-A426-01A-22R-A24H-07
## ENSG00000141736.14                         6562                        20613
##                    TCGA-E2-A1BC-11A-32R-A12P-07 TCGA-E2-A1LH-11A-22R-A14D-07
## ENSG00000141736.14                          765                        23037
##                    TCGA-E2-A1LH-01A-11R-A14D-07 TCGA-E9-A1N8-01A-11R-A144-07
## ENSG00000141736.14                         6004                         5749
##                    TCGA-BH-AOHK-11A-11R-A089-07 TCGA-EW-A2FR-01A-11R-A21T-07
## ENSG00000141736.14                        19695                       622145
##                    TCGA-E2-A154-01A-11R-A115-07 TCGA-A2-AOT4-01A-31R-A084-07
## ENSG00000141736.14                        17862                        17131
##                    TCGA-E9-A1NH-01A-11R-A14D-07 TCGA-AR-A2LO-01A-31R-A18M-07
## ENSG00000141736.14                        19716                        16135
##                    TCGA-Z7-A8R6-01A-11R-A41B-07 TCGA-AC-A3YJ-01A-11R-A22U-07
## ENSG00000141736.14                        14744                        15372
##                    TCGA-A2-A04U-01A-11R-A115-07 TCGA-A7-A26I-01A-11R-A169-07
## ENSG00000141736.14                         3814                        20269
##                    TCGA-A8-A09A-01A-11R-A00Z-07 TCGA-AO-A126-01A-11R-A10J-07
## ENSG00000141736.14                        41853                        15164
##                    TCGA-EW-A1J6-01A-11R-A13Q-07 TCGA-LQ-A4E4-01A-11R-A266-07
## ENSG00000141736.14                         7547                        14356
##                    TCGA-E2-A1IL-01A-11R-A14D-07 TCGA-D8-A1XW-01A-11R-A14M-07
## ENSG00000141736.14                        23310                        14725
##                    TCGA-A8-A07C-01A-11R-A034-07 TCGA-EW-A2FV-01A-11R-A17B-07
## ENSG00000141736.14                        16121                        22038
##                    TCGA-E2-A1AZ-01A-11R-A12P-07 TCGA-AC-A2FM-11B-32R-A19W-07
## ENSG00000141736.14                         6643                        20042
##                    TCGA-BH-A1FJ-01A-11R-A13Q-07 TCGA-EW-A1J1-01A-11R-A13Q-07
## ENSG00000141736.14                        23775                        28239
##                    TCGA-A7-AODB-11A-33R-A089-07 TCGA-A7-AODB-01A-11R-A277-07
## ENSG00000141736.14                         1604                         9471
##                    TCGA-A7-A56D-01A-11R-A27Q-07 TCGA-AC-A3OD-01A-11R-A21T-07
## ENSG00000141736.14                         8670                        11252
##                    TCGA-E2-A109-01A-11R-A10J-07 TCGA-BH-AODK-11A-13R-A089-07
## ENSG00000141736.14                        38099                        28590
```

```
##                    TCGA-E2-A15O-01A-11R-A115-07 TCGA-BH-A0DK-01A-21R-A056-07
## ENSG00000141736.14                        13852                         7008
##                    TCGA-B6-A0WW-01A-11R-A109-07 TCGA-E2-A2P5-01A-11R-A19W-07
## ENSG00000141736.14                        18290                        11633
##                    TCGA-E9-A3Q9-01A-11R-A21T-07 TCGA-E2-A158-01A-11R-A12D-07
## ENSG00000141736.14                         9005                        15166
##                    TCGA-A1-A0SD-01A-11R-A115-07 TCGA-E9-A5FL-01A-11R-A27Q-07
## ENSG00000141736.14                        40159                         4779
##                    TCGA-BH-A1FU-11A-23R-A14D-07 TCGA-AR-A1AT-01A-11R-A12P-07
## ENSG00000141736.14                        12209                       397787
##                    TCGA-A1-A0SE-01A-11R-A084-07 TCGA-AN-A0XN-01A-21R-A109-07
## ENSG00000141736.14                        29159                         5611
##                    TCGA-A2-A0YG-01A-21R-A109-07 TCGA-E9-A1RF-01A-11R-A157-07
## ENSG00000141736.14                       411151                        33575
##                    TCGA-B6-A0RQ-01A-11R-A115-07 TCGA-A7-A6VY-01A-12R-A33J-07
## ENSG00000141736.14                        29882                         5003
##                    TCGA-A7-A13D-01B-04R-A277-07 TCGA-BH-A203-11A-42R-A169-07
## ENSG00000141736.14                         4663                         9071
##                    TCGA-GM-A2DD-01A-11R-A180-07 TCGA-E9-A2JT-01A-22R-A18M-07
## ENSG00000141736.14                        15615                         2639
##                    TCGA-EW-A1OW-01A-21R-A144-07 TCGA-AN-A03X-01A-21R-A00Z-07
## ENSG00000141736.14                         9191                        14430
##                    TCGA-EW-A1PB-01A-11R-A144-07 TCGA-AC-A2FO-01A-11R-A180-07
## ENSG00000141736.14                        10388                        35617
##                    TCGA-BH-A18M-01A-11R-A12D-07 TCGA-A8-A090-01A-11R-A00Z-07
## ENSG00000141736.14                        15346                        32874
##                    TCGA-AC-A2FM-01A-11R-A19W-07 TCGA-BH-A1FJ-11B-42R-A13Q-07
## ENSG00000141736.14                        71701                         7839
##                    TCGA-A8-A099-01A-11R-A00Z-07 TCGA-AO-A12B-01A-11R-A10J-07
## ENSG00000141736.14                        37882                         8664
##                    TCGA-E2-A15F-01A-11R-A115-07 TCGA-AO-A0JF-01A-11R-A056-07
## ENSG00000141736.14                        16758                        19770
##                    TCGA-E2-A576-01A-11R-A31O-07 TCGA-AN-A0AR-01A-11R-A00Z-07
## ENSG00000141736.14                        26760                         6760
##                    TCGA-AC-A3OD-01B-06R-A22O-07 TCGA-LL-A5YN-01A-11R-A28M-07
## ENSG00000141736.14                         2472                        26795
##                    TCGA-E9-A1RO-01A-22R-A16F-07 TCGA-AN-A03Y-01A-21R-A00Z-07
## ENSG00000141736.14                        31135                        13708
##                    TCGA-A7-A4SC-01A-12R-A266-07 TCGA-OL-A5D6-01A-21R-A27Q-07
## ENSG00000141736.14                        38634                        34530
##                    TCGA-EW-A1IZ-01A-11R-A13Q-07 TCGA-BH-A0EB-01A-11R-A034-07
## ENSG00000141736.14                        16594                        21113
##                    TCGA-C8-A3M8-01A-11R-A213-07 TCGA-AO-A129-01A-21R-A10J-07
## ENSG00000141736.14                        73369                         7679
##                    TCGA-D8-A27P-01A-11R-A16F-07 TCGA-E2-A152-01A-11R-A12D-07
## ENSG00000141736.14                        31524                       684776
##                    TCGA-BH-A1FU-01A-11R-A14D-07 TCGA-E9-A1NI-01A-11R-A14D-07
## ENSG00000141736.14                       287582                        13612
##                    TCGA-D8-A27M-01A-11R-A16F-07 TCGA-A2-A0YF-01A-21R-A109-07
## ENSG00000141736.14                         8680                        25389
##                    TCGA-E9-A1RG-01A-11R-A14D-07 TCGA-B6-A0X4-01A-11R-A109-07
## ENSG00000141736.14                        18136                         8290
##                    TCGA-AO-A0J4-01A-11R-A034-07 TCGA-A7-A425-01A-11R-A24H-07
## ENSG00000141736.14                        16476                        19304
```

```
##                    TCGA-A8-A08X-01A-21R-A00Z-07 TCGA-OL-A66N-01A-12R-A31O-07
## ENSG00000141736.14                       293287                        29693
##                    TCGA-B6-A1KF-01A-11R-A13Q-07 TCGA-AC-A2FE-01A-11R-A19W-07
## ENSG00000141736.14                        87217                        29450
##                    TCGA-A2-A0CM-01A-31R-A034-07 TCGA-GM-A5PX-01A-12R-A28M-07
## ENSG00000141736.14                         5447                        22508
##                    TCGA-AR-A1AR-01A-31R-A137-07 TCGA-AO-A0J6-01A-11R-A034-07
## ENSG00000141736.14                         6548                        16373
##                    TCGA-E2-A156-01A-11R-A12D-07 TCGA-D8-A73W-01A-22R-A352-07
## ENSG00000141736.14                        15344                        15649
##                    TCGA-E2-A107-01A-11R-A10J-07 TCGA-E2-A158-11A-22R-A12D-07
## ENSG00000141736.14                        38656                         3862
##                    TCGA-AC-A23G-01A-11R-A213-07 TCGA-A2-A1G4-01A-11R-A13Q-07
## ENSG00000141736.14                         9607                        14301
##                    TCGA-AO-A1KO-01A-31R-A13Q-07 TCGA-A2-A3XS-01A-11R-A22U-07
## ENSG00000141736.14                        10769                        20347
##                    TCGA-B6-A0RH-01A-21R-A115-07 TCGA-D8-A1JH-01A-11R-A13Q-07
## ENSG00000141736.14                        90541                        30992
##                    TCGA-AO-A03P-01A-11R-A00Z-07 TCGA-OL-A5DA-01A-11R-A27Q-07
## ENSG00000141736.14                        24238                        25897
##                    TCGA-A8-A07B-01A-11R-A00Z-07 TCGA-A2-A25A-01A-12R-A16F-07
## ENSG00000141736.14                        50494                        14655
##                    TCGA-A2-A04R-01A-41R-A109-07 TCGA-A2-A04T-01A-21R-A034-07
## ENSG00000141736.14                        20915                        15000
##                    TCGA-AR-A1AX-01A-11R-A12P-07 TCGA-C8-A26V-01A-11R-A16F-07
## ENSG00000141736.14                        97458                        12827
##                    TCGA-BH-A18R-11A-42R-A12D-07 TCGA-AN-A0XR-01A-11R-A109-07
## ENSG00000141736.14                        15607                         8311
##                    TCGA-BH-A0HQ-01A-11R-A034-07 TCGA-LL-A5YM-01A-11R-A28M-07
## ENSG00000141736.14                        33804                        30713
##                    TCGA-D8-A1X5-01A-11R-A14D-07 TCGA-A7-A0DB-01A-11R-A00Z-07
## ENSG00000141736.14                       107027                        23395
##                    TCGA-EW-A1P6-01A-11R-A144-07 TCGA-A7-A0DB-01C-02R-A277-07
## ENSG00000141736.14                        12522                         1646
##                    TCGA-LL-A73Y-01A-11R-A33J-07 TCGA-E2-A1L8-01A-11R-A13Q-07
## ENSG00000141736.14                        11589                        17449
##                    TCGA-AR-A1AM-01A-41R-A22K-07 TCGA-E2-A1LG-01A-21R-A14M-07
## ENSG00000141736.14                        42523                        32771
##                    TCGA-E2-A14W-01A-11R-A12D-07 TCGA-C8-A27B-01A-11R-A169-07
## ENSG00000141736.14                        15751                         4332
##                    TCGA-EW-A1P1-01A-31R-A14D-07 TCGA-AN-A0FN-01A-11R-A034-07
## ENSG00000141736.14                         9973                        11944
##                    TCGA-BH-A0AW-01A-11R-A056-07 TCGA-AR-A0TX-01A-11R-A084-07
## ENSG00000141736.14                       219209                       457857
##                    TCGA-E9-A3QA-01A-61R-A22K-07 TCGA-A2-A0SU-01A-11R-A084-07
## ENSG00000141736.14                        11550                        22951
##                    TCGA-GM-A2DA-01A-11R-A18M-07 TCGA-OL-A5D7-01A-11R-A27Q-07
## ENSG00000141736.14                        42201                         4416
##                    TCGA-E9-A1RF-11A-32R-A157-07 TCGA-BH-A0EE-01A-11R-A034-07
## ENSG00000141736.14                         1421                       120725
##                    TCGA-BH-A0H6-01A-21R-A056-07 TCGA-OL-A66L-01A-12R-A31O-07
## ENSG00000141736.14                        15097                        16576
##                    TCGA-A7-A13D-01A-13R-A12P-07 TCGA-A7-A13D-01A-13R-A277-07
## ENSG00000141736.14                        29349                        16272
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##                    TCGA-E9-A5FK-01A-11R-A27Q-07 TCGA-D8-A1JL-01A-11R-A13Q-07
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##                    TCGA-BH-A203-01A-12R-A169-07 TCGA-EW-A1PE-01A-11R-A144-07
## ENSG00000141736.14                        10897                        23287
##                    TCGA-AR-A0TP-01A-11R-A084-07 TCGA-EW-A6SA-01A-21R-A32P-07
## ENSG00000141736.14                         8340                        20438
##                    TCGA-BH-A18M-11A-33R-A12D-07 TCGA-BH-A18R-01A-11R-A12D-07
## ENSG00000141736.14                        12201                       156876
##                    TCGA-BH-A0BP-01A-11R-A115-07 TCGA-E9-A1ND-01A-11R-A144-07
## ENSG00000141736.14                         6463                         5754
##                    TCGA-E9-A1ND-11A-43R-A144-07 TCGA-E2-A1IH-01A-11R-A13Q-07
## ENSG00000141736.14                         6758                        17604
##                    TCGA-S3-A6ZH-01A-22R-A32P-07 TCGA-BH-A0C1-01B-11R-A12D-07
## ENSG00000141736.14                        42060                        24567
##                    TCGA-A2-A4RX-01A-11R-A266-07 TCGA-AC-A5EH-01A-11R-A28M-07
## ENSG00000141736.14                         9384                        17204
##                    TCGA-AR-A0TR-01A-11R-A084-07 TCGA-E2-A106-01A-11R-A10J-07
## ENSG00000141736.14                        35331                        26121
##                    TCGA-AQ-A1H2-01A-11R-A13Q-07 TCGA-A2-A0CX-01A-21R-A00Z-07
## ENSG00000141736.14                        39092                       267883
##                    TCGA-AR-A0TZ-01A-12R-A084-07 TCGA-A2-A25F-01A-11R-A169-07
## ENSG00000141736.14                        48238                         4337
##                    TCGA-E2-A15D-01A-11R-A115-07 TCGA-E2-A14X-01A-11R-A115-07
## ENSG00000141736.14                        29404                        17384
##                    TCGA-D8-A1Y1-01A-21R-A14M-07 TCGA-A8-A092-01A-11R-A00Z-07
## ENSG00000141736.14                        24222                        45078
##                    TCGA-E2-A573-01A-11R-A29R-07 TCGA-EW-A1IY-01A-11R-A13Q-07
## ENSG00000141736.14                         7280                         7551
##                    TCGA-C8-A133-01A-32R-A12D-07 TCGA-BH-A1EV-11A-24R-A137-07
## ENSG00000141736.14                        12515                        55794
##                    TCGA-B6-A2IU-01A-32R-A18M-07 TCGA-D8-A147-01A-11R-A115-07
## ENSG00000141736.14                        20284                        14254
##                    TCGA-AC-A3EH-01A-22R-A22K-07 TCGA-A8-A09Z-01A-11R-A00Z-07
## ENSG00000141736.14                        52891                        32312
##                    TCGA-BH-A1FR-01A-11R-A13Q-07 TCGA-E2-A10B-01A-11R-A10J-07
## ENSG00000141736.14                         9541                        19539
##                    TCGA-E2-A2P6-01A-11R-A19W-07 TCGA-E9-A227-01A-11R-A157-07
## ENSG00000141736.14                        31991                        15378
##                    TCGA-C8-A132-01A-31R-A115-07 TCGA-AQ-A1H3-01A-31R-A13Q-07
## ENSG00000141736.14                       396603                        56352
##                    TCGA-A2-A0YE-01A-11R-A109-07 TCGA-A8-A08L-01A-11R-A00Z-07
## ENSG00000141736.14                        18473                        38857
##                    TCGA-S3-AA17-01A-11R-A41B-07 TCGA-BH-A0BC-01A-22R-A084-07
## ENSG00000141736.14                        17369                         9628
##                    TCGA-AR-A2LK-01A-11R-A180-07 TCGA-AO-A03R-01A-21R-A034-07
## ENSG00000141736.14                        20310                         2483
##                    TCGA-BH-A1ET-01A-11R-A137-07 TCGA-OL-A6VO-01A-12R-A33J-07
## ENSG00000141736.14                        32420                        14166
##                    TCGA-AR-A1AU-01A-11R-A12P-07 TCGA-AO-A1KP-01A-11R-A13Q-07
## ENSG00000141736.14                        20071                         9959
##                    TCGA-A8-A08R-01A-11R-A034-07 TCGA-BH-A0E1-11A-13R-A089-07
## ENSG00000141736.14                        36045                        11141
##                    TCGA-AR-A0TV-01A-21R-A084-07 TCGA-E9-A1RD-01A-11R-A157-07
## ENSG00000141736.14                        10597                        25680
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##                    TCGA-A8-A09G-01A-21R-A00Z-07 TCGA-AR-A1AI-01A-11R-A12P-07
## ENSG00000141736.14                       176140                       14041
##                    TCGA-BH-A1FC-11A-32R-A13Q-07 TCGA-BH-A1FC-01A-11R-A13Q-07
## ENSG00000141736.14                        10340                        6394
##                    TCGA-C8-A12N-01A-11R-A115-07 TCGA-B6-A40C-01A-11R-A239-07
## ENSG00000141736.14                        10259                       35302
##                    TCGA-AC-A3TN-01A-11R-A22K-07 TCGA-A2-A0CP-01A-11R-A034-07
## ENSG00000141736.14                        37714                       10046
##                    TCGA-E2-A14O-01A-31R-A115-07 TCGA-AQ-A04J-01A-02R-A034-07
## ENSG00000141736.14                        17462                       11578
##                    TCGA-S3-A6ZF-01A-32R-A32P-07 TCGA-A2-A0EU-01A-22R-A056-07
## ENSG00000141736.14                        22003                       15111
##                    TCGA-C8-A12O-01A-11R-A115-07 TCGA-A2-A0CW-01A-21R-A115-07
## ENSG00000141736.14                        16251                       31044
##                    TCGA-E9-A1RA-01A-11R-A14D-07 TCGA-A1-A0SF-01A-11R-A144-07
## ENSG00000141736.14                        14666                       19182
##                    TCGA-AC-A3W5-01A-11R-A22K-07 TCGA-EW-A1PA-01A-11R-A144-07
## ENSG00000141736.14                       469350                       10148
##                    TCGA-D8-A73X-01A-11R-A32P-07 TCGA-C8-A8HR-01A-11R-A36F-07
## ENSG00000141736.14                        20759                       13501
##                    TCGA-A0-A0J3-01A-11R-A034-07 TCGA-D8-A27L-01A-11R-A16F-07
## ENSG00000141736.14                        13678                       27520
##                    TCGA-A8-A09D-01A-11R-A00Z-07 TCGA-EW-A1OX-01A-11R-A144-07
## ENSG00000141736.14                        12643                       13877
##                    TCGA-AR-A1AY-01A-21R-A12P-07 TCGA-LD-A66U-01A-11R-A31O-07
## ENSG00000141736.14                        10171                       26722
##                    TCGA-C8-A12Z-01A-11R-A115-07 TCGA-LD-A7W6-01A-81R-A352-07
## ENSG00000141736.14                       771526                       25114
##                    TCGA-E9-A1N9-11A-71R-A14D-07 TCGA-E9-A228-01A-31R-A157-07
## ENSG00000141736.14                        19125                      113624
##                    TCGA-A7-A6VW-01A-21R-A33J-07 TCGA-BH-A0HP-01A-12R-A084-07
## ENSG00000141736.14                         8297                       19876
##                    TCGA-EW-A1P0-01A-11R-A144-07 TCGA-A8-A094-01A-11R-A00Z-07
## ENSG00000141736.14                         1143                        8145
##                    TCGA-BH-A1FR-11B-42R-A13Q-07 TCGA-D8-A4Z1-01A-21R-A266-07
## ENSG00000141736.14                        10290                       21201
##                    TCGA-A2-A1FZ-01A-51R-A14D-07 TCGA-BH-A18F-01A-11R-A12D-07
## ENSG00000141736.14                        23043                        8873
##                    TCGA-BH-A0HO-01A-11R-A034-07 TCGA-B6-A0RI-01A-11R-A056-07
## ENSG00000141736.14                        22277                       16740
##                    TCGA-S3-AA15-01A-11R-A41B-07 TCGA-E9-A245-01A-22R-A16F-07
## ENSG00000141736.14                         3056                        9608
##                    TCGA-AN-A041-01A-11R-A034-07 TCGA-A0-A125-01A-11R-A10J-07
## ENSG00000141736.14                        58437                       16576
##                    TCGA-BH-A0BA-11A-22R-A19E-07 TCGA-AN-A0FV-01A-11R-A00Z-07
## ENSG00000141736.14                        10989                      519623
##                    TCGA-AN-A0FK-01A-11R-A034-07 TCGA-B6-A0I2-01A-11R-A034-07
## ENSG00000141736.14                        14702                        9992
##                    TCGA-BH-A0DV-01A-21R-A12P-07 TCGA-OL-A66J-01A-11R-A29R-07
## ENSG00000141736.14                        25102                       32526
##                    TCGA-D8-A73U-01A-11R-A33J-07 TCGA-D8-A1XQ-01A-11R-A14M-07
## ENSG00000141736.14                         8668                       12565
##                    TCGA-A2-A0T5-01A-21R-A084-07 TCGA-AR-A2LQ-01A-22R-A18M-07
## ENSG00000141736.14                        27237                       23639
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##                    TCGA-HN-A2OB-01A-21R-A27Q-07 TCGA-AR-A1AO-01A-11R-A12P-07
## ENSG00000141736.14                        33595                        10224
##                    TCGA-BH-A0BR-01A-21R-A115-07 TCGA-A8-A0A1-01A-11R-A00Z-07
## ENSG00000141736.14                        14499                        12615
##                    TCGA-BH-A0DG-11A-43R-A12P-07 TCGA-GM-A4E0-01A-12R-A266-07
## ENSG00000141736.14                        19990                         6850
##                    TCGA-E9-A3HO-01A-11R-A213-07 TCGA-E9-A1N5-01A-11R-A14D-07
## ENSG00000141736.14                        18138                         6894
##                    TCGA-AR-A1AQ-01A-11R-A12P-07 TCGA-A8-A086-01A-11R-A00Z-07
## ENSG00000141736.14                         9212                        15016
##                    TCGA-BH-A0DS-01A-11R-A056-07 TCGA-E9-A295-01A-11R-A16F-07
## ENSG00000141736.14                        16668                        15363
##                    TCGA-BH-A42U-01A-12R-A24H-07 TCGA-AO-A0JJ-01A-11R-A056-07
## ENSG00000141736.14                        25084                        11921
##                    TCGA-AQ-A04H-01B-11R-A10J-07 TCGA-E9-A2JS-01A-11R-A180-07
## ENSG00000141736.14                       113705                        34540
##                    TCGA-AN-A049-01A-21R-A00Z-07 TCGA-A8-A08T-01A-21R-A00Z-07
## ENSG00000141736.14                        37495                        26413
##                    TCGA-A8-A079-01A-21R-A00Z-07 TCGA-GM-A2DC-01A-11R-A18M-07
## ENSG00000141736.14                        14698                        22365
##                    TCGA-E9-A226-01A-21R-A157-07 TCGA-AR-A254-01A-21R-A169-07
## ENSG00000141736.14                        10344                       449575
##                    TCGA-BH-A0DG-01A-21R-A12P-07 TCGA-E9-A1RH-11A-34R-A169-07
## ENSG00000141736.14                        11470                         3645
##                    TCGA-D8-A1J8-01A-11R-A13Q-07 TCGA-E9-A1RH-01A-21R-A169-07
## ENSG00000141736.14                         5540                        12622
##                    TCGA-A8-A0A7-01A-11R-A00Z-07 TCGA-AC-A6IW-01A-12R-A33J-07
## ENSG00000141736.14                       154513                         7626
##                    TCGA-A2-A0SW-01A-11R-A084-07 TCGA-XX-A899-01A-11R-A36F-07
## ENSG00000141736.14                         3094                        22720
##                    TCGA-PE-A5DE-01A-11R-A27Q-07 TCGA-E2-A150-01A-11R-A12D-07
## ENSG00000141736.14                         9352                         9252
##                    TCGA-A8-A0A6-01A-12R-A056-07 TCGA-AN-A04D-01A-21R-A034-07
## ENSG00000141736.14                        21939                        14936
##                    TCGA-C8-A1HM-01A-12R-A137-07 TCGA-A7-A4SD-01A-11R-A266-07
## ENSG00000141736.14                         9571                        10767
##                    TCGA-BH-A0BA-01A-11R-A056-07 TCGA-BH-A0BC-11A-22R-A089-07
## ENSG00000141736.14                        24092                        11080
##                    TCGA-BH-A1ET-11B-23R-A137-07 TCGA-A2-A0CT-01A-31R-A056-07
## ENSG00000141736.14                        18605                        36018
##                    TCGA-D8-A1JF-01A-11R-A13Q-07 TCGA-BH-A0E1-01A-11R-A056-07
## ENSG00000141736.14                        25385                        12330
##                    TCGA-A8-A097-01A-11R-A034-07 TCGA-LD-A74U-01A-13R-A33J-07
## ENSG00000141736.14                       129284                        14057
##                    TCGA-AO-A0JE-01A-11R-A056-07 TCGA-A2-A4S3-01A-21R-A266-07
## ENSG00000141736.14                       120763                         9678
##                    TCGA-E9-A1RD-11A-33R-A157-07 TCGA-AO-A03T-01A-21R-A034-07
## ENSG00000141736.14                         6555                         8581
##                    TCGA-A8-A09M-01A-11R-A00Z-07 TCGA-E9-A22B-01A-11R-A157-07
## ENSG00000141736.14                        22202                        20236
##                    TCGA-E9-A1N9-01A-11R-A14D-07 TCGA-BH-A1EV-01A-11R-A137-07
## ENSG00000141736.14                         9225                       292516
##                    TCGA-AO-A1KR-01A-12R-A144-07 TCGA-OL-A5RU-01A-11R-A28M-07
## ENSG00000141736.14                        10613                        33437
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##                    TCGA-E9-A1N5-11A-41R-A14D-07 TCGA-C8-A12U-01A-11R-A115-07
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##                    TCGA-LL-A5YO-01A-21R-A28M-07 TCGA-A8-A08A-01A-11R-A32Y-07
## ENSG00000141736.14                        10455                        26191
##                    TCGA-PL-A8LX-01A-11R-A41B-07 TCGA-E9-A249-01A-11R-A169-07
## ENSG00000141736.14                        25551                        36628
##                    TCGA-A2-A3XZ-01A-42R-A239-07 TCGA-OL-A66P-01A-11R-A31O-07
## ENSG00000141736.14                       610336                        22978
##                    TCGA-C8-A26W-01A-11R-A16F-07 TCGA-A7-A13H-01A-11R-A22K-07
## ENSG00000141736.14                        73993                        40096
##                    TCGA-A2-A4S1-01A-21R-A266-07 TCGA-BH-A0E2-01A-11R-A056-07
## ENSG00000141736.14                         205                         8856
##                    TCGA-BH-A0HY-01A-11R-A056-07 TCGA-EW-A1PF-01A-11R-A144-07
## ENSG00000141736.14                       410137                        17028
##                    TCGA-AC-A3W6-01A-12R-A22K-07 TCGA-B6-A40B-01A-11R-A239-07
## ENSG00000141736.14                        10057                        38801
##                    TCGA-A2-A0D0-01A-11R-A00Z-07 TCGA-A2-A0D3-01A-11R-A115-07
## ENSG00000141736.14                         6183                        33934
##                    TCGA-BH-A42T-01A-11R-A24H-07 TCGA-A2-A04N-01A-11R-A115-07
## ENSG00000141736.14                        65588                        16389
##                    TCGA-GI-A2C9-01A-11R-A21T-07 TCGA-GI-A2C9-11A-22R-A21T-07
## ENSG00000141736.14                        11003                        12040
##                    TCGA-E2-A15P-01A-11R-A115-07 TCGA-AO-A03L-01A-41R-A056-07
## ENSG00000141736.14                        51459                        36330
##                    TCGA-3C-AALI-01A-11R-A41B-07 TCGA-E9-A22E-01A-11R-A157-07
## ENSG00000141736.14                       367620                         5421
##                    TCGA-A2-A04V-01A-21R-A034-07 TCGA-E9-A1NG-01A-21R-A14M-07
## ENSG00000141736.14                        24927                        16789
##                    TCGA-D8-A1JK-01A-11R-A13Q-07 TCGA-BH-A0DO-01B-11R-A12D-07
## ENSG00000141736.14                        10214                        23445
##                    TCGA-AR-A24P-01A-11R-A169-07 TCGA-AC-A2QJ-01A-12R-A19W-07
## ENSG00000141736.14                        37635                         3844
##                    TCGA-AO-A0J2-01A-11R-A034-07 TCGA-BH-A1FG-11B-12R-A13Q-07
## ENSG00000141736.14                        31317                         3625
##                    TCGA-BH-A0AY-01A-21R-A00Z-07 TCGA-A2-A0T0-01A-22R-A084-07
## ENSG00000141736.14                         8735                        10778
##                    TCGA-A2-A04W-01A-31R-A115-07 TCGA-B6-A0IG-01A-11R-A034-07
## ENSG00000141736.14                       179229                       364758
##                    TCGA-AN-A0XS-01A-22R-A109-07 TCGA-D8-A1X9-01A-12R-A157-07
## ENSG00000141736.14                        22876                        52468
##                    TCGA-5T-A9QA-01A-11R-A41B-07 TCGA-E9-A1NE-01A-21R-A14M-07
## ENSG00000141736.14                        22634                        17678
##                    TCGA-A2-A0EM-01A-11R-A034-07 TCGA-AC-A8OQ-01A-11R-A41B-07
## ENSG00000141736.14                        21360                        10947
##                    TCGA-AC-A23C-01A-12R-A169-07 TCGA-AN-A0XV-01A-11R-A109-07
## ENSG00000141736.14                       141748                        13072
##                    TCGA-PE-A5DD-01A-12R-A27Q-07 TCGA-BH-A0H5-01A-21R-A115-07
## ENSG00000141736.14                         9270                        17022
##                    TCGA-A2-A1FX-01A-11R-A13Q-07 TCGA-E2-A105-01A-11R-A10J-07
## ENSG00000141736.14                        21905                        31493
##                    TCGA-A8-A09R-01A-11R-A00Z-07 TCGA-C8-A1HE-01A-11R-A13Q-07
## ENSG00000141736.14                        48757                        43702
##                    TCGA-A8-A09E-01A-11R-A00Z-07 TCGA-A7-A0DC-11A-41R-A089-07
## ENSG00000141736.14                        38670                         9309
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##                    TCGA-A8-A08F-01A-11R-A00Z-07 TCGA-LL-A6FP-01A-11R-A31O-07
## ENSG00000141736.14                        16779                        33899
##                    TCGA-5L-AAT1-01A-12R-A41B-07 TCGA-A8-A07E-01A-11R-A034-07
## ENSG00000141736.14                        16060                        14232
##                    TCGA-GI-A2C8-11A-22R-A16F-07 TCGA-A8-A09K-01A-11R-A00Z-07
## ENSG00000141736.14                         3371                        20408
##                    TCGA-A8-A09I-01A-22R-A034-07 TCGA-AO-A0JI-01A-21R-A056-07
## ENSG00000141736.14                       995067                        28077
##                    TCGA-B6-A1KC-01B-11R-A157-07 TCGA-C8-A1HO-01A-11R-A13Q-07
## ENSG00000141736.14                        15026                         6325
##                    TCGA-E2-A1L7-01A-11R-A144-07 TCGA-C8-A274-01A-11R-A16F-07
## ENSG00000141736.14                         4348                        32263
##                    TCGA-D8-A1XU-01A-11R-A14M-07 TCGA-AR-A24Q-01A-12R-A169-07
## ENSG00000141736.14                        16224                        15256
##                    TCGA-BH-A18L-01A-32R-A12D-07 TCGA-A2-A0YM-01A-11R-A109-07
## ENSG00000141736.14                        43443                        14475
##                    TCGA-BH-A0B2-01A-11R-A10J-07 TCGA-E9-A22A-01A-11R-A157-07
## ENSG00000141736.14                        26914                         8177
##                    TCGA-A8-A08B-01A-11R-A00Z-07 TCGA-BH-A0B0-01A-21R-A115-07
## ENSG00000141736.14                       119209                        10006
##                    TCGA-BH-A1FE-01A-11R-A13Q-07 TCGA-A1-A0SH-01A-11R-A084-07
## ENSG00000141736.14                        28046                        51040
##                    TCGA-A8-A095-01A-11R-A00Z-07 TCGA-BH-A0BW-01A-11R-A115-07
## ENSG00000141736.14                        15368                        19198
##                    TCGA-AN-A0FF-01A-11R-A034-07 TCGA-A2-A3XX-01A-21R-A239-07
## ENSG00000141736.14                        14016                        21954
##                    TCGA-BH-A1EN-01A-11R-A13Q-07 TCGA-BH-A1EN-11A-23R-A13Q-07
## ENSG00000141736.14                       552724                        27760
##                    TCGA-A2-A3XW-01A-11R-A239-07 TCGA-A8-A08S-01A-11R-A034-07
## ENSG00000141736.14                        28311                       142803
##                    TCGA-E2-A14V-01A-11R-A12D-07 TCGA-A8-A09C-01A-11R-A00Z-07
## ENSG00000141736.14                       236513                        15713
##                    TCGA-D8-A1Y2-01A-11R-A157-07 TCGA-GM-A5PV-01A-11R-A28M-07
## ENSG00000141736.14                        26291                        27196
##                    TCGA-E2-A1B1-01A-21R-A12P-07 TCGA-B6-A0WX-01A-11R-A109-07
## ENSG00000141736.14                       132242                        10866
##                    TCGA-EW-A1J2-01A-21R-A13Q-07 TCGA-AR-A255-01A-11R-A169-07
## ENSG00000141736.14                        54974                        24890
##                    TCGA-UL-AAZ6-01A-11R-A41B-07 TCGA-A2-A0CL-01A-11R-A115-07
## ENSG00000141736.14                       395481                         9196
##                    TCGA-AC-A3HN-01A-11R-A213-07 TCGA-E2-A108-01A-13R-A10J-07
## ENSG00000141736.14                        38380                         6952
##                    TCGA-BH-A1F8-11B-21R-A13Q-07 TCGA-BH-A1F8-01A-11R-A13Q-07
## ENSG00000141736.14                         8059                        20684
##                    TCGA-A8-A08I-01A-11R-A00Z-07 TCGA-A8-A08J-01A-11R-A00Z-07
## ENSG00000141736.14                        17389                       379098
##                    TCGA-A8-A08P-01A-11R-A00Z-07 TCGA-C8-A12M-01A-11R-A115-07
## ENSG00000141736.14                        29885                        25707
##                    TCGA-A2-A0EQ-01A-11R-A034-07 TCGA-BH-A0W7-01A-11R-A115-07
## ENSG00000141736.14                        15761                        10658
##                    TCGA-D8-A27W-01A-11R-A16F-07 TCGA-BH-A0HN-01A-11R-A10U-07
## ENSG00000141736.14                        45723                        23510
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##                    TCGA-A2-A1FW-01A-11R-A13Q-07 TCGA-E9-A1RI-11A-41R-A169-07
## ENSG00000141736.14                        10997                         1778
##                    TCGA-BH-A0DP-01A-21R-A056-07 TCGA-A7-A26G-01A-21R-A169-07
## ENSG00000141736.14                        15709                         4171
##                    TCGA-3C-AALK-01A-11R-A41B-07 TCGA-C8-A12Q-01A-11R-A115-07
## ENSG00000141736.14                        87332                       547420
##                    TCGA-BH-A0B5-11A-23R-A12P-07 TCGA-BH-A1F6-01A-11R-A13Q-07
## ENSG00000141736.14                         1146                         7633
##                    TCGA-A0-A03M-01B-11R-A10J-07 TCGA-C8-A131-01A-11R-A115-07
## ENSG00000141736.14                        53320                         8274
##                    TCGA-AC-A6IV-01A-12R-A33J-07 TCGA-A2-A0T2-01A-11R-A084-07
## ENSG00000141736.14                        21421                         8237
##                    TCGA-E9-A1NC-01A-21R-A26B-07 TCGA-XX-A89A-01A-11R-A36F-07
## ENSG00000141736.14                         7867                         6900
##                    TCGA-E9-A5UO-01A-11R-A28M-07 TCGA-LL-A740-01A-21R-A32P-07
## ENSG00000141736.14                        32455                        37030
##                    TCGA-E9-A1N6-11A-32R-A144-07 TCGA-AN-A046-01A-21R-A034-07
## ENSG00000141736.14                         9472                        21359
##                    TCGA-E9-A1N6-01A-11R-A144-07 TCGA-BH-A18G-01A-11R-A12D-07
## ENSG00000141736.14                        32291                         2914
##                    TCGA-B6-A0RO-01A-22R-A084-07 TCGA-E9-A54Y-01A-11R-A466-07
## ENSG00000141736.14                        31899                        13451
##                    TCGA-BH-A1ES-01A-11R-A137-07 TCGA-BH-A1ES-06A-12R-A24H-07
## ENSG00000141736.14                         8677                        32245
##                    TCGA-E9-A1R5-01A-11R-A14M-07 TCGA-B6-A0IC-01A-11R-A034-07
## ENSG00000141736.14                        37231                        33799
##                    TCGA-A2-A0SY-01A-31R-A084-07 TCGA-AC-A2BK-01A-11R-A21T-07
## ENSG00000141736.14                        75675                         5125
##                    TCGA-C8-A26X-01A-31R-A16F-07 TCGA-A8-A07U-01A-11R-A034-07
## ENSG00000141736.14                        29846                        17215
##                    TCGA-A8-A07J-01A-11R-A00Z-07 TCGA-BH-A18H-01A-11R-A12D-07
## ENSG00000141736.14                        18130                         8982
##                    TCGA-E2-A14P-01A-31R-A12D-07 TCGA-AN-A04A-01A-21R-A034-07
## ENSG00000141736.14                       465371                        11645
##                    TCGA-BH-A202-01A-11R-A14M-07 TCGA-WT-AB41-01A-11R-A41B-07
## ENSG00000141736.14                       185097                         7258
##                    TCGA-BH-A0RX-01A-21R-A084-07 TCGA-A8-A07L-01A-11R-A00Z-07
## ENSG00000141736.14                         6933                        20447
##                    TCGA-D8-A140-01A-11R-A115-07 TCGA-AR-A5QQ-01A-11R-A28M-07
## ENSG00000141736.14                        42816                         3858
##                    TCGA-BH-A0HX-01A-21R-A056-07 TCGA-C8-A27A-01A-11R-A169-07
## ENSG00000141736.14                        19474                        15464
##                    TCGA-A2-A25D-01A-12R-A16F-07 TCGA-BH-A201-01A-11R-A14M-07
## ENSG00000141736.14                        27106                        21866
##                    TCGA-A8-A085-01A-11R-A00Z-07 TCGA-AR-A5QP-01A-11R-A28M-07
## ENSG00000141736.14                        42373                        17170
##                    TCGA-BH-A0W3-01A-11R-A109-07 TCGA-AR-A24V-01A-21R-A169-07
## ENSG00000141736.14                        27136                        21342
##                    TCGA-A8-A09V-01A-11R-A034-07 TCGA-BH-A0HK-01A-11R-A056-07
## ENSG00000141736.14                        25816                        16329
##                    TCGA-E2-A14Q-01A-11R-A12D-07 TCGA-D8-A1X8-01A-11R-A14M-07
## ENSG00000141736.14                        15887                        30007
##                    TCGA-BH-A0H5-11A-62R-A115-07 TCGA-A7-A0CD-01A-11R-A00Z-07
## ENSG00000141736.14                        16977                        56778
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##                    TCGA-A8-A06T-01A-11R-A00Z-07 TCGA-A8-A06U-01A-11R-A00Z-07
## ENSG00000141736.14                        47189                        33642
##                    TCGA-AQ-A0Y5-01A-11R-A14M-07 TCGA-AN-A0XW-01A-11R-A109-07
## ENSG00000141736.14                       408733                        12397
##                    TCGA-BH-A0B1-01A-12R-A056-07 TCGA-AC-A62V-01A-11R-A31O-07
## ENSG00000141736.14                        16084                        15018
##                    TCGA-AC-A2QH-01B-04R-A22O-07 TCGA-A7-A4SB-01A-21R-A266-07
## ENSG00000141736.14                          490                        24722
##                    TCGA-A2-A0SV-01A-11R-A084-07 TCGA-BH-A0BS-01A-11R-A12P-07
## ENSG00000141736.14                        24302                        36222
##                    TCGA-PE-A5DC-01A-12R-A27Q-07 TCGA-BH-A0HA-01A-11R-A12P-07
## ENSG00000141736.14                        19872                        14383
##                    TCGA-A8-A06P-01A-11R-A00Z-07 TCGA-BH-A0HA-11A-31R-A12P-07
## ENSG00000141736.14                        21451                        13044
##                    TCGA-A2-A0CO-01A-13R-A22K-07 TCGA-A2-A259-01A-11R-A16F-07
## ENSG00000141736.14                        28689                        12818
##                    TCGA-A8-A09T-01A-11R-A00Z-07 TCGA-E9-A6HE-01A-11R-A31O-07
## ENSG00000141736.14                         8484                        30311
##                    TCGA-A2-A0CR-01A-11R-A22K-07 TCGA-AR-A24M-01A-11R-A169-07
## ENSG00000141736.14                        10262                        30282
##                    TCGA-AC-A4ZE-01A-11R-A41B-07 TCGA-A2-A25B-01A-11R-A169-07
## ENSG00000141736.14                        31073                         9404
##                    TCGA-BH-A1FE-11B-14R-A13Q-07 TCGA-BH-A0BM-01A-11R-A056-07
## ENSG00000141736.14                         7381                        10222
##                    TCGA-AR-A0U0-01A-11R-A109-07 TCGA-AR-A2LL-01A-11R-A180-07
## ENSG00000141736.14                        12266                        51737
##                    TCGA-B6-A0IK-01A-12R-A056-07 TCGA-A2-A0D4-01A-11R-A00Z-07
## ENSG00000141736.14                        14034                        40543
##                    TCGA-E9-A1R4-01A-21R-A14D-07 TCGA-AC-A2QH-01A-11R-A18M-07
## ENSG00000141736.14                        14231                         2186
##                    TCGA-A0-A1KQ-01A-11R-A13Q-07 TCGA-A8-A06Q-01A-11R-A034-07
## ENSG00000141736.14                        11395                        26178
##                    TCGA-AC-A8OP-01A-11R-A36F-07 TCGA-BH-A18L-11A-42R-A12D-07
## ENSG00000141736.14                        23588                        11561
##                    TCGA-E2-A1IG-11A-22R-A144-07 TCGA-LL-A442-01A-11R-A24H-07
## ENSG00000141736.14                         7885                        37611
##                    TCGA-E2-A155-01A-11R-A12D-07 TCGA-C8-A134-01A-11R-A115-07
## ENSG00000141736.14                        56880                        13823
##                    TCGA-BH-A0BS-11A-11R-A12P-07 TCGA-AR-A2LE-01A-11R-A180-07
## ENSG00000141736.14                        22037                        19292
##                    TCGA-BH-A28Q-01A-11R-A16F-07 TCGA-AR-A252-01A-11R-A169-07
## ENSG00000141736.14                        43657                         5230
##                    TCGA-E2-A572-01A-13R-A31O-07 TCGA-BH-A0DI-01A-21R-A12P-07
## ENSG00000141736.14                        17458                        23439
##                    TCGA-E2-A1IO-01A-11R-A144-07 TCGA-AN-A0FD-01A-11R-A034-07
## ENSG00000141736.14                        27501                         9118
##                    TCGA-E9-A1NA-01A-11R-A144-07 TCGA-A1-A0SN-01A-11R-A144-07
## ENSG00000141736.14                        14458                       501389
##                    TCGA-D8-A1XL-01A-11R-A14M-07 TCGA-A0-A03O-01A-11R-A00Z-07
## ENSG00000141736.14                        50328                        14408
##                    TCGA-B6-A0IP-01A-11R-A034-07 TCGA-BH-A0W5-01A-11R-A109-07
## ENSG00000141736.14                        37809                        15354
##                    TCGA-B6-A0IM-01A-11R-A034-07 TCGA-A0-A128-01A-11R-A10J-07
## ENSG00000141736.14                        40759                        20681
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##                    TCGA-AO-A124-01A-11R-A10J-07 TCGA-BH-A0BZ-01A-31R-A12P-07
## ENSG00000141736.14                        28326                        23585
##                    TCGA-OL-A66K-01A-11R-A29R-07 TCGA-A2-A0ET-01A-31R-A034-07
## ENSG00000141736.14                        32517                        31750
##                    TCGA-BH-A1F5-01A-12R-A13Q-07 TCGA-A2-A0YT-01A-11R-A109-07
## ENSG00000141736.14                         9360                        23788
##                    TCGA-A2-A0EO-01A-11R-A034-07 TCGA-D8-A27G-01A-11R-A16F-07
## ENSG00000141736.14                        36287                        66899
##                    TCGA-E9-A5UP-01A-11R-A28M-07 TCGA-E9-A1NF-11A-73R-A14D-07
## ENSG00000141736.14                        69227                         1969
##                    TCGA-E9-A1NF-01A-11R-A14D-07 TCGA-B6-A0RT-01A-21R-A084-07
## ENSG00000141736.14                         7497                        10156
##                    TCGA-A1-A0SP-01A-11R-A084-07 TCGA-C8-A278-01A-11R-A169-07
## ENSG00000141736.14                         8843                       201820
##                    TCGA-AC-A2BM-01A-11R-A21T-07 TCGA-A2-A0CQ-01A-21R-A034-07
## ENSG00000141736.14                         8929                        19248
##                    TCGA-E9-A1QZ-01A-21R-A169-07 TCGA-BH-A6R8-01A-21R-A33J-07
## ENSG00000141736.14                        25940                        25091
##                    TCGA-AR-A1AS-01A-11R-A12P-07 TCGA-D8-A27K-01A-11R-A16F-07
## ENSG00000141736.14                        15713                        37685
##                    TCGA-LD-A9QF-01A-32R-A41B-07 TCGA-BH-A0H9-01A-11R-A056-07
## ENSG00000141736.14                       384081                        18163
##                    TCGA-E9-A1RI-01A-11R-A169-07 TCGA-BH-A209-11A-42R-A157-07
## ENSG00000141736.14                        38426                        11080
##                    TCGA-BH-A0DP-11A-12R-A089-07 TCGA-BH-A0BQ-11A-33R-A115-07
## ENSG00000141736.14                        39933                        13579
##                    TCGA-W8-A86G-01A-21R-A36F-07 TCGA-E2-A574-01A-11R-A29R-07
## ENSG00000141736.14                        21310                         7547
##                    TCGA-A1-A0SG-01A-11R-A144-07 TCGA-AO-A0J9-01A-11R-A034-07
## ENSG00000141736.14                        30694                         6452
##                    TCGA-A8-A06N-01A-11R-A00Z-07 TCGA-B6-A0WY-01A-11R-A109-07
## ENSG00000141736.14                        55363                        27272
##                    TCGA-EW-A424-01A-11R-A24H-07 TCGA-AR-A1AL-01A-21R-A12P-07
## ENSG00000141736.14                        22195                        22298
##                    TCGA-D8-A1X6-01A-11R-A14M-07 TCGA-E2-A1L6-01A-11R-A13Q-07
## ENSG00000141736.14                        20901                        20767
##                    TCGA-E2-A15S-01A-11R-A115-07 TCGA-A2-A04Q-01A-21R-A034-07
## ENSG00000141736.14                        29043                         8648
##                    TCGA-GM-A3XG-01A-31R-A24H-07 TCGA-A8-A0A9-01A-11R-A00Z-07
## ENSG00000141736.14                        32529                        38356
##                    TCGA-Z7-A8R5-01A-42R-A41B-07 TCGA-C8-A12T-01A-11R-A115-07
## ENSG00000141736.14                        12893                        98397
##                    TCGA-OL-A97C-01A-32R-A41B-07 TCGA-A8-A06X-01A-21R-A00Z-07
## ENSG00000141736.14                         8074                        17066
##                    TCGA-E2-A1IF-01A-11R-A144-07 TCGA-AR-A0U4-01A-11R-A109-07
## ENSG00000141736.14                        12308                         4637
##                    TCGA-BH-A0DT-01A-21R-A12D-07 TCGA-A7-A3RF-01A-11R-A22K-07
## ENSG00000141736.14                        24460                         2406
##                    TCGA-A2-A0ER-01A-21R-A034-07 TCGA-AO-A12A-01A-21R-A115-07
## ENSG00000141736.14                        24932                        18323
##                    TCGA-AR-A5QM-01A-11R-A27Q-07 TCGA-B6-A0RG-01A-11R-A056-07
## ENSG00000141736.14                        15031                        19104
##                    TCGA-PL-A8LY-01A-11R-A41B-07 TCGA-A2-A4S2-01A-12R-A266-07
## ENSG00000141736.14                        15596                        16308
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##                    TCGA-A7-A0CH-01A-21R-A00Z-07 TCGA-A7-A0CH-11A-32R-A089-07
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##                    TCGA-BH-A1EW-11B-33R-A137-07 TCGA-E9-A24A-01A-11R-A169-07
## ENSG00000141736.14                        25271                        14547
##                    TCGA-AR-A24W-01A-11R-A169-07 TCGA-E9-A1N4-01A-11R-A14M-07
## ENSG00000141736.14                        15613                        16742
##                    TCGA-A1-A0SO-01A-22R-A084-07 TCGA-BH-A18N-01A-11R-A12D-07
## ENSG00000141736.14                        36614                        26356
##                    TCGA-A7-A0CG-01A-12R-A056-07 TCGA-D8-A1XO-01A-11R-A14M-07
## ENSG00000141736.14                        16328                        25177
##                    TCGA-D8-A27E-01A-11R-A16F-07 TCGA-BH-A18V-11A-52R-A12D-07
## ENSG00000141736.14                        20966                        13477
##                    TCGA-BH-A0E6-01A-11R-A034-07 TCGA-C8-A12W-01A-11R-A115-07
## ENSG00000141736.14                        10747                        22246
##                    TCGA-BH-A18V-06A-11R-A213-07 TCGA-AR-A2LR-01A-12R-A18M-07
## ENSG00000141736.14                        20988                         9514
##                    TCGA-A8-A091-01A-11R-A00Z-07 TCGA-GI-A2C8-01A-11R-A16F-07
## ENSG00000141736.14                        37543                         1947
##                    TCGA-D8-A146-01A-31R-A115-07 TCGA-A8-A07I-01A-11R-A00Z-07
## ENSG00000141736.14                        14215                       281350
##                    TCGA-A7-A0DA-01A-31R-A115-07 TCGA-EW-A6SD-01A-12R-A33J-07
## ENSG00000141736.14                        15472                       320171
##                    TCGA-D8-A1XR-01A-11R-A14M-07 TCGA-E2-A1LE-01A-12R-A19W-07
## ENSG00000141736.14                        22904                       431221
##                    TCGA-E2-A1IG-01A-11R-A144-07 TCGA-3C-AAAU-01A-11R-A41B-07
## ENSG00000141736.14                        23058                        21360
##                    TCGA-E9-A1R2-01A-11R-A14D-07 TCGA-D8-A1XB-01A-11R-A14D-07
## ENSG00000141736.14                        29827                        14726
##                    TCGA-OL-A5RY-01A-21R-A28M-07 TCGA-LL-A9Q3-01A-11R-A41B-07
## ENSG00000141736.14                      262479                         24516
##                    TCGA-D8-A27R-01A-11R-A16F-07 TCGA-BH-A1FD-01A-11R-A13Q-07
## ENSG00000141736.14                        31551                         6563
##                    TCGA-BH-A1FD-11B-21R-A13Q-07 TCGA-E9-A22G-01A-11R-A157-07
## ENSG00000141736.14                        10746                        10669
##                    TCGA-AN-A0AK-01A-21R-A00Z-07 TCGA-D8-A1JJ-01A-31R-A14M-07
## ENSG00000141736.14                        27159                         9047
##                    TCGA-A2-A4RW-01A-21R-A266-07 TCGA-A8-A07S-01A-11R-A034-07
## ENSG00000141736.14                        18158                        23006
##                    TCGA-AR-A0TS-01A-11R-A115-07 TCGA-D8-A1JP-01A-11R-A13Q-07
## ENSG00000141736.14                        28157                        10571
##                    TCGA-AR-A24T-01A-11R-A169-07 TCGA-B6-A1KI-01A-11R-A14M-07
## ENSG00000141736.14                        22472                        42316
##                    TCGA-S3-AA11-01A-31R-A41B-07 TCGA-A1-A0SB-01A-11R-A144-07
## ENSG00000141736.14                        18092                        16001
##                    TCGA-BH-A0EI-01A-11R-A115-07 TCGA-AQ-A7U7-01A-22R-A352-07
## ENSG00000141736.14                        18799                        10526
##                    TCGA-AC-A2FF-11A-13R-A17B-07 TCGA-AC-A2FG-01A-11R-A17B-07
## ENSG00000141736.14                        26153                        36832
##                    TCGA-A8-A0A2-01A-11R-A034-07 TCGA-D8-A1JG-01B-11R-A13Q-07
## ENSG00000141736.14                        27930                        84849
##                    TCGA-BH-A0BZ-11A-61R-A12P-07 TCGA-GM-A2DF-01A-11R-A180-07
## ENSG00000141736.14                        13062                         5629
##                    TCGA-BH-A1FL-01A-11R-A13Q-07 TCGA-A8-A06Y-01A-21R-A00Z-07
## ENSG00000141736.14                        35332                        22653
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##                    TCGA-BH-A0AZ-01A-21R-A12P-07 TCGA-AR-A0TT-01A-31R-A084-07
## ENSG00000141736.14                        12382                        10482
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## ENSG00000141736.14                       256749                        23644
##                    TCGA-A2-A3XV-01A-21R-A239-07 TCGA-B6-A0I6-01A-11R-A034-07
## ENSG00000141736.14                        77243                         8971
##                    TCGA-A7-A0D9-11A-53R-A089-07 TCGA-B6-A0WS-01A-11R-A115-07
## ENSG00000141736.14                         2435                        14204
##                    TCGA-B6-A0I5-01A-11R-A034-07 TCGA-AR-A250-01A-31R-A169-07
## ENSG00000141736.14                        20325                       211505
##                    TCGA-BH-A1EU-11A-23R-A137-07 TCGA-S3-A6ZG-01A-22R-A32P-07
## ENSG00000141736.14                         8456                        25917
##                    TCGA-BH-A1EU-01A-11R-A137-07 TCGA-AR-A1AW-01A-21R-A12P-07
## ENSG00000141736.14                        13966                        14807
##                    TCGA-EW-A1P5-01A-11R-A144-07 TCGA-AN-A0FJ-01A-11R-A00Z-07
## ENSG00000141736.14                        12919                         4849
##                    TCGA-A2-A0EW-01A-21R-A115-07 TCGA-BH-A0B8-01A-21R-A056-07
## ENSG00000141736.14                        20892                        20957
##                    TCGA-BH-A0B8-11A-41R-A089-07 TCGA-BH-A18P-01A-11R-A12D-07
## ENSG00000141736.14                         1625                        34616
##                    TCGA-AC-A6NO-01A-12R-A33J-07 TCGA-E2-A15H-01A-11R-A12D-07
## ENSG00000141736.14                        29947                       171924
##                    TCGA-BH-A0E7-01A-11R-A034-07 TCGA-E9-A1NG-11A-52R-A14M-07
## ENSG00000141736.14                        27502                         4322
##                    TCGA-A0-A0JL-01A-11R-A056-07 TCGA-BH-A0DO-11A-22R-A12D-07
## ENSG00000141736.14                        24049                        14525
##                    TCGA-AC-A8OS-01A-12R-A41B-07 TCGA-E2-A14Z-01A-11R-A115-07
## ENSG00000141736.14                        11923                        10242
##                    TCGA-HN-A2NL-01A-11R-A18M-07 TCGA-EW-A1PG-01A-11R-A144-07
## ENSG00000141736.14                         1920                        13679
##                    TCGA-D8-A3Z5-01A-41R-A24H-07 TCGA-AR-A1AH-01A-11R-A12D-07
## ENSG00000141736.14                        50323                        19528
##                    TCGA-A8-A075-01A-11R-A084-07 TCGA-BH-A0AV-01A-31R-A115-07
## ENSG00000141736.14                        14004                        11013
##                    TCGA-BH-A1FM-11B-23R-A13Q-07 TCGA-A2-A0YJ-01A-11R-A109-07
## ENSG00000141736.14                        17362                       125457
##                    TCGA-B6-A402-01A-11R-A239-07 TCGA-BH-A0DE-01A-11R-A115-07
## ENSG00000141736.14                        12248                        32292
##                    TCGA-A2-A1G1-01A-21R-A13Q-07 TCGA-OL-A5S0-01A-11R-A28M-07
## ENSG00000141736.14                       668719                        12770
##                    TCGA-BH-A0E9-01B-11R-A115-07 TCGA-BH-A0GY-01A-11R-A056-07
## ENSG00000141736.14                        30969                        11331
##                    TCGA-GM-A2DH-01A-11R-A180-07 TCGA-BH-A1EO-11A-31R-A137-07
## ENSG00000141736.14                        24847                         3577
##                    TCGA-A2-A0T7-01A-21R-A084-07 TCGA-B6-A0I1-01A-11R-A21T-07
## ENSG00000141736.14                        26407                        20552
##                    TCGA-BH-A0B9-01A-11R-A056-07 TCGA-A2-A0CK-01A-11R-A22K-07
## ENSG00000141736.14                         7153                        26814
##                    TCGA-A0-A12G-01A-11R-A10J-07 TCGA-E2-A1L9-01A-11R-A13Q-07
## ENSG00000141736.14                        27213                        18954
##                    TCGA-WT-AB44-01A-11R-A41B-07 TCGA-D8-A1XF-01A-11R-A14D-07
## ENSG00000141736.14                         9371                        24733
##                    TCGA-BH-A8G0-01A-11R-A352-07 TCGA-AR-A251-01A-12R-A169-07
## ENSG00000141736.14                        19762                        14580
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## ENSG00000141736.14                        14680                        400201
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## ENSG00000141736.14                        21698                           798
##                    TCGA-A7-A13E-11A-61R-A12P-07 TCGA-D8-A1X7-01A-11R-A14M-07
## ENSG00000141736.14                         6040                          6671
##                    TCGA-E9-A1R6-01A-11R-A14D-07 TCGA-A2-A04P-01A-31R-A034-07
## ENSG00000141736.14                        42172                          4870
##                    TCGA-E2-A570-01A-11R-A29R-07 TCGA-AC-A8OR-01A-21R-A41B-07
## ENSG00000141736.14                        16687                          8585
##                    TCGA-A8-A09W-01A-11R-A00Z-07 TCGA-BH-A0DH-11A-31R-A089-07
## ENSG00000141736.14                        17953                         12783
##                    TCGA-A2-A1FV-01A-11R-A13Q-07 TCGA-S3-AA0Z-01A-11R-A41B-07
## ENSG00000141736.14                        19742                         19853
##                    TCGA-D8-A1JT-01A-31R-A13Q-07 TCGA-LL-A6FQ-01A-11R-A31O-07
## ENSG00000141736.14                        12477                         36066
##                    TCGA-BH-A0AY-11A-23R-A089-07 TCGA-E2-A1IU-01A-11R-A14D-07
## ENSG00000141736.14                         8337                         27429
##                    TCGA-E9-A3X8-01A-31R-A22U-07 TCGA-AC-A3QQ-01B-06R-A22O-07
## ENSG00000141736.14                         9925                         34948
##                    TCGA-D8-A1XK-01A-21R-A14M-07 TCGA-BH-A0AU-01A-11R-A12P-07
## ENSG00000141736.14                         9438                         25672
##                    TCGA-BH-A0BD-01A-11R-A034-07 TCGA-AO-A0J5-01A-11R-A034-07
## ENSG00000141736.14                        15524                         26651
##                    TCGA-PL-A8LV-01A-21R-A41B-07 TCGA-EW-A2FW-01A-11R-A17B-07
## ENSG00000141736.14                         8233                         24643
##                    TCGA-C8-A1HI-01A-11R-A137-07 TCGA-E9-A247-01A-11R-A169-07
## ENSG00000141736.14                        43156                         29463
##                    TCGA-BH-A28O-01A-11R-A22K-07 TCGA-OL-A6VQ-01A-12R-A41B-07
## ENSG00000141736.14                        21596                          9513
##                    TCGA-B6-A0WV-01A-11R-A109-07 TCGA-AR-A2LM-01A-11R-A180-07
## ENSG00000141736.14                         9365                         40575
##                    TCGA-A2-A0EP-01A-52R-A22U-07 TCGA-AR-A0TW-01A-11R-A084-07
## ENSG00000141736.14                         7362                         25162
##                    TCGA-3C-AALJ-01A-31R-A41B-07 TCGA-B6-A0RM-01A-11R-A084-07
## ENSG00000141736.14                        12131                          8725
##                    TCGA-AO-A0JD-01A-11R-A056-07 TCGA-E2-A15L-01A-11R-A12D-07
## ENSG00000141736.14                        12827                         24451
##                    TCGA-A7-A6VV-01A-22R-A33J-07 TCGA-B6-A0IA-01A-11R-A034-07
## ENSG00000141736.14                        14322                         12215
##                    TCGA-BH-A1FE-06A-11R-A213-07 TCGA-A8-A07F-01A-11R-A00Z-07
## ENSG00000141736.14                         4149                         21757
##                    TCGA-E2-A15T-01A-11R-A115-07 TCGA-C8-A12Y-01A-11R-A12D-07
## ENSG00000141736.14                        35227                         24192
##                    TCGA-EW-A1PC-01B-11R-A21T-07 TCGA-A2-A3XT-01A-11R-A22U-07
## ENSG00000141736.14                        44744                          4756
##                    TCGA-C8-A138-01A-11R-A115-07 TCGA-BH-A0B6-01A-11R-A19W-07
## ENSG00000141736.14                        24894                          9503
##                    TCGA-BH-A1FB-01A-11R-A13Q-07 TCGA-C8-A1HN-01A-11R-A137-07
## ENSG00000141736.14                        20204                         39154
##                    TCGA-BH-A1FB-11A-33R-A13Q-07 TCGA-A8-A084-01A-21R-A00Z-07
## ENSG00000141736.14                        15289                         15604
##                    TCGA-D8-A1Y3-01A-11R-A157-07 TCGA-BH-A5J0-01A-11R-A27Q-07
## ENSG00000141736.14                        19336                          9864
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##                    TCGA-D8-A1XD-01A-11R-A14D-07 TCGA-BH-A0HL-01A-11R-A10U-07
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##                    TCGA-AR-A0U3-01A-11R-A109-07 TCGA-B6-A0IE-01A-11R-A034-07
## ENSG00000141736.14                         9673                        11700
##                    TCGA-A0-A0J8-01A-21R-A034-07 TCGA-A7-A6VX-01A-12R-A33J-07
## ENSG00000141736.14                        41898                        16292
##                    TCGA-AC-A23E-01A-11R-A157-07 TCGA-AC-A3W7-01A-11R-A22K-07
## ENSG00000141736.14                        22428                        30279
##                    TCGA-BH-A0BF-01A-21R-A12P-07 TCGA-BH-A0H9-11A-22R-A466-07
## ENSG00000141736.14                         9256                         1295
##                    TCGA-A8-A09B-01A-11R-A00Z-07 TCGA-EW-A1P8-01A-11R-A144-07
## ENSG00000141736.14                        28994                         9803
##                    TCGA-BH-A0GZ-01A-11R-A056-07 TCGA-D8-A1XC-01A-11R-A14D-07
## ENSG00000141736.14                        13160                         7098
##                    TCGA-A2-A3KD-01A-12R-A213-07 TCGA-C8-A1HK-01A-21R-A13Q-07
## ENSG00000141736.14                        31227                       255024
##                    TCGA-A0-A12H-01A-11R-A115-07 TCGA-AC-A2FK-01A-12R-A180-07
## ENSG00000141736.14                        14609                        26639
##                    TCGA-A2-A0CV-01A-31R-A115-07 TCGA-EW-A1IX-01A-12R-A144-07
## ENSG00000141736.14                        19399                        32516
##                    TCGA-AN-A0XP-01A-11R-A109-07 TCGA-EW-A6S9-01A-22R-A33J-07
## ENSG00000141736.14                        30279                       389808
##                    TCGA-A7-A4SF-01A-11R-A266-07 TCGA-AR-A24H-01A-11R-A169-07
## ENSG00000141736.14                        67008                        54476
##                    TCGA-EW-A1J5-01A-11R-A13Q-07 TCGA-E2-A1LB-11A-22R-A144-07
## ENSG00000141736.14                        52017                        22789
##                    TCGA-E2-A1LB-01A-11R-A144-07 TCGA-A8-A07R-01A-21R-A034-07
## ENSG00000141736.14                       572819                        20734
##                    TCGA-EW-A6SB-01A-12R-A32P-07 TCGA-E2-A15I-11A-32R-A137-07
## ENSG00000141736.14                         4719                         1990
##                    TCGA-E2-A15I-01A-21R-A137-07 TCGA-EW-A3E8-01B-11R-A24H-07
## ENSG00000141736.14                        27273                        47964
##                    TCGA-BH-A0BV-01A-11R-A00Z-07 TCGA-BH-A0BV-11A-31R-A089-07
## ENSG00000141736.14                        16643                        23230
##                    TCGA-AR-A24S-01A-11R-A169-07 TCGA-C8-A12X-01A-11R-A115-07
## ENSG00000141736.14                        11889                         6268
##                    TCGA-A2-A0EV-01A-11R-A034-07 TCGA-A0-A12C-01A-11R-A10J-07
## ENSG00000141736.14                        38651                        52692
##                    TCGA-A2-A0YI-01A-31R-A10J-07 TCGA-A2-A04X-01A-21R-A034-07
## ENSG00000141736.14                        21376                       200874
##                    TCGA-BH-A0DT-11A-12R-A12D-07 TCGA-E2-A15A-01A-11R-A12D-07
## ENSG00000141736.14                        23207                        28871
##                    TCGA-B6-A408-01A-12R-A24H-07 TCGA-E2-A1B0-01A-11R-A12P-07
## ENSG00000141736.14                        24510                       651053
##                    TCGA-A2-A0D2-01A-21R-A034-07 TCGA-BH-A1EW-01A-11R-A137-07
## ENSG00000141736.14                         9828                        23179
##                    TCGA-AN-A04C-01A-21R-A034-07 TCGA-A2-A0EX-01A-21R-A034-07
## ENSG00000141736.14                       443539                        32953
##                    TCGA-4H-AAAK-01A-12R-A41B-07 TCGA-A7-A0CE-01A-11R-A00Z-07
## ENSG00000141736.14                        29303                        23420
##                    TCGA-AC-A3TM-01A-11R-A22K-07 TCGA-GM-A2DI-01A-31R-A18M-07
## ENSG00000141736.14                        20421                         5161
##                    TCGA-BH-A1F2-11A-32R-A13Q-07 TCGA-C8-A8HQ-01A-11R-A36F-07
## ENSG00000141736.14                        12305                        24036
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##                    TCGA-A7-A3IY-01A-21R-A21T-07 TCGA-E2-A1II-01A-11R-A144-07
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##                    TCGA-A7-A13G-11A-51R-A13Q-07 TCGA-D8-A27T-01A-11R-A16F-07
## ENSG00000141736.14                         1366                        31914
##                    TCGA-A2-A0YH-01A-11R-A109-07 TCGA-D8-A1JI-01A-11R-A13Q-07
## ENSG00000141736.14                        21339                        14744
##                    TCGA-AN-A0FT-01A-11R-A034-07 TCGA-AO-A03V-01A-11R-A115-07
## ENSG00000141736.14                       409479                        15196
##                    TCGA-GM-A3NY-01A-11R-A21T-07 TCGA-E2-A10A-01A-21R-A115-07
## ENSG00000141736.14                        15557                        20165
##                    TCGA-D8-A3Z6-01A-11R-A239-07 TCGA-A8-A08C-01A-11R-A00Z-07
## ENSG00000141736.14                        25571                        23749
##                    TCGA-BH-A1EX-01A-11R-A13Q-07 TCGA-BH-A0C3-01A-21R-A12P-07
## ENSG00000141736.14                        36211                        15864
##                    TCGA-AR-A1AN-01A-11R-A12P-07 TCGA-AO-A0JM-01A-21R-A056-07
## ENSG00000141736.14                        11703                        99974
##                    TCGA-UU-A93S-01A-21R-A41B-07 TCGA-OL-A5RZ-01A-11R-A28M-07
## ENSG00000141736.14                       502475                       437519
##                    TCGA-GM-A2DN-01A-11R-A180-07 TCGA-A2-A0EN-01A-13R-A084-07
## ENSG00000141736.14                        17276                        21348
##                    TCGA-E2-A15M-01A-11R-A12D-07 TCGA-E9-A1RC-11A-33R-A157-07
## ENSG00000141736.14                        52468                         2063
##                    TCGA-A8-A08O-01A-21R-A056-07 TCGA-AN-A0XL-01A-11R-A10J-07
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##                    TCGA-AO-A12F-01A-11R-A115-07 TCGA-B6-A401-01A-11R-A239-07
## ENSG00000141736.14                        16649                        33924
##                    TCGA-BH-A0DH-01A-11R-A084-07 TCGA-AN-A0FW-01A-11R-A034-07
## ENSG00000141736.14                        27058                        10238
##                    TCGA-D8-A1XM-01A-21R-A14M-07 TCGA-E9-A1RB-01A-11R-A157-07
## ENSG00000141736.14                        52290                         8162
##                    TCGA-PL-A8LZ-01A-31R-A36F-07 TCGA-MS-A51U-01A-31R-A266-07
## ENSG00000141736.14                         3762                        23306
##                    TCGA-BH-A0C7-01B-11R-A115-07 TCGA-E2-A1L7-11A-33R-A144-07
## ENSG00000141736.14                        45720                        17606
##                    TCGA-A8-A07G-01A-11R-A034-07 TCGA-GM-A2DO-01A-11R-A18M-07
## ENSG00000141736.14                        40762                         4723
##                    TCGA-BH-A0C0-11A-21R-A089-07 TCGA-BH-A0C0-01A-21R-A056-07
## ENSG00000141736.14                        26234                        23817
##                    TCGA-D8-A143-01A-11R-A115-07 TCGA-D8-A1XV-01A-11R-A14M-07
## ENSG00000141736.14                         9784                         4477
##                    TCGA-E2-A1LI-01A-12R-A157-07 TCGA-AO-A03N-01B-11R-A10J-07
## ENSG00000141736.14                        19057                         5871
##                    TCGA-B6-A0X5-01A-21R-A109-07 TCGA-E2-A15R-01A-11R-A115-07
## ENSG00000141736.14                         7783                        36799
##                    TCGA-E9-A1NA-11A-33R-A144-07 TCGA-BH-A0BM-11A-12R-A089-07
## ENSG00000141736.14                        11454                        17496
##                    TCGA-AN-A0FZ-01A-11R-A034-07 TCGA-AC-A2FF-01A-11R-A17B-07
## ENSG00000141736.14                        54497                        34813
##                    TCGA-BH-A0BW-11A-12R-A115-07 TCGA-AN-A0FL-01A-11R-A034-07
## ENSG00000141736.14                        10135                         7510
##                    TCGA-B6-A0WT-01A-11R-A109-07 TCGA-C8-A12V-01A-11R-A115-07
## ENSG00000141736.14                        10332                         4130
##                    TCGA-A7-A26F-01B-04R-A22O-07 TCGA-A7-A26F-01A-21R-A169-07
## ENSG00000141736.14                         6121                        11501
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## ENSG00000141736.14                        24643                        22617
##                    TCGA-AR-A24K-01A-11R-A169-07 TCGA-BH-A0H0-01A-11R-A056-07
## ENSG00000141736.14                        25468                        30870
##                    TCGA-EW-A1PH-01A-11R-A14M-07 TCGA-BH-A18T-01A-11R-A12D-07
## ENSG00000141736.14                        19466                        20412
##                    TCGA-AR-A0U2-01A-11R-A109-07 TCGA-BH-A209-01A-11R-A157-07
## ENSG00000141736.14                        22616                         6945
##                    TCGA-AR-A2LN-01A-21R-A18M-07 TCGA-E2-A14S-01A-11R-A12D-07
## ENSG00000141736.14                        13092                        22311
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## ENSG00000141736.14                        16720                        13856
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## ENSG00000141736.14                        38616                        14190
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## ENSG00000141736.14                        18658                         1023
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## ENSG00000141736.14                        62314                       171298
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## ENSG00000141736.14                        22348                        12563
##                    TCGA-A2-A0EY-01A-11R-A034-07 TCGA-GM-A2DK-01A-21R-A180-07
## ENSG00000141736.14                      287952                         20397
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## ENSG00000141736.14                        23299                        33434
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## ENSG00000141736.14                        19229                         3845
##                    TCGA-E2-A15A-06A-11R-A12D-07 TCGA-E2-A14N-01A-31R-A137-07
## ENSG00000141736.14                        25948                         2573
##                    TCGA-BH-A18I-01A-11R-A12D-07 TCGA-AC-A6IX-06A-11R-A32P-07
## ENSG00000141736.14                        19148                        20151
##                    TCGA-AC-A6IX-01A-12R-A32P-07 TCGA-E2-A10C-01A-21R-A10J-07
## ENSG00000141736.14                        16512                        15778
##                    TCGA-A2-A04Y-01A-21R-A034-07 TCGA-D8-A1JA-01A-11R-A13Q-07
## ENSG00000141736.14                        23617                       239915
##                    TCGA-E9-A1N4-11A-33R-A14M-07 TCGA-A7-A0CE-11A-21R-A089-07
## ENSG00000141736.14                        19563                        22182
##                    TCGA-LL-A441-01A-11R-A24H-07 TCGA-BH-A18N-11A-43R-A12D-07
## ENSG00000141736.14                        12148                        17199
##                    TCGA-BH-A1F2-01A-31R-A13Q-07 TCGA-A8-A06O-01A-11R-A00Z-07
## ENSG00000141736.14                      149277                         27395
##                    TCGA-A7-A26J-01B-02R-A277-07 TCGA-B6-A0RU-01A-11R-A084-07
## ENSG00000141736.14                         4117                        10114
##                    TCGA-A7-A26J-01A-11R-A277-07 TCGA-A7-A26J-01A-11R-A169-07
## ENSG00000141736.14                        16198                        33771
##                    TCGA-C8-A12K-01A-21R-A115-07 TCGA-BH-A18V-01A-11R-A12D-07
## ENSG00000141736.14                         6020                        16444
##                    TCGA-A7-A13G-01B-04R-A22O-07 TCGA-D8-A1XS-01A-11R-A14M-07
## ENSG00000141736.14                         8987                        45726
##                    TCGA-A7-A3J0-01A-11R-A213-07 TCGA-E2-A15C-01A-31R-A12D-07
## ENSG00000141736.14                         9909                        30529
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##                    TCGA-AR-A1AV-01A-21R-A12P-07 TCGA-A8-A08H-01A-21R-A00Z-07
## ENSG00000141736.14                        34243                         5219
##                    TCGA-OL-A6VR-01A-32R-A33J-07 TCGA-BH-A0DD-01A-31R-A12P-07
## ENSG00000141736.14                        24112                       111914
##                    TCGA-BH-A18Q-01A-12R-A12D-07 TCGA-A7-A4SE-01A-11R-A266-07
## ENSG00000141736.14                        10271                         6073
##                    TCGA-BH-A0HB-01A-11R-A056-07 TCGA-BH-A1FN-01A-11R-A13Q-07
## ENSG00000141736.14                        11831                         4591
##                    TCGA-BH-A0BO-01A-23R-A12D-07 TCGA-D8-A1JS-01A-11R-A13Q-07
## ENSG00000141736.14                        20536                        30320
##                    TCGA-E9-A1R7-01A-11R-A14M-07 TCGA-D8-A27H-01A-11R-A16F-07
## ENSG00000141736.14                        25516                        14360
##                    TCGA-E9-A1R7-11A-42R-A14M-07 TCGA-B6-A400-01A-11R-A239-07
## ENSG00000141736.14                         4212                        15888
##                    TCGA-B6-A1KN-01A-11R-A13Q-07 TCGA-E2-A15G-01A-11R-A12D-07
## ENSG00000141736.14                       481455                        20581
##                    TCGA-A2-A1G6-01A-11R-A13Q-07 TCGA-A2-A25E-01A-11R-A169-07
## ENSG00000141736.14                        17004                        24484
##                    TCGA-A7-A26H-01A-11R-A169-07 TCGA-D8-A1JB-01A-11R-A13Q-07
## ENSG00000141736.14                        48770                        24640
##                    TCGA-BH-A208-01A-11R-A157-07 TCGA-A2-A4S0-01A-21R-A266-07
## ENSG00000141736.14                        10756                         7632
##                    TCGA-A8-A08Z-01A-21R-A00Z-07 TCGA-OL-A66H-01A-11R-A29R-07
## ENSG00000141736.14                        20029                        16011
##                    TCGA-EW-A1J3-01A-11R-A13Q-07 TCGA-AQ-A04L-01B-21R-A10J-07
## ENSG00000141736.14                       170528                        19228
##                    TCGA-EW-A423-01A-11R-A24H-07 TCGA-GM-A2DM-01A-11R-A180-07
## ENSG00000141736.14                        31523                        11315
##                    TCGA-E2-A153-01A-12R-A12D-07 TCGA-BH-A0WA-01A-11R-A109-07
## ENSG00000141736.14                        18420                        11979
##                    TCGA-E2-A1IK-01A-11R-A144-07 TCGA-BH-A0DL-01A-11R-A115-07
## ENSG00000141736.14                         3105                         7099
##                    TCGA-A2-A25C-01A-11R-A169-07 TCGA-A2-A0CY-01A-12R-A034-07
## ENSG00000141736.14                        38886                         7839
##                    TCGA-D8-A141-01A-11R-A115-07 TCGA-AN-A0AS-01A-11R-A00Z-07
## ENSG00000141736.14                        19450                         9853
##                    TCGA-BH-A18S-01A-11R-A12D-07 TCGA-BH-A18S-11A-43R-A12D-07
## ENSG00000141736.14                        41295                        11929
##                    TCGA-BH-A6R9-01A-21R-A32P-07 TCGA-BH-A1EO-01A-11R-A137-07
## ENSG00000141736.14                         5003                         7934
##                    TCGA-BH-A42V-01A-11R-A24H-07 TCGA-AQ-A54N-01A-11R-A266-07
## ENSG00000141736.14                        37250                         3878
##                    TCGA-E9-A1RC-01A-11R-A157-07 TCGA-E2-A9RU-01A-11R-A41B-07
## ENSG00000141736.14                        24589                         7948
##                    TCGA-AC-A3QP-01A-11R-A22U-07 TCGA-D8-A145-01A-11R-A115-07
## ENSG00000141736.14                        25953                        19118
##                    TCGA-A7-A13E-01A-11R-A12P-07 TCGA-OK-A5Q2-01A-11R-A27Q-07
## ENSG00000141736.14                        22436                        44066
##                    TCGA-A7-A13E-01A-11R-A277-07 TCGA-BH-A0DZ-01A-11R-A00Z-07
## ENSG00000141736.14                        15379                        51868
##                    TCGA-A8-A07P-01A-11R-A00Z-07 TCGA-C8-A275-01A-21R-A16F-07
## ENSG00000141736.14                        92811                       668276
##                    TCGA-AR-A24X-01A-11R-A169-07 TCGA-A8-A07O-01A-11R-A00Z-07
## ENSG00000141736.14                        20427                        11357
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##                    TCGA-B6-A0RS-01A-11R-A084-07 TCGA-E2-A1LS-11A-32R-A157-07
## ENSG00000141736.14                        57200                         1440
##                    TCGA-E2-A10F-01A-11R-A10J-07 TCGA-A7-A2KD-01A-31R-A21T-07
## ENSG00000141736.14                        18399                       457638
##                    TCGA-BH-A0HI-01A-11R-A084-07 TCGA-EW-A1OZ-01A-11R-A144-07
## ENSG00000141736.14                        21861                       423008
##                    TCGA-E2-A14T-01A-11R-A115-07 TCGA-AR-A1AK-01A-21R-A12P-07
## ENSG00000141736.14                        36447                        21729
##                    TCGA-A7-A3J1-01A-11R-A213-07 TCGA-E2-A1B6-01A-31R-A12P-07
## ENSG00000141736.14                        48890                        11847
##                    TCGA-GM-A3XL-01A-11R-A22U-07 TCGA-C8-A3M7-01A-12R-A21T-07
## ENSG00000141736.14                         7203                        14861
##                    TCGA-A2-A0CU-01A-12R-A034-07 TCGA-OL-A66I-01A-21R-A29R-07
## ENSG00000141736.14                        37737                         9805
##                    TCGA-AN-A0G0-01A-11R-A034-07 TCGA-LL-A440-01A-11R-A24H-07
## ENSG00000141736.14                        22425                        19952
##                    TCGA-A8-A09N-01A-11R-A00Z-07 TCGA-BH-A0DV-11A-22R-A12P-07
## ENSG00000141736.14                        11632                        23281
##                    TCGA-AC-A3BB-01A-21R-A19W-07 TCGA-B6-A0IO-01A-11R-A034-07
## ENSG00000141736.14                        20383                        15712
##                    TCGA-C8-A1HF-01A-11R-A137-07 TCGA-A8-A07Z-01A-11R-A00Z-07
## ENSG00000141736.14                       578487                        15755
##                    TCGA-A2-A0ES-01A-11R-A115-07 TCGA-BH-A0DQ-11A-12R-A089-07
## ENSG00000141736.14                        22780                        24488
##                    TCGA-BH-A0DQ-01A-11R-A084-07 TCGA-A7-A26E-01A-11R-A169-07
## ENSG00000141736.14                        11115                        34447
##                    TCGA-A7-A26E-01B-06R-A277-07 TCGA-A7-A26E-01A-11R-A277-07
## ENSG00000141736.14                         3594                         9536
##                    TCGA-B6-A0IQ-01A-11R-A034-07 TCGA-LL-A73Z-01A-11R-A32P-07
## ENSG00000141736.14                        10086                        12620
##                    TCGA-AN-A0AL-01A-11R-A00Z-07 TCGA-GM-A2DB-01A-31R-A18M-07
## ENSG00000141736.14                         6847                         5138
##                    TCGA-A2-A0T3-01A-21R-A115-07 TCGA-AO-A03U-01B-21R-A10J-07
## ENSG00000141736.14                        21074                        13813
##                    TCGA-BH-A0W4-01A-11R-A109-07 TCGA-AR-A5QN-01A-12R-A28M-07
## ENSG00000141736.14                        28502                        30811
##                    TCGA-A8-A0A4-01A-11R-A00Z-07 TCGA-AO-A12E-01A-11R-A10J-07
## ENSG00000141736.14                        22871                        30472
##                    TCGA-E2-A1LL-01A-11R-A144-07 TCGA-LL-A7SZ-01A-32R-A352-07
## ENSG00000141736.14                         4822                         4583
##                    TCGA-E2-A1IE-01A-11R-A13Q-07 TCGA-AR-A24L-01A-11R-A169-07
## ENSG00000141736.14                        41056                        30767
##                    TCGA-S3-AA10-01A-21R-A41B-07 TCGA-A2-A3XY-01A-11R-A239-07
## ENSG00000141736.14                        10008                        16869
##                    TCGA-B6-A0RN-01A-12R-A084-07 TCGA-BH-A1F0-01A-11R-A137-07
## ENSG00000141736.14                        19549                        16307
##                    TCGA-EW-A1P3-01A-11R-A144-07 TCGA-A1-A0SJ-01A-11R-A084-07
## ENSG00000141736.14                        25040                        25938
##                    TCGA-A7-A4SA-01A-11R-A266-07 TCGA-BH-A0EA-01A-11R-A115-07
## ENSG00000141736.14                        21731                        23635
##                    TCGA-AO-A1KT-01A-11R-A13Q-07 TCGA-BH-A0BG-01A-11R-A115-07
## ENSG00000141736.14                        25605                        14791
##                    TCGA-BH-A8FZ-01A-11R-A352-07 TCGA-E2-A1LK-01A-21R-A14D-07
## ENSG00000141736.14                        41508                        13490
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##                    TCGA-OL-A5RV-01A-12R-A28M-07 TCGA-B6-A409-01A-11R-A24H-07
## ENSG00000141736.14                        36782                        13818
##                    TCGA-AC-A2FB-01A-11R-A17B-07 TCGA-B6-A0X0-01A-21R-A115-07
## ENSG00000141736.14                       240925                        18908
##                    TCGA-LL-A6FR-01A-12R-A31O-07 TCGA-D8-A1JC-01A-11R-A13Q-07
## ENSG00000141736.14                        12181                        26158
##                    TCGA-BH-A0DX-01A-11R-A115-07 TCGA-A1-A0SI-01A-11R-A144-07
## ENSG00000141736.14                         8589                        16187
##                    TCGA-A7-A0D9-01A-31R-A056-07 TCGA-E2-A15K-06A-11R-A12P-07
## ENSG00000141736.14                        13873                        28202
##                    TCGA-AR-A24Z-01A-11R-A169-07 TCGA-EW-A1OY-01A-11R-A144-07
## ENSG00000141736.14                        17524                        18504
##                    TCGA-C8-A130-01A-31R-A115-07 TCGA-E2-A14Y-01A-21R-A12D-07
## ENSG00000141736.14                        23254                       206912
##                    TCGA-D8-A27N-01A-11R-A16F-07 TCGA-AR-A0TY-01A-12R-A115-07
## ENSG00000141736.14                       168974                        16190
##                    TCGA-E2-A1BC-01A-11R-A12P-07 TCGA-E9-A243-01A-21R-A169-07
## ENSG00000141736.14                        12891                         7210
##                    TCGA-BH-A0BJ-11A-23R-A089-07 TCGA-BH-A0BJ-01A-11R-A056-07
## ENSG00000141736.14                        15187                        23932
##                    TCGA-AR-A24O-01A-11R-A169-07 TCGA-AR-A24R-01A-11R-A169-07
## ENSG00000141736.14                        12534                        17720
##                    TCGA-LL-A7T0-01A-31R-A352-07 TCGA-AC-A5XS-01A-11R-A29R-07
## ENSG00000141736.14                        29759                        13394
##                    TCGA-BH-A0DD-11A-23R-A12P-07 TCGA-BH-A18Q-11A-34R-A12D-07
## ENSG00000141736.14                         3795                        20738
##                    TCGA-E2-A1B4-01A-11R-A12P-07 TCGA-C8-A26Z-01A-11R-A16F-07
## ENSG00000141736.14                        40104                        41813
##                    TCGA-E2-A10E-01A-21R-A10J-07 TCGA-BH-A0HU-01A-11R-A034-07
## ENSG00000141736.14                        30336                        21323
##                    TCGA-OL-A5D8-01A-11R-A27Q-07 TCGA-C8-A1HG-01A-11R-A137-07
## ENSG00000141736.14                        22312                         2141
##                    TCGA-BH-A1FN-11A-34R-A13Q-07 TCGA-GM-A3NW-01A-21R-A22K-07
## ENSG00000141736.14                         9078                        32566
##                    TCGA-A8-A09X-01A-11R-A00Z-07 TCGA-E2-A1BD-01A-11R-A12P-07
## ENSG00000141736.14                        17192                        23326
##                    TCGA-D8-A1JN-01A-11R-A13Q-07 TCGA-A7-A5ZV-01A-11R-A28M-07
## ENSG00000141736.14                        13515                        13935
##                    TCGA-BH-A0HW-01A-11R-A034-07 TCGA-BH-A2L8-01A-11R-A18M-07
## ENSG00000141736.14                        60866                        16139
##                    TCGA-BH-A208-11A-51R-A157-07 TCGA-C8-A26Y-01A-11R-A16F-07
## ENSG00000141736.14                        15897                        15960
##                    TCGA-A1-A0SM-01A-11R-A084-07 TCGA-BH-A0DL-11A-13R-A115-07
## ENSG00000141736.14                       582474                        15461
##                    TCGA-E2-A153-11A-31R-A12D-07 TCGA-A7-A5ZX-01A-12R-A29R-07
## ENSG00000141736.14                        19017                        34829
##                    TCGA-BH-A0C3-11A-23R-A12P-07 TCGA-C8-A12L-01A-11R-A115-07
## ENSG00000141736.14                        14826                        53922
##                    TCGA-B6-A0I9-01A-11R-A034-07 TCGA-JL-A3YX-01A-11R-A22U-07
## ENSG00000141736.14                       645735                        25145
##                    TCGA-AC-A23H-01A-11R-A157-07 TCGA-OL-A5RX-01A-11R-A28M-07
## ENSG00000141736.14                       333649                        31423
##                    TCGA-AC-A23H-11A-12R-A157-07 TCGA-BH-A1FM-01A-11R-A13Q-07
## ENSG00000141736.14                         9958                         5250
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##                    TCGA-B6-A0RP-01A-21R-A084-07 TCGA-EW-A1IW-01A-11R-A13Q-07
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##                    TCGA-AO-A0JG-01A-31R-A084-07 TCGA-C8-A273-01A-11R-A16F-07
## ENSG00000141736.14                        21246                         13090
##                    TCGA-A8-A082-01A-11R-A00Z-07 TCGA-B6-A0X7-01A-11R-A10J-07
## ENSG00000141736.14                        14100                         29017
##                    TCGA-E2-A15M-11A-22R-A12D-07 TCGA-BH-A8FY-01A-11R-A36F-07
## ENSG00000141736.14                        12022                         10993
##                    TCGA-E2-A56Z-01A-12R-A29R-07 TCGA-D8-A1XJ-01A-11R-A14M-07
## ENSG00000141736.14                        14778                        768469
##                    TCGA-D8-A142-01A-11R-A115-07 TCGA-BH-A0DZ-11A-22R-A089-07
## ENSG00000141736.14                        23461                        109747
##                    TCGA-E2-A1LA-01A-11R-A144-07 TCGA-E9-A229-01A-31R-A157-07
## ENSG00000141736.14                        23911                         25765
##                    TCGA-E9-A22D-01A-11R-A157-07 TCGA-E9-A1R3-01A-31R-A14M-07
## ENSG00000141736.14                       151335                         19499
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## ENSG00000141736.14                        20086                         12346
##                    TCGA-E2-A1LS-01A-12R-A157-07 TCGA-LL-A50Y-01A-11R-A266-07
## ENSG00000141736.14                         2774                         27657
##                    TCGA-LL-A8F5-01A-11R-A36F-07 TCGA-A2-A0CS-01A-11R-A115-07
## ENSG00000141736.14                         5199                         18602
##                    TCGA-A2-A0YK-01A-22R-A109-07 TCGA-B6-A3ZX-01A-11R-A239-07
## ENSG00000141736.14                        13119                          9110
##                    TCGA-AR-A0TU-01A-31R-A109-07 TCGA-AR-A1AP-01A-11R-A12P-07
## ENSG00000141736.14                         5005                         19752
##                    TCGA-S3-AA14-01A-11R-A41B-07 TCGA-E9-A1RB-11A-33R-A157-07
## ENSG00000141736.14                       132576                          8093
##                    TCGA-AO-A0J7-01A-11R-A034-07 TCGA-AC-A5XU-01A-11R-A28M-07
## ENSG00000141736.14                        22616                         13906
##                    TCGA-EW-A6SC-01A-12R-A32P-07 TCGA-AN-A0AM-01A-11R-A034-07
## ENSG00000141736.14                        14980                          8393
##                    TCGA-AQ-A54O-01A-11R-A266-07 TCGA-AN-A0AJ-01A-11R-A00Z-07
## ENSG00000141736.14                         8333                         20870
##                    TCGA-C8-A135-01A-11R-A115-07 TCGA-BH-A0AU-11A-11R-A12P-07
## ENSG00000141736.14                       221100                         15357
##                    TCGA-AR-A2LH-01A-31R-A18M-07 TCGA-OL-A5RW-01A-11R-A28M-07
## ENSG00000141736.14                        13411                          9176
##                    TCGA-C8-A1HJ-01A-11R-A13Q-07 TCGA-C8-A1HL-01A-11R-A137-07
## ENSG00000141736.14                         1419                         29286
##                    TCGA-A8-A07W-01A-11R-A00Z-07 TCGA-D8-A1XY-01A-11R-A14M-07
## ENSG00000141736.14                        21336                         27847
##                    TCGA-BH-A5IZ-01A-11R-A27Q-07 TCGA-A2-A3XU-01A-12R-A22U-07
## ENSG00000141736.14                        13797                          7890
##                    TCGA-EW-A1P7-01A-21R-A144-07 TCGA-E9-A54X-01A-11R-A266-07
## ENSG00000141736.14                        24570                          8981
##                    TCGA-AC-A2FB-11A-13R-A17B-07 TCGA-AN-A0XT-01A-11R-A109-07
## ENSG00000141736.14                        14687                         10621
##                    TCGA-AO-A0JB-01A-11R-A32Y-07 TCGA-D8-A1JM-01A-11R-A13Q-07
## ENSG00000141736.14                         2739                         16057
##                    TCGA-AC-A2B8-01A-11R-A17B-07 TCGA-B6-A0I8-01A-11R-A034-07
## ENSG00000141736.14                        26557                         63638
##                    TCGA-AN-A0AT-01A-11R-A034-07 TCGA-A2-A0YC-01A-11R-A109-07
## ENSG00000141736.14                         6943                         52589
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##                    TCGA-BH-A0H3-01A-11R-A12P-07 TCGA-BH-A204-11A-53R-A157-07
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## ENSG00000141736.14                        13745                        42175
##                    TCGA-A2-A0D1-01A-11R-A034-07 TCGA-B6-A0WZ-01A-11R-A109-07
## ENSG00000141736.14                       186770                        20966
##                    TCGA-B6-A0RE-01A-11R-A056-07 TCGA-A7-A13F-01A-11R-A12P-07
## ENSG00000141736.14                         8495                        30898
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## ENSG00000141736.14                         7835                        20234
##                    TCGA-E2-A14U-01A-11R-A22K-07 TCGA-A8-A076-01A-21R-A00Z-07
## ENSG00000141736.14                        47425                       203174
##                    TCGA-BH-A0B7-11A-34R-A115-07 TCGA-BH-A18U-01A-21R-A12D-07
## ENSG00000141736.14                        15602                       380982
##                    TCGA-A8-A06Z-01A-11R-A00Z-07 TCGA-BH-A0BT-11A-21R-A12P-07
## ENSG00000141736.14                        47463                        21700
##                    TCGA-BH-A0H7-01A-13R-A056-07 TCGA-BH-A0E0-01A-11R-A056-07
## ENSG00000141736.14                        16651                         7048
##                    TCGA-A7-A5ZW-01A-12R-A29R-07 TCGA-E9-A244-01A-11R-A169-07
## ENSG00000141736.14                        20287                        17200
##                    TCGA-D8-A1JD-01A-11R-A13Q-07 TCGA-BH-A18K-11A-13R-A12D-07
## ENSG00000141736.14                        31637                        20885
##                    TCGA-E9-A248-01A-11R-A169-07 TCGA-A2-A0T1-01A-21R-A084-07
## ENSG00000141736.14                        17035                       401142
##                    TCGA-BH-A0AZ-11A-22R-A12P-07 TCGA-AN-A0FX-01A-11R-A034-07
## ENSG00000141736.14                        25431                        16336
##                    TCGA-BH-A1FH-01A-12R-A13Q-07 TCGA-A7-A3IZ-01A-11R-A213-07
## ENSG00000141736.14                        13196                        31295
##                    TCGA-D8-A13Y-01A-11R-A115-07 TCGA-A2-A0SX-01A-12R-A084-07
## ENSG00000141736.14                         2577                        13796
##                    TCGA-B6-A0IN-01A-11R-A034-07 TCGA-A2-A3Y0-01A-11R-A239-07
## ENSG00000141736.14                        37151                         7440
##                    TCGA-BH-A0B3-01A-11R-A056-07 TCGA-AN-A0XO-01A-11R-A109-07
## ENSG00000141736.14                        11703                        37441
##                    TCGA-E2-A15K-11A-13R-A12P-07 TCGA-BH-A0HF-01A-11R-A056-07
## ENSG00000141736.14                        11069                        12750
##                    TCGA-GM-A2D9-01A-11R-A18M-07 TCGA-A2-A0CZ-01A-11R-A034-07
## ENSG00000141736.14                         9712                        14557
##                    TCGA-D8-A1JE-01A-11R-A13Q-07 TCGA-D8-A1XT-01A-11R-A14M-07
## ENSG00000141736.14                        11679                       542614
##                    TCGA-EW-A2FS-01A-11R-A17B-07 TCGA-A7-A0CJ-01A-21R-A00Z-07
## ENSG00000141736.14                        21655                         9919
##                    TCGA-BH-A1FG-01A-11R-A13Q-07 TCGA-E2-A159-01A-11R-A115-07
## ENSG00000141736.14                        19912                        11774
##                    TCGA-AC-A62Y-01A-11R-A29R-07 TCGA-A7-A13G-01A-11R-A13Q-07
## ENSG00000141736.14                        11045                        15216
##                    TCGA-AN-A0XU-01A-11R-A109-07 TCGA-JL-A3YW-01A-12R-A239-07
## ENSG00000141736.14                        17485                        21072
##                    TCGA-BH-A1EY-01A-11R-A13Q-07 TCGA-BH-A204-01A-11R-A157-07
## ENSG00000141736.14                        10084                        17763
##                    TCGA-LL-A5YP-01A-21R-A28M-07 TCGA-BH-AB28-01A-31R-A41B-07
## ENSG00000141736.14                         6922                        45944
##                    TCGA-B6-A0IJ-01A-11R-A034-07 TCGA-E2-A15E-01A-11R-A12D-07
## ENSG00000141736.14                         8376                        43796
```

```
##                     TCGA-D8-A1XZ-01A-11R-A14M-07 TCGA-E2-A15E-06A-11R-A12D-07
## ENSG00000141736.14                        11609                        73711
##                     TCGA-A1-A0SQ-01A-21R-A144-07 TCGA-A7-A13F-11A-42R-A12P-07
## ENSG00000141736.14                        15271                         6871
##                     TCGA-GM-A3XN-01A-12R-A22U-07 TCGA-AC-A7VC-01A-11R-A352-07
## ENSG00000141736.14                        17948                         2383
##                     TCGA-S3-AA12-01A-11R-A41B-07 TCGA-A2-A3KC-01A-11R-A213-07
## ENSG00000141736.14                        30358                        43379
##                     TCGA-AN-A0FY-01A-11R-A034-07 TCGA-BH-A0B7-01A-12R-A115-07
## ENSG00000141736.14                        15712                       598132
##                     TCGA-E2-A15J-01A-11R-A12P-07 TCGA-BH-A1F0-11B-23R-A137-07
## ENSG00000141736.14                        29852                        13629
##                     TCGA-AR-A1AJ-01A-21R-A12P-07 TCGA-A8-A0AB-01A-11R-A034-07
## ENSG00000141736.14                         5013                        57781
##                     TCGA-E9-A1RE-01A-11R-A157-07 TCGA-BH-A18U-11A-23R-A12D-07
## ENSG00000141736.14                        42923                        12690
##                     TCGA-BH-A0H7-11A-13R-A089-07 TCGA-BH-A0BT-01A-11R-A12P-07
## ENSG00000141736.14                        15161                        40481
##                     TCGA-OL-A66O-01A-11R-A31O-07 TCGA-AC-A3YI-01A-21R-A239-07
## ENSG00000141736.14                        17626                        27214
##                     TCGA-BH-A0E0-11A-13R-A089-07 TCGA-E2-A14R-01A-11R-A115-07
## ENSG00000141736.14                        32978                         9862
##                     TCGA-LL-A5YL-01A-12R-A29R-07 TCGA-BH-A18K-01A-11R-A12D-07
## ENSG00000141736.14                        25319                        16028
##                     TCGA-A2-A0ST-01A-12R-A084-07 TCGA-D8-A27F-01A-11R-A16F-07
## ENSG00000141736.14                        10494                        15426
##                     TCGA-EW-A1P4-01A-21R-A144-07 TCGA-B6-A0IB-01A-11R-A034-07
## ENSG00000141736.14                         8384                        13280
##                     TCGA-A0-A0JC-01A-11R-A056-07 TCGA-BH-A1FH-11B-42R-A13Q-07
## ENSG00000141736.14                         2302                         7111
##                     TCGA-B6-A0X1-01A-11R-A109-07 TCGA-E9-A1N3-01A-12R-A157-07
## ENSG00000141736.14                        11240                        25170
##                     TCGA-BH-A0B3-11B-21R-A089-07 TCGA-AN-A0FS-01A-11R-A034-07
## ENSG00000141736.14                        18342                        36028
##                     TCGA-E2-A15K-01A-11R-A12P-07 TCGA-AC-A7VB-01A-11R-A352-07
## ENSG00000141736.14                        22515                        98009
##                     TCGA-LD-A7W5-01A-22R-A352-07 TCGA-B6-A0RL-01A-11R-A084-07
## ENSG00000141736.14                        16292                        46134
##                     TCGA-BH-A18P-11A-43R-A12D-07 TCGA-C8-A8HP-01A-11R-A36F-07
## ENSG00000141736.14                         8275                       314820
##                     TCGA-EW-A3U0-01A-11R-A22K-07 TCGA-D8-A1J9-01A-11R-A13Q-07
## ENSG00000141736.14                         9766                       758382
##                     TCGA-V7-A7HQ-01A-11R-A33J-07
## ENSG00000141736.14                        12795
```

```
dim(ERBB2_data)
```

```
## [1]    1 1231
```

```
# Transpose
ERBB2_data <- t(ERBB2_data) # transpose the data
head(ERBB2_data)
```

```
##                               ENSG00000141736.14
```

```
## TCGA-GM-A2DL-01A-11R-A18M-07               7613
## TCGA-AC-A2QI-01A-12R-A19W-07              41006
## TCGA-A8-A06R-01A-11R-A00Z-07               6858
## TCGA-EW-A1PD-01A-11R-A144-07              52772
## TCGA-AO-A12D-01A-11R-A115-07             453696
## TCGA-AR-A24N-01A-11R-A169-07              24483
```

```
dim(ERBB2_data)
```

```
## [1] 1231    1
```

```
# Rename the column
colnames(ERBB2_data) <- "ERBB2"
head(ERBB2_data)
```

```
##                               ERBB2
## TCGA-GM-A2DL-01A-11R-A18M-07   7613
## TCGA-AC-A2QI-01A-12R-A19W-07  41006
## TCGA-A8-A06R-01A-11R-A00Z-07   6858
## TCGA-EW-A1PD-01A-11R-A144-07  52772
## TCGA-AO-A12D-01A-11R-A115-07 453696
## TCGA-AR-A24N-01A-11R-A169-07  24483
```

```
dim(ERBB2_data)
```

```
## [1] 1231    1
```

```
class(ERBB2_data)
```

```
## [1] "matrix" "array"
```

```
# Change matrix to data frame
ERBB2_data_df <- as.data.frame(ERBB2_data)
head(ERBB2_data_df)
```

```
##                               ERBB2
## TCGA-GM-A2DL-01A-11R-A18M-07   7613
## TCGA-AC-A2QI-01A-12R-A19W-07  41006
## TCGA-A8-A06R-01A-11R-A00Z-07   6858
## TCGA-EW-A1PD-01A-11R-A144-07  52772
## TCGA-AO-A12D-01A-11R-A115-07 453696
## TCGA-AR-A24N-01A-11R-A169-07  24483
```

```
# Convert to tibble
ERBB2_data_tbl <- tibble(ERBB2_data_df)

head (ERBB2_data_tbl)
```

```
## # A tibble: 6 x 1
##    ERBB2
##    <int>
## 1   7613
## 2  41006
## 3   6858
## 4  52772
## 5 453696
## 6  24483
```

```
# Summary statistics
ERBB2_summary <- ERBB2_data_tbl %>%
  summarize(
    mean = mean(ERBB2, na.rm = TRUE),
    median = median(ERBB2, na.rm = TRUE),
    sd = sd(ERBB2, na.rm = TRUE),
    min = min(ERBB2, na.rm = TRUE),
    max = max(ERBB2, na.rm = TRUE)
  )

ERBB2_summary
```

```
## # A tibble: 1 x 5
##     mean median      sd   min    max
##    <dbl>  <int>   <dbl> <int>  <int>
## 1 46982.  19125 105288.   205 995067
```

2b. Covariate Summary Statistics

```
# Choose one covariate from metadata
colnames(metadata) # the molecular BRCA subtype seems the most relevant to me
```

```
##  [1] "patient"
##  [2] "sample"
##  [3] "shortLetterCode"
##  [4] "definition"
##  [5] "sample_submitter_id"
##  [6] "sample_type_id"
##  [7] "tumor_descriptor"
##  [8] "sample_id"
##  [9] "sample_type"
## [10] "composition"
## [11] "days_to_collection"
## [12] "state"
## [13] "initial_weight"
## [14] "preservation_method"
## [15] "pathology_report_uuid"
## [16] "submitter_id"
## [17] "oct_embedded"
## [18] "specimen_type"
## [19] "is_ffpe"
```

```
## [20] "tissue_type"
## [21] "synchronous_malignancy"
## [22] "ajcc_pathologic_stage"
## [23] "days_to_diagnosis"
## [24] "laterality"
## [25] "treatments"
## [26] "tissue_or_organ_of_origin"
## [27] "age_at_diagnosis"
## [28] "primary_diagnosis"
## [29] "metastasis_at_diagnosis"
## [30] "prior_malignancy"
## [31] "year_of_diagnosis"
## [32] "prior_treatment"
## [33] "diagnosis_is_primary_disease"
## [34] "method_of_diagnosis"
## [35] "ajcc_staging_system_edition"
## [36] "ajcc_pathologic_t"
## [37] "morphology"
## [38] "ajcc_pathologic_n"
## [39] "ajcc_pathologic_m"
## [40] "classification_of_tumor"
## [41] "diagnosis_id"
## [42] "icd_10_code"
## [43] "site_of_resection_or_biopsy"
## [44] "sites_of_involvement"
## [45] "days_to_last_follow_up"
## [46] "follow_ups_disease_response"
## [47] "race"
## [48] "gender"
## [49] "ethnicity"
## [50] "vital_status"
## [51] "age_at_index"
## [52] "days_to_birth"
## [53] "demographic_id"
## [54] "age_is_obfuscated"
## [55] "country_of_residence_at_enrollment"
## [56] "bcr_patient_barcode"
## [57] "primary_site"
## [58] "project_id"
## [59] "disease_type"
## [60] "name"
## [61] "releasable"
## [62] "released"
## [63] "figo_stage"
## [64] "figo_staging_edition_year"
## [65] "days_to_death"
## [66] "tumor_of_origin"
## [67] "days_to_sample_procurement"
## [68] "last_known_disease_status"
## [69] "tumor_grade"
## [70] "progression_or_recurrence"
## [71] "alcohol_history"
## [72] "exposure_id"
## [73] "year_of_birth"
```

```
## [74] "paper_patient"
## [75] "paper_Tumor.Type"
## [76] "paper_Included_in_previous_marker_papers"
## [77] "paper_vital_status"
## [78] "paper_days_to_birth"
## [79] "paper_days_to_death"
## [80] "paper_days_to_last_followup"
## [81] "paper_age_at_initial_pathologic_diagnosis"
## [82] "paper_pathologic_stage"
## [83] "paper_Tumor_Grade"
## [84] "paper_BRCA_Pathology"
## [85] "paper_BRCA_Subtype_PAM50"
## [86] "paper_MSI_status"
## [87] "paper_HPV_Status"
## [88] "paper_tobacco_smoking_history"
## [89] "paper_CNV.Clusters"
## [90] "paper_Mutation.Clusters"
## [91] "paper_DNA.Methylation.Clusters"
## [92] "paper_mRNA.Clusters"
## [93] "paper_miRNA.Clusters"
## [94] "paper_lncRNA.Clusters"
## [95] "paper_Protein.Clusters"
## [96] "paper_PARADIGM.Clusters"
## [97] "paper_Pan.Gyn.Clusters"
```

```
# PAM50 looks at the RNA expression pattern of 50 genes and
# uses that pattern to classify tumors into molecular subtypes

PAM50_counts <- table(metadata$paper_BRCA_Subtype_PAM50) # identify categories in this group
# Used ChatGPT's assistance to learn table()
PAM50_counts
```

```
##
##  Basal   Her2   LumA   LumB Normal
##    197     82    571    209     40
```

```
# Calculate percentage of subtypes
PAM50_percent <- PAM50_counts / sum(PAM50_counts) * 100
PAM50_percent
```

```
##
##     Basal      Her2      LumA      LumB    Normal
## 17.925387  7.461328 51.956324 19.017288  3.639672
```

```
# Combine in a data frame
PAM50_summary <- data.frame(
  Subtype = names(PAM50_counts),
  Count = as.numeric(PAM50_counts),
  Percent = round(as.numeric(PAM50_percent), 1) # round to one decimal place
)

PAM50_summary
```

```
##   Subtype Count Percent
## 1   Basal   197    17.9
## 2    Her2    82     7.5
## 3    LumA   571    52.0
## 4    LumB   209    19.0
## 5  Normal    40     3.6
```

```
# Create PAM50 table

PAM50_table <- data.frame(
  Covariate = "PAM50 subtype",
  Statistic = paste(PAM50_summary$Subtype, "n (%)"),
  Value = paste0(
    PAM50_summary$Count,
    " (",
    PAM50_summary$Percent,
    "%)"
  )
)

PAM50_table
```

```
##       Covariate    Statistic       Value
## 1 PAM50 subtype  Basal n (%) 197 (17.9%)
## 2 PAM50 subtype   Her2 n (%)   82 (7.5%)
## 3 PAM50 subtype   LumA n (%)   571 (52%)
## 4 PAM50 subtype   LumB n (%)   209 (19%)
## 5 PAM50 subtype Normal n (%)   40 (3.6%)
```

```
# Choose age at initial pathologic diagnosis as the second covariate
summary(metadata$paper_age_at_initial_pathologic_diagnosis)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##   26.00   49.00   58.00   58.42   67.00   90.00     132
```

```
# Check the variable type
class(metadata$paper_age_at_initial_pathologic_diagnosis)
```

```
## [1] "integer"
```

```
# Check for missing values
sum(is.na(metadata$paper_age_at_initial_pathologic_diagnosis))
```

```
## [1] 132
```

```
# Calculate summary statistics for age at initial pathologic diagnosis

age_summary <- metadata %>%
  summarize(
    mean = mean(paper_age_at_initial_pathologic_diagnosis, na.rm = TRUE),
    median = median(paper_age_at_initial_pathologic_diagnosis, na.rm = TRUE),
```

```
    sd = sd(paper_age_at_initial_pathologic_diagnosis, na.rm = TRUE),
    min = min(paper_age_at_initial_pathologic_diagnosis, na.rm = TRUE),
    max = max(paper_age_at_initial_pathologic_diagnosis, na.rm = TRUE)
  )

age_summary
```

```
##       mean median       sd min max
## 1 58.42493     58 13.14604  26  90
```

```
# Create age table

age_table <- data.frame(
  Covariate = "Age at initial pathologic diagnosis",
  Statistic = c("Mean", "Median", "SD", "Minimum", "Maximum"),
  Value = c(
    round(age_summary$mean, 2),
    age_summary$median,
    round(age_summary$sd, 2),
    age_summary$min,
    age_summary$max
  )
)

age_table
```

```
##                             Covariate Statistic Value
## 1 Age at initial pathologic diagnosis      Mean 58.42
## 2 Age at initial pathologic diagnosis    Median 58.00
## 3 Age at initial pathologic diagnosis        SD 13.15
## 4 Age at initial pathologic diagnosis   Minimum 26.00
## 5 Age at initial pathologic diagnosis   Maximum 90.00
```

```
# Combine the covariates
covariate_summary <- rbind(PAM50_table, age_table)

# Remove repeated covariate names for cleaner table display
covariate_summary$Covariate[c(2:5, 7:10)] <- ""

covariate_summary
```

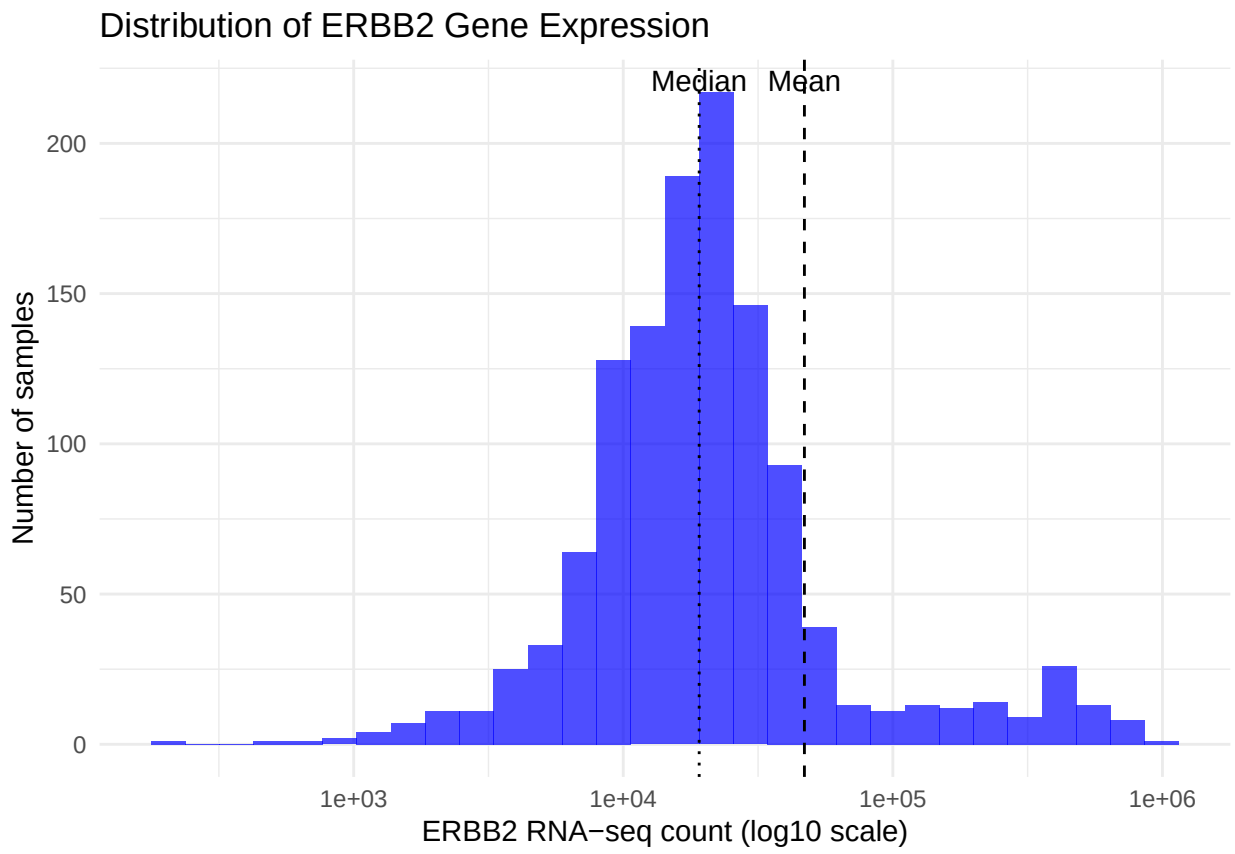
```
##                               Covariate    Statistic       Value
## 1                         PAM50 subtype  Basal n (%) 197 (17.9%)
## 2                                        Her2 n (%)   82 (7.5%)
## 3                                        LumA n (%)   571 (52%)
## 4                                        LumB n (%)   209 (19%)
## 5                                     Normal n (%)   40 (3.6%)
## 6  Age at initial pathologic diagnosis         Mean       58.42
## 7                                            Median          58
## 8                                                SD       13.15
## 9                                           Minimum          26
## 10                                          Maximum          90
```

3. Data visualization

3a. Histogram — distribution of ERBB2 expression

```
ggplot(
  ERBB2_data_tbl, # data to be used
  aes(x = ERBB2)) + # map variable
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7) + # choose plot

  # add mean and median lines based on peer feedback
geom_vline(
  xintercept = mean(ERBB2_data_tbl$ERBB2, na.rm = TRUE),
  linetype = "dashed"
) +
geom_vline(
  xintercept = median(ERBB2_data_tbl$ERBB2, na.rm = TRUE),
  linetype = "dotted"
) +
  # Annotate the mean and median lines
  annotate(
  "text",
  x = mean(ERBB2_data_tbl$ERBB2, na.rm = TRUE),
  y = Inf, # put the label on top of the plot
  label = "Mean",
  vjust = 1.5 # move it slightly downward
) +
annotate(
  "text",
  x = median(ERBB2_data_tbl$ERBB2, na.rm = TRUE),
  y = Inf,
  label = "Median",
  vjust = 1.5
) +
  scale_x_log10() + # ERBB2 expression is strongly right-skewed,
# so a log10 scale is used to better visualize the distribution
  labs(
    x = "ERBB2 RNA-seq count (log10 scale)",
    y = "Number of samples",
    title = "Distribution of ERBB2 Gene Expression"
  ) +
  theme_minimal()
```




```
# Used ChatGPT's assistance to learn geom_vline, annotate, & scale_x_log10
```

3b. Scatterplot — relationship between ERBB2 and CDK4

```
# Select a second gene - CDK4 gene Ensembl ID ENSG00000135446
grep("ENSG00000135446", rownames(counts), value = TRUE)
```

```
## [1] "ENSG00000135446.17"
```

```
CDK4_data <- counts["ENSG00000135446.17", ] # from the data frame of counts,
# I'm selecting the row with my gene of interest
head(CDK4_data)
```

```
##                    TCGA-GM-A2DL-01A-11R-A18M-07 TCGA-AC-A2QI-01A-12R-A19W-07
## ENSG00000135446.17                         2709                         3738
##                    TCGA-A8-A06R-01A-11R-A00Z-07 TCGA-EW-A1PD-01A-11R-A144-07
## ENSG00000135446.17                         1722                         3313
##                    TCGA-AO-A12D-01A-11R-A115-07 TCGA-AR-A24N-01A-11R-A169-07
## ENSG00000135446.17                         1459                         3656
##                    TCGA-AR-A24U-01A-11R-A169-07 TCGA-D8-A1JU-01A-11R-A13Q-07
## ENSG00000135446.17                         2361                         3590
##                    TCGA-A8-AOAD-01A-11R-A056-07 TCGA-E2-A3DX-01A-21R-A213-07
## ENSG00000135446.17                         2532                         2784
```

```
##                    TCGA-BH-A18J-11A-31R-A12D-07 TCGA-A2-A0YL-01A-21R-A109-07
## ENSG00000135446.17                         1820                         3552
##                    TCGA-C8-A12P-01A-11R-A115-07 TCGA-AO-A0JA-01A-11R-A056-07
## ENSG00000135446.17                         2312                         3378
##                    TCGA-EW-A1OV-01A-11R-A144-07 TCGA-5L-AAT0-01A-12R-A41B-07
## ENSG00000135446.17                         2932                         2006
##                    TCGA-BH-A18J-01A-11R-A12D-07 TCGA-D8-A27V-01A-12R-A17B-07
## ENSG00000135446.17                         3149                         5215
##                    TCGA-A7-A26I-01B-06R-A22O-07 TCGA-A8-A096-01A-11R-A00Z-07
## ENSG00000135446.17                          560                         3649
##                    TCGA-AR-A256-01A-11R-A169-07 TCGA-A7-A426-01A-22R-A24H-07
## ENSG00000135446.17                         4202                         3621
##                    TCGA-E2-A1BC-11A-32R-A12P-07 TCGA-E2-A1LH-11A-22R-A14D-07
## ENSG00000135446.17                         1371                         1938
##                    TCGA-E2-A1LH-01A-11R-A14D-07 TCGA-E9-A1N8-01A-11R-A144-07
## ENSG00000135446.17                          469                         5762
##                    TCGA-BH-A0HK-11A-11R-A089-07 TCGA-EW-A2FR-01A-11R-A21T-07
## ENSG00000135446.17                         2795                         3846
##                    TCGA-E2-A154-01A-11R-A115-07 TCGA-A2-A0T4-01A-31R-A084-07
## ENSG00000135446.17                         2139                         2673
##                    TCGA-E9-A1NH-01A-11R-A14D-07 TCGA-AR-A2LO-01A-31R-A18M-07
## ENSG00000135446.17                         2092                         2572
##                    TCGA-Z7-A8R6-01A-11R-A41B-07 TCGA-AC-A3YJ-01A-11R-A22U-07
## ENSG00000135446.17                         3335                         2185
##                    TCGA-A2-A04U-01A-11R-A115-07 TCGA-A7-A26I-01A-11R-A169-07
## ENSG00000135446.17                         7194                         4570
##                    TCGA-A8-A09A-01A-11R-A00Z-07 TCGA-AO-A126-01A-11R-A10J-07
## ENSG00000135446.17                         2057                         1685
##                    TCGA-EW-A1J6-01A-11R-A13Q-07 TCGA-LQ-A4E4-01A-11R-A266-07
## ENSG00000135446.17                          729                         1936
##                    TCGA-E2-A1IL-01A-11R-A14D-07 TCGA-D8-A1XW-01A-11R-A14M-07
## ENSG00000135446.17                         1941                         3636
##                    TCGA-A8-A07C-01A-11R-A034-07 TCGA-EW-A2FV-01A-11R-A17B-07
## ENSG00000135446.17                         8106                         3530
##                    TCGA-E2-A1AZ-01A-11R-A12P-07 TCGA-AC-A2FM-11B-32R-A19W-07
## ENSG00000135446.17                         1446                         2929
##                    TCGA-BH-A1FJ-01A-11R-A13Q-07 TCGA-EW-A1J1-01A-11R-A13Q-07
## ENSG00000135446.17                         4233                         3723
##                    TCGA-A7-A0DB-11A-33R-A089-07 TCGA-A7-A0DB-01A-11R-A277-07
## ENSG00000135446.17                         1803                         1113
##                    TCGA-A7-A56D-01A-11R-A27Q-07 TCGA-AC-A3OD-01A-11R-A21T-07
## ENSG00000135446.17                         1087                         1702
##                    TCGA-E2-A109-01A-11R-A10J-07 TCGA-BH-A0DK-11A-13R-A089-07
## ENSG00000135446.17                         3928                         2556
##                    TCGA-E2-A15O-01A-11R-A115-07 TCGA-BH-A0DK-01A-21R-A056-07
## ENSG00000135446.17                         3966                         2236
##                    TCGA-B6-A0WW-01A-11R-A109-07 TCGA-E2-A2P5-01A-11R-A19W-07
## ENSG00000135446.17                         1489                         5644
##                    TCGA-E9-A3Q9-01A-11R-A21T-07 TCGA-E2-A158-01A-11R-A12D-07
## ENSG00000135446.17                         1749                         3567
##                    TCGA-A1-A0SD-01A-11R-A115-07 TCGA-E9-A5FL-01A-11R-A27Q-07
## ENSG00000135446.17                         2643                          632
##                    TCGA-BH-A1FU-11A-23R-A14D-07 TCGA-AR-A1AT-01A-11R-A12P-07
## ENSG00000135446.17                         1762                         2192
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##                    TCGA-A1-A0SE-01A-11R-A084-07 TCGA-AN-A0XN-01A-21R-A109-07
## ENSG00000135446.17                         3811                         2008
##                    TCGA-A2-A0YG-01A-21R-A109-07 TCGA-E9-A1RF-01A-11R-A157-07
## ENSG00000135446.17                         2937                         3244
##                    TCGA-B6-A0RQ-01A-11R-A115-07 TCGA-A7-A6VY-01A-12R-A33J-07
## ENSG00000135446.17                         3119                         4891
##                    TCGA-A7-A13D-01B-04R-A277-07 TCGA-BH-A203-11A-42R-A169-07
## ENSG00000135446.17                          407                         1881
##                    TCGA-GM-A2DD-01A-11R-A180-07 TCGA-E9-A2JT-01A-22R-A18M-07
## ENSG00000135446.17                         1739                         1230
##                    TCGA-EW-A1OW-01A-21R-A144-07 TCGA-AN-A03X-01A-21R-A00Z-07
## ENSG00000135446.17                         5226                         1572
##                    TCGA-EW-A1PB-01A-11R-A144-07 TCGA-AC-A2FO-01A-11R-A180-07
## ENSG00000135446.17                         3324                         2437
##                    TCGA-BH-A18M-01A-11R-A12D-07 TCGA-A8-A090-01A-11R-A00Z-07
## ENSG00000135446.17                         2955                         2515
##                    TCGA-AC-A2FM-01A-11R-A19W-07 TCGA-BH-A1FJ-11B-42R-A13Q-07
## ENSG00000135446.17                         3673                         2064
##                    TCGA-A8-A099-01A-11R-A00Z-07 TCGA-AO-A12B-01A-11R-A10J-07
## ENSG00000135446.17                         2537                         1745
##                    TCGA-E2-A15F-01A-11R-A115-07 TCGA-AO-A0JF-01A-11R-A056-07
## ENSG00000135446.17                         2083                         2590
##                    TCGA-E2-A576-01A-11R-A31O-07 TCGA-AN-A0AR-01A-11R-A00Z-07
## ENSG00000135446.17                         2258                         6443
##                    TCGA-AC-A3OD-01B-06R-A22O-07 TCGA-LL-A5YN-01A-11R-A28M-07
## ENSG00000135446.17                          243                         2414
##                    TCGA-E9-A1RO-01A-22R-A16F-07 TCGA-AN-A03Y-01A-21R-A00Z-07
## ENSG00000135446.17                         4031                         3311
##                    TCGA-A7-A4SC-01A-12R-A266-07 TCGA-OL-A5D6-01A-21R-A27Q-07
## ENSG00000135446.17                         1155                         2869
##                    TCGA-EW-A1IZ-01A-11R-A13Q-07 TCGA-BH-A0EB-01A-11R-A034-07
## ENSG00000135446.17                         1973                         3047
##                    TCGA-C8-A3M8-01A-11R-A213-07 TCGA-AO-A129-01A-21R-A10J-07
## ENSG00000135446.17                         2869                         2365
##                    TCGA-D8-A27P-01A-11R-A16F-07 TCGA-E2-A152-01A-11R-A12D-07
## ENSG00000135446.17                         2457                         3091
##                    TCGA-BH-A1FU-01A-11R-A14D-07 TCGA-E9-A1NI-01A-11R-A14D-07
## ENSG00000135446.17                         2824                         2294
##                    TCGA-D8-A27M-01A-11R-A16F-07 TCGA-A2-A0YF-01A-21R-A109-07
## ENSG00000135446.17                         3595                         2321
##                    TCGA-E9-A1RG-01A-11R-A14D-07 TCGA-B6-A0X4-01A-11R-A109-07
## ENSG00000135446.17                         3308                         1548
##                    TCGA-AO-A0J4-01A-11R-A034-07 TCGA-A7-A425-01A-11R-A24H-07
## ENSG00000135446.17                         3805                         3022
##                    TCGA-A8-A08X-01A-21R-A00Z-07 TCGA-OL-A66N-01A-12R-A31O-07
## ENSG00000135446.17                         2354                         2785
##                    TCGA-B6-A1KF-01A-11R-A13Q-07 TCGA-AC-A2FE-01A-11R-A19W-07
## ENSG00000135446.17                         2800                         3198
##                    TCGA-A2-A0CM-01A-31R-A034-07 TCGA-GM-A5PX-01A-12R-A28M-07
## ENSG00000135446.17                         2526                         2503
##                    TCGA-AR-A1AR-01A-31R-A137-07 TCGA-AO-A0J6-01A-11R-A034-07
## ENSG00000135446.17                         3905                         7078
##                    TCGA-E2-A156-01A-11R-A12D-07 TCGA-D8-A73W-01A-22R-A352-07
## ENSG00000135446.17                         1381                         2207
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##                    TCGA-E2-A107-01A-11R-A10J-07 TCGA-E2-A158-11A-22R-A12D-07
## ENSG00000135446.17                         4860                         1243
##                    TCGA-AC-A23G-01A-11R-A213-07 TCGA-A2-A1G4-01A-11R-A13Q-07
## ENSG00000135446.17                         1600                         3604
##                    TCGA-AO-A1KO-01A-31R-A13Q-07 TCGA-A2-A3XS-01A-11R-A22U-07
## ENSG00000135446.17                         2625                         4890
##                    TCGA-B6-A0RH-01A-21R-A115-07 TCGA-D8-A1JH-01A-11R-A13Q-07
## ENSG00000135446.17                         2758                         3367
##                    TCGA-AO-A03P-01A-11R-A00Z-07 TCGA-OL-A5DA-01A-11R-A27Q-07
## ENSG00000135446.17                         4512                         2125
##                    TCGA-A8-A07B-01A-11R-A00Z-07 TCGA-A2-A25A-01A-12R-A16F-07
## ENSG00000135446.17                         1596                         4980
##                    TCGA-A2-A04R-01A-41R-A109-07 TCGA-A2-A04T-01A-21R-A034-07
## ENSG00000135446.17                         2358                         9202
##                    TCGA-AR-A1AX-01A-11R-A12P-07 TCGA-C8-A26V-01A-11R-A16F-07
## ENSG00000135446.17                         2804                         7927
##                    TCGA-BH-A18R-11A-42R-A12D-07 TCGA-AN-A0XR-01A-11R-A109-07
## ENSG00000135446.17                         2558                         2645
##                    TCGA-BH-A0HQ-01A-11R-A034-07 TCGA-LL-A5YM-01A-11R-A28M-07
## ENSG00000135446.17                         2075                         2085
##                    TCGA-D8-A1X5-01A-11R-A14D-07 TCGA-A7-A0DB-01A-11R-A00Z-07
## ENSG00000135446.17                         1833                         2933
##                    TCGA-EW-A1P6-01A-11R-A144-07 TCGA-A7-A0DB-01C-02R-A277-07
## ENSG00000135446.17                         1921                          185
##                    TCGA-LL-A73Y-01A-11R-A33J-07 TCGA-E2-A1L8-01A-11R-A13Q-07
## ENSG00000135446.17                         2515                         2723
##                    TCGA-AR-A1AM-01A-41R-A22K-07 TCGA-E2-A1LG-01A-21R-A14M-07
## ENSG00000135446.17                         2993                         2233
##                    TCGA-E2-A14W-01A-11R-A12D-07 TCGA-C8-A27B-01A-11R-A169-07
## ENSG00000135446.17                         4473                         5450
##                    TCGA-EW-A1P1-01A-31R-A14D-07 TCGA-AN-A0FN-01A-11R-A034-07
## ENSG00000135446.17                         2214                         2883
##                    TCGA-BH-A0AW-01A-11R-A056-07 TCGA-AR-A0TX-01A-11R-A084-07
## ENSG00000135446.17                         1741                        12906
##                    TCGA-E9-A3QA-01A-61R-A22K-07 TCGA-A2-A0SU-01A-11R-A084-07
## ENSG00000135446.17                         9354                         2267
##                    TCGA-GM-A2DA-01A-11R-A18M-07 TCGA-OL-A5D7-01A-11R-A27Q-07
## ENSG00000135446.17                          415                         4025
##                    TCGA-E9-A1RF-11A-32R-A157-07 TCGA-BH-A0EE-01A-11R-A034-07
## ENSG00000135446.17                         1642                         2478
##                    TCGA-BH-A0H6-01A-21R-A056-07 TCGA-OL-A66L-01A-12R-A31O-07
## ENSG00000135446.17                         2193                         2060
##                    TCGA-A7-A13D-01A-13R-A12P-07 TCGA-A7-A13D-01A-13R-A277-07
## ENSG00000135446.17                         3783                         1924
##                    TCGA-E9-A5FK-01A-11R-A27Q-07 TCGA-D8-A1JL-01A-11R-A13Q-07
## ENSG00000135446.17                         1813                         3861
##                    TCGA-BH-A203-01A-12R-A169-07 TCGA-EW-A1PE-01A-11R-A144-07
## ENSG00000135446.17                         3520                         2498
##                    TCGA-AR-A0TP-01A-11R-A084-07 TCGA-EW-A6SA-01A-21R-A32P-07
## ENSG00000135446.17                         6849                         5035
##                    TCGA-BH-A18M-11A-33R-A12D-07 TCGA-BH-A18R-01A-11R-A12D-07
## ENSG00000135446.17                         1857                         6643
##                    TCGA-BH-A0BP-01A-11R-A115-07 TCGA-E9-A1ND-01A-11R-A144-07
## ENSG00000135446.17                         2383                         2256
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##                    TCGA-E9-A1ND-11A-43R-A144-07 TCGA-E2-A1IH-01A-11R-A13Q-07
## ENSG00000135446.17                         2250                         2743
##                    TCGA-S3-A6ZH-01A-22R-A32P-07 TCGA-BH-A0C1-01B-11R-A12D-07
## ENSG00000135446.17                         3999                         5270
##                    TCGA-A2-A4RX-01A-11R-A266-07 TCGA-AC-A5EH-01A-11R-A28M-07
## ENSG00000135446.17                         1041                         3797
##                    TCGA-AR-A0TR-01A-11R-A084-07 TCGA-E2-A106-01A-11R-A10J-07
## ENSG00000135446.17                         4346                         3656
##                    TCGA-AQ-A1H2-01A-11R-A13Q-07 TCGA-A2-A0CX-01A-21R-A00Z-07
## ENSG00000135446.17                         4323                         4579
##                    TCGA-AR-A0TZ-01A-12R-A084-07 TCGA-A2-A25F-01A-11R-A169-07
## ENSG00000135446.17                         5547                         4192
##                    TCGA-E2-A15D-01A-11R-A115-07 TCGA-E2-A14X-01A-11R-A115-07
## ENSG00000135446.17                         2607                         4543
##                    TCGA-D8-A1Y1-01A-21R-A14M-07 TCGA-A8-A092-01A-11R-A00Z-07
## ENSG00000135446.17                         5010                         5804
##                    TCGA-E2-A573-01A-11R-A29R-07 TCGA-EW-A1IY-01A-11R-A13Q-07
## ENSG00000135446.17                         2974                         2195
##                    TCGA-C8-A133-01A-32R-A12D-07 TCGA-BH-A1EV-11A-24R-A137-07
## ENSG00000135446.17                         2467                         3289
##                    TCGA-B6-A2IU-01A-32R-A18M-07 TCGA-D8-A147-01A-11R-A115-07
## ENSG00000135446.17                         1139                         4974
##                    TCGA-AC-A3EH-01A-22R-A22K-07 TCGA-A8-A09Z-01A-11R-A00Z-07
## ENSG00000135446.17                         4506                         2199
##                    TCGA-BH-A1FR-01A-11R-A13Q-07 TCGA-E2-A10B-01A-11R-A10J-07
## ENSG00000135446.17                         1740                         4435
##                    TCGA-E2-A2P6-01A-11R-A19W-07 TCGA-E9-A227-01A-11R-A157-07
## ENSG00000135446.17                         1596                         2476
##                    TCGA-C8-A132-01A-31R-A115-07 TCGA-AQ-A1H3-01A-31R-A13Q-07
## ENSG00000135446.17                         2746                         3125
##                    TCGA-A2-A0YE-01A-11R-A109-07 TCGA-A8-A08L-01A-11R-A00Z-07
## ENSG00000135446.17                         4620                         6985
##                    TCGA-S3-AA17-01A-11R-A41B-07 TCGA-BH-A0BC-01A-22R-A084-07
## ENSG00000135446.17                         2579                         4083
##                    TCGA-AR-A2LK-01A-11R-A180-07 TCGA-AO-A03R-01A-21R-A034-07
## ENSG00000135446.17                         3178                         1054
##                    TCGA-BH-A1ET-01A-11R-A137-07 TCGA-OL-A6VO-01A-12R-A33J-07
## ENSG00000135446.17                         4512                         4103
##                    TCGA-AR-A1AU-01A-11R-A12P-07 TCGA-AO-A1KP-01A-11R-A13Q-07
## ENSG00000135446.17                         2936                         1631
##                    TCGA-A8-A08R-01A-11R-A034-07 TCGA-BH-A0E1-11A-13R-A089-07
## ENSG00000135446.17                         5250                         2580
##                    TCGA-AR-A0TV-01A-21R-A084-07 TCGA-E9-A1RD-01A-11R-A157-07
## ENSG00000135446.17                         3383                         3348
##                    TCGA-A8-A09G-01A-21R-A00Z-07 TCGA-AR-A1AI-01A-11R-A12P-07
## ENSG00000135446.17                         3431                         4547
##                    TCGA-BH-A1FC-11A-32R-A13Q-07 TCGA-BH-A1FC-01A-11R-A13Q-07
## ENSG00000135446.17                         1399                         7978
##                    TCGA-C8-A12N-01A-11R-A115-07 TCGA-B6-A40C-01A-11R-A239-07
## ENSG00000135446.17                         2431                         3232
##                    TCGA-AC-A3TN-01A-11R-A22K-07 TCGA-A2-A0CP-01A-11R-A034-07
## ENSG00000135446.17                         2264                         1750
##                    TCGA-E2-A14O-01A-31R-A115-07 TCGA-AQ-A04J-01A-02R-A034-07
## ENSG00000135446.17                         3775                         3573
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##                     TCGA-S3-A6ZF-01A-32R-A32P-07 TCGA-A2-A0EU-01A-22R-A056-07
## ENSG00000135446.17                         2801                         2617
##                     TCGA-C8-A12O-01A-11R-A115-07 TCGA-A2-A0CW-01A-21R-A115-07
## ENSG00000135446.17                         3457                         4934
##                     TCGA-E9-A1RA-01A-11R-A14D-07 TCGA-A1-A0SF-01A-11R-A144-07
## ENSG00000135446.17                         3315                         3329
##                     TCGA-AC-A3W5-01A-11R-A22K-07 TCGA-EW-A1PA-01A-11R-A144-07
## ENSG00000135446.17                         2689                         2193
##                     TCGA-D8-A73X-01A-11R-A32P-07 TCGA-C8-A8HR-01A-11R-A36F-07
## ENSG00000135446.17                         2452                         2953
##                     TCGA-AO-A0J3-01A-11R-A034-07 TCGA-D8-A27L-01A-11R-A16F-07
## ENSG00000135446.17                         2832                         3930
##                     TCGA-A8-A09D-01A-11R-A00Z-07 TCGA-EW-A1OX-01A-11R-A144-07
## ENSG00000135446.17                         7105                         2202
##                     TCGA-AR-A1AY-01A-21R-A12P-07 TCGA-LD-A66U-01A-11R-A31O-07
## ENSG00000135446.17                         7968                         4516
##                     TCGA-C8-A12Z-01A-11R-A115-07 TCGA-LD-A7W6-01A-81R-A352-07
## ENSG00000135446.17                         2070                         2281
##                     TCGA-E9-A1N9-11A-71R-A14D-07 TCGA-E9-A228-01A-31R-A157-07
## ENSG00000135446.17                         1946                         4218
##                     TCGA-A7-A6VW-01A-21R-A33J-07 TCGA-BH-A0HP-01A-12R-A084-07
## ENSG00000135446.17                         5650                         2906
##                     TCGA-EW-A1P0-01A-11R-A144-07 TCGA-A8-A094-01A-11R-A00Z-07
## ENSG00000135446.17                         1742                         2303
##                     TCGA-BH-A1FR-11B-42R-A13Q-07 TCGA-D8-A4Z1-01A-21R-A266-07
## ENSG00000135446.17                         1767                         1660
##                     TCGA-A2-A1FZ-01A-51R-A14D-07 TCGA-BH-A18F-01A-11R-A12D-07
## ENSG00000135446.17                         2071                         3333
##                     TCGA-BH-A0HO-01A-11R-A034-07 TCGA-B6-A0RI-01A-11R-A056-07
## ENSG00000135446.17                         3113                         2900
##                     TCGA-S3-AA15-01A-11R-A41B-07 TCGA-E9-A245-01A-22R-A16F-07
## ENSG00000135446.17                         2288                         1556
##                     TCGA-AN-A041-01A-11R-A034-07 TCGA-AO-A125-01A-11R-A10J-07
## ENSG00000135446.17                         3048                         1180
##                     TCGA-BH-A0BA-11A-22R-A19E-07 TCGA-AN-A0FV-01A-11R-A00Z-07
## ENSG00000135446.17                         2008                         4254
##                     TCGA-AN-A0FK-01A-11R-A034-07 TCGA-B6-A0I2-01A-11R-A034-07
## ENSG00000135446.17                         2743                         2529
##                     TCGA-BH-A0DV-01A-21R-A12P-07 TCGA-OL-A66J-01A-11R-A29R-07
## ENSG00000135446.17                         3028                         1873
##                     TCGA-D8-A73U-01A-11R-A33J-07 TCGA-D8-A1XQ-01A-11R-A14M-07
## ENSG00000135446.17                         1994                         3824
##                     TCGA-A2-A0T5-01A-21R-A084-07 TCGA-AR-A2LQ-01A-22R-A18M-07
## ENSG00000135446.17                         3440                         2169
##                     TCGA-HN-A2OB-01A-21R-A27Q-07 TCGA-AR-A1AO-01A-11R-A12P-07
## ENSG00000135446.17                         2401                         3519
##                     TCGA-BH-A0BR-01A-21R-A115-07 TCGA-A8-A0A1-01A-11R-A00Z-07
## ENSG00000135446.17                         3275                         1709
##                     TCGA-BH-A0DG-11A-43R-A12P-07 TCGA-GM-A4E0-01A-12R-A266-07
## ENSG00000135446.17                         2508                          786
##                     TCGA-E9-A3HO-01A-11R-A213-07 TCGA-E9-A1N5-01A-11R-A14D-07
## ENSG00000135446.17                         2459                         2287
##                     TCGA-AR-A1AQ-01A-11R-A12P-07 TCGA-A8-A086-01A-11R-A00Z-07
## ENSG00000135446.17                         3158                         3610
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##                    TCGA-BH-A0DS-01A-11R-A056-07 TCGA-E9-A295-01A-11R-A16F-07
## ENSG00000135446.17                         1991                         2582
##                    TCGA-BH-A42U-01A-12R-A24H-07 TCGA-A0-A0JJ-01A-11R-A056-07
## ENSG00000135446.17                         3237                         3093
##                    TCGA-AQ-A04H-01B-11R-A10J-07 TCGA-E9-A2JS-01A-11R-A180-07
## ENSG00000135446.17                         4782                         3598
##                    TCGA-AN-A049-01A-21R-A00Z-07 TCGA-A8-A08T-01A-21R-A00Z-07
## ENSG00000135446.17                         2654                         2594
##                    TCGA-A8-A079-01A-21R-A00Z-07 TCGA-GM-A2DC-01A-11R-A18M-07
## ENSG00000135446.17                         4045                         1066
##                    TCGA-E9-A226-01A-21R-A157-07 TCGA-AR-A254-01A-21R-A169-07
## ENSG00000135446.17                         2570                         4310
##                    TCGA-BH-A0DG-01A-21R-A12P-07 TCGA-E9-A1RH-11A-34R-A169-07
## ENSG00000135446.17                         2806                         2092
##                    TCGA-D8-A1J8-01A-11R-A13Q-07 TCGA-E9-A1RH-01A-21R-A169-07
## ENSG00000135446.17                         2593                         4539
##                    TCGA-A8-A0A7-01A-11R-A00Z-07 TCGA-AC-A6IW-01A-12R-A33J-07
## ENSG00000135446.17                         2800                         7577
##                    TCGA-A2-A0SW-01A-11R-A084-07 TCGA-XX-A899-01A-11R-A36F-07
## ENSG00000135446.17                         3125                         2410
##                    TCGA-PE-A5DE-01A-11R-A27Q-07 TCGA-E2-A150-01A-11R-A12D-07
## ENSG00000135446.17                         2151                         2118
##                    TCGA-A8-A0A6-01A-12R-A056-07 TCGA-AN-A04D-01A-21R-A034-07
## ENSG00000135446.17                         1475                         5743
##                    TCGA-C8-A1HM-01A-12R-A137-07 TCGA-A7-A4SD-01A-11R-A266-07
## ENSG00000135446.17                        10214                         2588
##                    TCGA-BH-A0BA-01A-11R-A056-07 TCGA-BH-A0BC-11A-22R-A089-07
## ENSG00000135446.17                         2773                         2276
##                    TCGA-BH-A1ET-11B-23R-A137-07 TCGA-A2-A0CT-01A-31R-A056-07
## ENSG00000135446.17                         2984                         2774
##                    TCGA-D8-A1JF-01A-11R-A13Q-07 TCGA-BH-A0E1-01A-11R-A056-07
## ENSG00000135446.17                         3112                         2638
##                    TCGA-A8-A097-01A-11R-A034-07 TCGA-LD-A74U-01A-13R-A33J-07
## ENSG00000135446.17                         3817                         3035
##                    TCGA-A0-A0JE-01A-11R-A056-07 TCGA-A2-A4S3-01A-21R-A266-07
## ENSG00000135446.17                         3865                         1773
##                    TCGA-E9-A1RD-11A-33R-A157-07 TCGA-A0-A03T-01A-21R-A034-07
## ENSG00000135446.17                         2518                          794
##                    TCGA-A8-A09M-01A-11R-A00Z-07 TCGA-E9-A22B-01A-11R-A157-07
## ENSG00000135446.17                         3333                         2911
##                    TCGA-E9-A1N9-01A-11R-A14D-07 TCGA-BH-A1EV-01A-11R-A137-07
## ENSG00000135446.17                         3215                         2827
##                    TCGA-A0-A1KR-01A-12R-A144-07 TCGA-OL-A5RU-01A-11R-A28M-07
## ENSG00000135446.17                         1520                         1854
##                    TCGA-E9-A1N5-11A-41R-A14D-07 TCGA-C8-A12U-01A-11R-A115-07
## ENSG00000135446.17                         1879                         2036
##                    TCGA-LL-A5YO-01A-21R-A28M-07 TCGA-A8-A08A-01A-11R-A32Y-07
## ENSG00000135446.17                         2110                         3800
##                    TCGA-PL-A8LX-01A-11R-A41B-07 TCGA-E9-A249-01A-11R-A169-07
## ENSG00000135446.17                         3186                         4543
##                    TCGA-A2-A3XZ-01A-42R-A239-07 TCGA-OL-A66P-01A-11R-A31O-07
## ENSG00000135446.17                         3815                         3099
##                    TCGA-C8-A26W-01A-11R-A16F-07 TCGA-A7-A13H-01A-11R-A22K-07
## ENSG00000135446.17                         4839                         2747
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##                    TCGA-A2-A4S1-01A-21R-A266-07 TCGA-BH-A0E2-01A-11R-A056-07
## ENSG00000135446.17                          962                         7800
##                    TCGA-BH-A0HY-01A-11R-A056-07 TCGA-EW-A1PF-01A-11R-A144-07
## ENSG00000135446.17                         2953                         3253
##                    TCGA-AC-A3W6-01A-12R-A22K-07 TCGA-B6-A40B-01A-11R-A239-07
## ENSG00000135446.17                         2703                         3561
##                    TCGA-A2-A0D0-01A-11R-A00Z-07 TCGA-A2-A0D3-01A-11R-A115-07
## ENSG00000135446.17                         4967                         4432
##                    TCGA-BH-A42T-01A-11R-A24H-07 TCGA-A2-A04N-01A-11R-A115-07
## ENSG00000135446.17                         5746                         2332
##                    TCGA-GI-A2C9-01A-11R-A21T-07 TCGA-GI-A2C9-11A-22R-A21T-07
## ENSG00000135446.17                         5358                         2660
##                    TCGA-E2-A15P-01A-11R-A115-07 TCGA-AO-A03L-01A-41R-A056-07
## ENSG00000135446.17                         3368                         4556
##                    TCGA-3C-AALI-01A-11R-A41B-07 TCGA-E9-A22E-01A-11R-A157-07
## ENSG00000135446.17                         2006                         2254
##                    TCGA-A2-A04V-01A-21R-A034-07 TCGA-E9-A1NG-01A-21R-A14M-07
## ENSG00000135446.17                         1694                         3244
##                    TCGA-D8-A1JK-01A-11R-A13Q-07 TCGA-BH-A0DO-01B-11R-A12D-07
## ENSG00000135446.17                         4183                         2112
##                    TCGA-AR-A24P-01A-11R-A169-07 TCGA-AC-A2QJ-01A-12R-A19W-07
## ENSG00000135446.17                         3637                         3065
##                    TCGA-AO-A0J2-01A-11R-A034-07 TCGA-BH-A1FG-11B-12R-A13Q-07
## ENSG00000135446.17                         7316                         1136
##                    TCGA-BH-A0AY-01A-21R-A00Z-07 TCGA-A2-A0T0-01A-22R-A084-07
## ENSG00000135446.17                         2451                         3553
##                    TCGA-A2-A04W-01A-31R-A115-07 TCGA-B6-A0IG-01A-11R-A034-07
## ENSG00000135446.17                         2229                         2523
##                    TCGA-AN-A0XS-01A-22R-A109-07 TCGA-D8-A1X9-01A-12R-A157-07
## ENSG00000135446.17                         2981                         4701
##                    TCGA-5T-A9QA-01A-11R-A41B-07 TCGA-E9-A1NE-01A-21R-A14M-07
## ENSG00000135446.17                         1580                         2959
##                    TCGA-A2-A0EM-01A-11R-A034-07 TCGA-AC-A8OQ-01A-11R-A41B-07
## ENSG00000135446.17                         2867                         2615
##                    TCGA-AC-A23C-01A-12R-A169-07 TCGA-AN-A0XV-01A-11R-A109-07
## ENSG00000135446.17                         3227                         1943
##                    TCGA-PE-A5DD-01A-12R-A27Q-07 TCGA-BH-A0H5-01A-21R-A115-07
## ENSG00000135446.17                         1885                         2595
##                    TCGA-A2-A1FX-01A-11R-A13Q-07 TCGA-E2-A105-01A-11R-A10J-07
## ENSG00000135446.17                         4787                         3399
##                    TCGA-A8-A09R-01A-11R-A00Z-07 TCGA-C8-A1HE-01A-11R-A13Q-07
## ENSG00000135446.17                         3695                         3032
##                    TCGA-A8-A09E-01A-11R-A00Z-07 TCGA-A7-A0DC-11A-41R-A089-07
## ENSG00000135446.17                         2803                         1623
##                    TCGA-A7-A0DC-01A-11R-A00Z-07 TCGA-A7-A0DC-01B-04R-A22O-07
## ENSG00000135446.17                         1883                          295
##                    TCGA-A8-A08F-01A-11R-A00Z-07 TCGA-LL-A6FP-01A-11R-A31O-07
## ENSG00000135446.17                         3065                         4197
##                    TCGA-5L-AAT1-01A-12R-A41B-07 TCGA-A8-A07E-01A-11R-A034-07
## ENSG00000135446.17                         1425                         4441
##                    TCGA-GI-A2C8-11A-22R-A16F-07 TCGA-A8-A09K-01A-11R-A00Z-07
## ENSG00000135446.17                         1024                         2395
##                    TCGA-A8-A09I-01A-22R-A034-07 TCGA-AO-A0JI-01A-21R-A056-07
## ENSG00000135446.17                         8443                         2857
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##                    TCGA-B6-A1KC-01B-11R-A157-07 TCGA-C8-A1HO-01A-11R-A13Q-07
## ENSG00000135446.17                         3012                         3135
##                    TCGA-E2-A1L7-01A-11R-A144-07 TCGA-C8-A274-01A-11R-A16F-07
## ENSG00000135446.17                         2726                         3522
##                    TCGA-D8-A1XU-01A-11R-A14M-07 TCGA-AR-A24Q-01A-12R-A169-07
## ENSG00000135446.17                         4590                         4976
##                    TCGA-BH-A18L-01A-32R-A12D-07 TCGA-A2-A0YM-01A-11R-A109-07
## ENSG00000135446.17                         3822                         5834
##                    TCGA-BH-A0B2-01A-11R-A10J-07 TCGA-E9-A22A-01A-11R-A157-07
## ENSG00000135446.17                         2721                         3362
##                    TCGA-A8-A08B-01A-11R-A00Z-07 TCGA-BH-A0B0-01A-21R-A115-07
## ENSG00000135446.17                         1268                         2619
##                    TCGA-BH-A1FE-01A-11R-A13Q-07 TCGA-A1-A0SH-01A-11R-A084-07
## ENSG00000135446.17                         4855                         3325
##                    TCGA-A8-A095-01A-11R-A00Z-07 TCGA-BH-A0BW-01A-11R-A115-07
## ENSG00000135446.17                         3890                         1515
##                    TCGA-AN-A0FF-01A-11R-A034-07 TCGA-A2-A3XX-01A-21R-A239-07
## ENSG00000135446.17                         3013                         4505
##                    TCGA-BH-A1EN-01A-11R-A13Q-07 TCGA-BH-A1EN-11A-23R-A13Q-07
## ENSG00000135446.17                         2369                         2794
##                    TCGA-A2-A3XW-01A-11R-A239-07 TCGA-A8-A08S-01A-11R-A034-07
## ENSG00000135446.17                         3979                         3961
##                    TCGA-E2-A14V-01A-11R-A12D-07 TCGA-A8-A09C-01A-11R-A00Z-07
## ENSG00000135446.17                         3552                         3676
##                    TCGA-D8-A1Y2-01A-11R-A157-07 TCGA-GM-A5PV-01A-11R-A28M-07
## ENSG00000135446.17                         3667                         2219
##                    TCGA-E2-A1B1-01A-21R-A12P-07 TCGA-B6-A0WX-01A-11R-A109-07
## ENSG00000135446.17                         3565                         3579
##                    TCGA-EW-A1J2-01A-21R-A13Q-07 TCGA-AR-A255-01A-11R-A169-07
## ENSG00000135446.17                         3642                         3450
##                    TCGA-UL-AAZ6-01A-11R-A41B-07 TCGA-A2-A0CL-01A-11R-A115-07
## ENSG00000135446.17                         4254                         3142
##                    TCGA-AC-A3HN-01A-11R-A213-07 TCGA-E2-A108-01A-13R-A10J-07
## ENSG00000135446.17                         3454                         2426
##                    TCGA-BH-A1F8-11B-21R-A13Q-07 TCGA-BH-A1F8-01A-11R-A13Q-07
## ENSG00000135446.17                         1787                         3576
##                    TCGA-A8-A08I-01A-11R-A00Z-07 TCGA-A8-A08J-01A-11R-A00Z-07
## ENSG00000135446.17                         5605                         5226
##                    TCGA-A8-A08P-01A-11R-A00Z-07 TCGA-C8-A12M-01A-11R-A115-07
## ENSG00000135446.17                         3118                         1609
##                    TCGA-A2-A0EQ-01A-11R-A034-07 TCGA-BH-A0W7-01A-11R-A115-07
## ENSG00000135446.17                         3749                         2769
##                    TCGA-D8-A27W-01A-11R-A16F-07 TCGA-BH-A0HN-01A-11R-A10U-07
## ENSG00000135446.17                         6765                         1618
##                    TCGA-A2-A1FW-01A-11R-A13Q-07 TCGA-E9-A1RI-11A-41R-A169-07
## ENSG00000135446.17                         4239                         1520
##                    TCGA-BH-A0DP-01A-21R-A056-07 TCGA-A7-A26G-01A-21R-A169-07
## ENSG00000135446.17                         2444                         4526
##                    TCGA-3C-AALK-01A-11R-A41B-07 TCGA-C8-A12Q-01A-11R-A115-07
## ENSG00000135446.17                         2600                         2125
##                    TCGA-BH-A0B5-11A-23R-A12P-07 TCGA-BH-A1F6-01A-11R-A13Q-07
## ENSG00000135446.17                         1727                         2526
##                    TCGA-AO-A03M-01B-11R-A10J-07 TCGA-C8-A131-01A-11R-A115-07
## ENSG00000135446.17                         4857                         1913
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##                    TCGA-AC-A6IV-01A-12R-A33J-07 TCGA-A2-A0T2-01A-11R-A084-07
## ENSG00000135446.17                         2657                         2333
##                    TCGA-E9-A1NC-01A-21R-A26B-07 TCGA-XX-A89A-01A-11R-A36F-07
## ENSG00000135446.17                         3230                         2115
##                    TCGA-E9-A5UO-01A-11R-A28M-07 TCGA-LL-A740-01A-21R-A32P-07
## ENSG00000135446.17                         4428                         3751
##                    TCGA-E9-A1N6-11A-32R-A144-07 TCGA-AN-A046-01A-21R-A034-07
## ENSG00000135446.17                         1971                         3496
##                    TCGA-E9-A1N6-01A-11R-A144-07 TCGA-BH-A18G-01A-11R-A12D-07
## ENSG00000135446.17                         3631                         1409
##                    TCGA-B6-A0RO-01A-22R-A084-07 TCGA-E9-A54Y-01A-11R-A466-07
## ENSG00000135446.17                         4653                          308
##                    TCGA-BH-A1ES-01A-11R-A137-07 TCGA-BH-A1ES-06A-12R-A24H-07
## ENSG00000135446.17                         1449                         3095
##                    TCGA-E9-A1R5-01A-11R-A14M-07 TCGA-B6-A0IC-01A-11R-A034-07
## ENSG00000135446.17                         2786                         4266
##                    TCGA-A2-A0SY-01A-31R-A084-07 TCGA-AC-A2BK-01A-11R-A21T-07
## ENSG00000135446.17                         3988                         8337
##                    TCGA-C8-A26X-01A-31R-A16F-07 TCGA-A8-A07U-01A-11R-A034-07
## ENSG00000135446.17                         3601                         2425
##                    TCGA-A8-A07J-01A-11R-A00Z-07 TCGA-BH-A18H-01A-11R-A12D-07
## ENSG00000135446.17                         2402                         2363
##                    TCGA-E2-A14P-01A-31R-A12D-07 TCGA-AN-A04A-01A-21R-A034-07
## ENSG00000135446.17                         3057                         3806
##                    TCGA-BH-A202-01A-11R-A14M-07 TCGA-WT-AB41-01A-11R-A41B-07
## ENSG00000135446.17                         3964                          922
##                    TCGA-BH-A0RX-01A-21R-A084-07 TCGA-A8-A07L-01A-11R-A00Z-07
## ENSG00000135446.17                         3275                         4574
##                    TCGA-D8-A140-01A-11R-A115-07 TCGA-AR-A5QQ-01A-11R-A28M-07
## ENSG00000135446.17                         2578                         4746
##                    TCGA-BH-A0HX-01A-21R-A056-07 TCGA-C8-A27A-01A-11R-A169-07
## ENSG00000135446.17                         3145                         4518
##                    TCGA-A2-A25D-01A-12R-A16F-07 TCGA-BH-A201-01A-11R-A14M-07
## ENSG00000135446.17                         3352                         3020
##                    TCGA-A8-A085-01A-11R-A00Z-07 TCGA-AR-A5QP-01A-11R-A28M-07
## ENSG00000135446.17                         4490                         2772
##                    TCGA-BH-A0W3-01A-11R-A109-07 TCGA-AR-A24V-01A-21R-A169-07
## ENSG00000135446.17                         2446                         3292
##                    TCGA-A8-A09V-01A-11R-A034-07 TCGA-BH-A0HK-01A-11R-A056-07
## ENSG00000135446.17                         2633                         2061
##                    TCGA-E2-A14Q-01A-11R-A12D-07 TCGA-D8-A1X8-01A-11R-A14M-07
## ENSG00000135446.17                         2681                         2590
##                    TCGA-BH-A0H5-11A-62R-A115-07 TCGA-A7-A0CD-01A-11R-A00Z-07
## ENSG00000135446.17                         1903                         4691
##                    TCGA-A8-A06T-01A-11R-A00Z-07 TCGA-A8-A06U-01A-11R-A00Z-07
## ENSG00000135446.17                         3469                         4181
##                    TCGA-AQ-A0Y5-01A-11R-A14M-07 TCGA-AN-A0XW-01A-11R-A109-07
## ENSG00000135446.17                         2712                         2417
##                    TCGA-BH-A0B1-01A-12R-A056-07 TCGA-AC-A62V-01A-11R-A31O-07
## ENSG00000135446.17                         2739                         4319
##                    TCGA-AC-A2QH-01B-04R-A22O-07 TCGA-A7-A4SB-01A-21R-A266-07
## ENSG00000135446.17                          883                         1051
##                    TCGA-A2-A0SV-01A-11R-A084-07 TCGA-BH-A0BS-01A-11R-A12P-07
## ENSG00000135446.17                         5154                         2742
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##                    TCGA-PE-A5DC-01A-12R-A27Q-07 TCGA-BH-A0HA-01A-11R-A12P-07
## ENSG00000135446.17                         2301                         1775
##                    TCGA-A8-A06P-01A-11R-A00Z-07 TCGA-BH-A0HA-11A-31R-A12P-07
## ENSG00000135446.17                         3413                         2577
##                    TCGA-A2-A0CO-01A-13R-A22K-07 TCGA-A2-A259-01A-11R-A16F-07
## ENSG00000135446.17                         2042                         3039
##                    TCGA-A8-A09T-01A-11R-A00Z-07 TCGA-E9-A6HE-01A-11R-A31O-07
## ENSG00000135446.17                         1498                         4433
##                    TCGA-A2-A0CR-01A-11R-A22K-07 TCGA-AR-A24M-01A-11R-A169-07
## ENSG00000135446.17                         1882                         4241
##                    TCGA-AC-A4ZE-01A-11R-A41B-07 TCGA-A2-A25B-01A-11R-A169-07
## ENSG00000135446.17                         2804                         4146
##                    TCGA-BH-A1FE-11B-14R-A13Q-07 TCGA-BH-A0BM-01A-11R-A056-07
## ENSG00000135446.17                         2171                          865
##                    TCGA-AR-A0U0-01A-11R-A109-07 TCGA-AR-A2LL-01A-11R-A180-07
## ENSG00000135446.17                         1790                         2905
##                    TCGA-B6-A0IK-01A-12R-A056-07 TCGA-A2-A0D4-01A-11R-A00Z-07
## ENSG00000135446.17                         2441                         4319
##                    TCGA-E9-A1R4-01A-21R-A14D-07 TCGA-AC-A2QH-01A-11R-A18M-07
## ENSG00000135446.17                         3297                         7461
##                    TCGA-AO-A1KQ-01A-11R-A13Q-07 TCGA-A8-A06Q-01A-11R-A034-07
## ENSG00000135446.17                         2739                         5035
##                    TCGA-AC-A8OP-01A-11R-A36F-07 TCGA-BH-A18L-11A-42R-A12D-07
## ENSG00000135446.17                         1976                         1808
##                    TCGA-E2-A1IG-11A-22R-A144-07 TCGA-LL-A442-01A-11R-A24H-07
## ENSG00000135446.17                         1459                         2624
##                    TCGA-E2-A155-01A-11R-A12D-07 TCGA-C8-A134-01A-11R-A115-07
## ENSG00000135446.17                         4157                         5710
##                    TCGA-BH-A0BS-11A-11R-A12P-07 TCGA-AR-A2LE-01A-11R-A180-07
## ENSG00000135446.17                         3256                         1428
##                    TCGA-BH-A28Q-01A-11R-A16F-07 TCGA-AR-A252-01A-11R-A169-07
## ENSG00000135446.17                         4634                         2663
##                    TCGA-E2-A572-01A-13R-A31O-07 TCGA-BH-A0DI-01A-21R-A12P-07
## ENSG00000135446.17                         1402                         2927
##                    TCGA-E2-A1IO-01A-11R-A144-07 TCGA-AN-A0FD-01A-11R-A034-07
## ENSG00000135446.17                         3800                         2423
##                    TCGA-E9-A1NA-01A-11R-A144-07 TCGA-A1-A0SN-01A-11R-A144-07
## ENSG00000135446.17                         4324                         4637
##                    TCGA-D8-A1XL-01A-11R-A14M-07 TCGA-AO-A03O-01A-11R-A00Z-07
## ENSG00000135446.17                         7702                         9975
##                    TCGA-B6-A0IP-01A-11R-A034-07 TCGA-BH-A0W5-01A-11R-A109-07
## ENSG00000135446.17                         3978                         2318
##                    TCGA-B6-A0IM-01A-11R-A034-07 TCGA-AO-A128-01A-11R-A10J-07
## ENSG00000135446.17                         4391                         4190
##                    TCGA-AO-A124-01A-11R-A10J-07 TCGA-BH-A0BZ-01A-31R-A12P-07
## ENSG00000135446.17                         6901                         4128
##                    TCGA-OL-A66K-01A-11R-A29R-07 TCGA-A2-A0ET-01A-31R-A034-07
## ENSG00000135446.17                         1787                         3135
##                    TCGA-BH-A1F5-01A-12R-A13Q-07 TCGA-A2-A0YT-01A-11R-A109-07
## ENSG00000135446.17                         1329                         8386
##                    TCGA-A2-A0EO-01A-11R-A034-07 TCGA-D8-A27G-01A-11R-A16F-07
## ENSG00000135446.17                         3296                         3487
##                    TCGA-E9-A5UP-01A-11R-A28M-07 TCGA-E9-A1NF-11A-73R-A14D-07
## ENSG00000135446.17                         1932                         1686
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##                    TCGA-E9-A1NF-01A-11R-A14D-07 TCGA-B6-A0RT-01A-21R-A084-07
## ENSG00000135446.17                         1771                         2865
##                    TCGA-A1-A0SP-01A-11R-A084-07 TCGA-C8-A278-01A-11R-A169-07
## ENSG00000135446.17                         6850                         4153
##                    TCGA-AC-A2BM-01A-11R-A21T-07 TCGA-A2-A0CQ-01A-21R-A034-07
## ENSG00000135446.17                         3739                         1976
##                    TCGA-E9-A1QZ-01A-21R-A169-07 TCGA-BH-A6R8-01A-21R-A33J-07
## ENSG00000135446.17                         3476                         3601
##                    TCGA-AR-A1AS-01A-11R-A12P-07 TCGA-D8-A27K-01A-11R-A16F-07
## ENSG00000135446.17                         3250                         3113
##                    TCGA-LD-A9QF-01A-32R-A41B-07 TCGA-BH-A0H9-01A-11R-A056-07
## ENSG00000135446.17                         2312                         2121
##                    TCGA-E9-A1RI-01A-11R-A169-07 TCGA-BH-A209-11A-42R-A157-07
## ENSG00000135446.17                         3218                         2679
##                    TCGA-BH-A0DP-11A-12R-A089-07 TCGA-BH-A0BQ-11A-33R-A115-07
## ENSG00000135446.17                         2884                         1970
##                    TCGA-W8-A86G-01A-21R-A36F-07 TCGA-E2-A574-01A-11R-A29R-07
## ENSG00000135446.17                         1532                         2158
##                    TCGA-A1-A0SG-01A-11R-A144-07 TCGA-A0-A0J9-01A-11R-A034-07
## ENSG00000135446.17                         2987                         1322
##                    TCGA-A8-A06N-01A-11R-A00Z-07 TCGA-B6-A0WY-01A-11R-A109-07
## ENSG00000135446.17                         4139                         3572
##                    TCGA-EW-A424-01A-11R-A24H-07 TCGA-AR-A1AL-01A-21R-A12P-07
## ENSG00000135446.17                         3964                         2796
##                    TCGA-D8-A1X6-01A-11R-A14M-07 TCGA-E2-A1L6-01A-11R-A13Q-07
## ENSG00000135446.17                         4417                         2623
##                    TCGA-E2-A15S-01A-11R-A115-07 TCGA-A2-A04Q-01A-21R-A034-07
## ENSG00000135446.17                         6128                         3205
##                    TCGA-GM-A3XG-01A-31R-A24H-07 TCGA-A8-A0A9-01A-11R-A00Z-07
## ENSG00000135446.17                         3074                         3712
##                    TCGA-Z7-A8R5-01A-42R-A41B-07 TCGA-C8-A12T-01A-11R-A115-07
## ENSG00000135446.17                         1856                         3442
##                    TCGA-OL-A97C-01A-32R-A41B-07 TCGA-A8-A06X-01A-21R-A00Z-07
## ENSG00000135446.17                         1703                         1731
##                    TCGA-E2-A1IF-01A-11R-A144-07 TCGA-AR-A0U4-01A-11R-A109-07
## ENSG00000135446.17                         2043                         3767
##                    TCGA-BH-A0DT-01A-21R-A12D-07 TCGA-A7-A3RF-01A-11R-A22K-07
## ENSG00000135446.17                         2476                         3357
##                    TCGA-A2-A0ER-01A-21R-A034-07 TCGA-A0-A12A-01A-21R-A115-07
## ENSG00000135446.17                         2391                         2574
##                    TCGA-AR-A5QM-01A-11R-A27Q-07 TCGA-B6-A0RG-01A-11R-A056-07
## ENSG00000135446.17                         2784                         1426
##                    TCGA-PL-A8LY-01A-11R-A41B-07 TCGA-A2-A4S2-01A-12R-A266-07
## ENSG00000135446.17                         2775                         1683
##                    TCGA-A7-A0CH-01A-21R-A00Z-07 TCGA-A7-A0CH-11A-32R-A089-07
## ENSG00000135446.17                         1906                         2288
##                    TCGA-BH-A1EW-11B-33R-A137-07 TCGA-E9-A24A-01A-11R-A169-07
## ENSG00000135446.17                         3381                         2146
##                    TCGA-AR-A24W-01A-11R-A169-07 TCGA-E9-A1N4-01A-11R-A14M-07
## ENSG00000135446.17                         2469                         4321
##                    TCGA-A1-A0SO-01A-22R-A084-07 TCGA-BH-A18N-01A-11R-A12D-07
## ENSG00000135446.17                         8465                         3042
##                    TCGA-A7-A0CG-01A-12R-A056-07 TCGA-D8-A1XO-01A-11R-A14M-07
## ENSG00000135446.17                         2653                         3613
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##                    TCGA-D8-A27E-01A-11R-A16F-07 TCGA-BH-A18V-11A-52R-A12D-07
## ENSG00000135446.17                         2916                         1691
##                    TCGA-BH-A0E6-01A-11R-A034-07 TCGA-C8-A12W-01A-11R-A115-07
## ENSG00000135446.17                         3219                         4628
##                    TCGA-BH-A18V-06A-11R-A213-07 TCGA-AR-A2LR-01A-12R-A18M-07
## ENSG00000135446.17                         3399                         3581
##                    TCGA-A8-A091-01A-11R-A00Z-07 TCGA-GI-A2C8-01A-11R-A16F-07
## ENSG00000135446.17                         4941                         3814
##                    TCGA-D8-A146-01A-31R-A115-07 TCGA-A8-A07I-01A-11R-A00Z-07
## ENSG00000135446.17                         2489                         1609
##                    TCGA-A7-A0DA-01A-31R-A115-07 TCGA-EW-A6SD-01A-12R-A33J-07
## ENSG00000135446.17                         3536                        23392
##                    TCGA-D8-A1XR-01A-11R-A14M-07 TCGA-E2-A1LE-01A-12R-A19W-07
## ENSG00000135446.17                         3580                         3271
##                    TCGA-E2-A1IG-01A-11R-A144-07 TCGA-3C-AAAU-01A-11R-A41B-07
## ENSG00000135446.17                         2241                         2946
##                    TCGA-E9-A1R2-01A-11R-A14D-07 TCGA-D8-A1XB-01A-11R-A14D-07
## ENSG00000135446.17                         2483                         2482
##                    TCGA-OL-A5RY-01A-21R-A28M-07 TCGA-LL-A9Q3-01A-11R-A41B-07
## ENSG00000135446.17                         2472                         2301
##                    TCGA-D8-A27R-01A-11R-A16F-07 TCGA-BH-A1FD-01A-11R-A13Q-07
## ENSG00000135446.17                         5095                         4830
##                    TCGA-BH-A1FD-11B-21R-A13Q-07 TCGA-E9-A22G-01A-11R-A157-07
## ENSG00000135446.17                         2175                         5113
##                    TCGA-AN-A0AK-01A-21R-A00Z-07 TCGA-D8-A1JJ-01A-31R-A14M-07
## ENSG00000135446.17                         4141                         4958
##                    TCGA-A2-A4RW-01A-21R-A266-07 TCGA-A8-A07S-01A-11R-A034-07
## ENSG00000135446.17                         1599                         3194
##                    TCGA-AR-A0TS-01A-11R-A115-07 TCGA-D8-A1JP-01A-11R-A13Q-07
## ENSG00000135446.17                         3468                         1798
##                    TCGA-AR-A24T-01A-11R-A169-07 TCGA-B6-A1KI-01A-11R-A14M-07
## ENSG00000135446.17                         4237                         3164
##                    TCGA-S3-AA11-01A-31R-A41B-07 TCGA-A1-A0SB-01A-11R-A144-07
## ENSG00000135446.17                         2578                         3128
##                    TCGA-BH-A0EI-01A-11R-A115-07 TCGA-AQ-A7U7-01A-22R-A352-07
## ENSG00000135446.17                         2685                         2101
##                    TCGA-AC-A2FF-11A-13R-A17B-07 TCGA-AC-A2FG-01A-11R-A17B-07
## ENSG00000135446.17                         4634                         3807
##                    TCGA-A8-A0A2-01A-11R-A034-07 TCGA-D8-A1JG-01B-11R-A13Q-07
## ENSG00000135446.17                         2938                         4742
##                    TCGA-BH-A0BZ-11A-61R-A12P-07 TCGA-GM-A2DF-01A-11R-A180-07
## ENSG00000135446.17                         1984                         1715
##                    TCGA-BH-A1FL-01A-11R-A13Q-07 TCGA-A8-A06Y-01A-21R-A00Z-07
## ENSG00000135446.17                         4238                         1637
##                    TCGA-BH-A0AZ-01A-21R-A12P-07 TCGA-AR-A0TT-01A-31R-A084-07
## ENSG00000135446.17                         1849                         2497
##                    TCGA-C8-A137-01A-11R-A115-07 TCGA-E9-A22H-01A-11R-A157-07
## ENSG00000135446.17                         2657                         3965
##                    TCGA-A2-A3XV-01A-21R-A239-07 TCGA-B6-A0I6-01A-11R-A034-07
## ENSG00000135446.17                         1919                         3188
##                    TCGA-A7-A0D9-11A-53R-A089-07 TCGA-B6-A0WS-01A-11R-A115-07
## ENSG00000135446.17                         1581                         3000
##                    TCGA-B6-A0I5-01A-11R-A034-07 TCGA-AR-A250-01A-31R-A169-07
## ENSG00000135446.17                          967                         3139
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##                    TCGA-BH-A1EU-11A-23R-A137-07 TCGA-S3-A6ZG-01A-22R-A32P-07
## ENSG00000135446.17                         1250                         3455
##                    TCGA-BH-A1EU-01A-11R-A137-07 TCGA-AR-A1AW-01A-21R-A12P-07
## ENSG00000135446.17                         4511                         2002
##                    TCGA-EW-A1P5-01A-11R-A144-07 TCGA-AN-A0FJ-01A-11R-A00Z-07
## ENSG00000135446.17                         3125                         2812
##                    TCGA-A2-A0EW-01A-21R-A115-07 TCGA-BH-A0B8-01A-21R-A056-07
## ENSG00000135446.17                         2178                         3007
##                    TCGA-BH-A0B8-11A-41R-A089-07 TCGA-BH-A18P-01A-11R-A12D-07
## ENSG00000135446.17                         1560                         3178
##                    TCGA-AC-A6NO-01A-12R-A33J-07 TCGA-E2-A15H-01A-11R-A12D-07
## ENSG00000135446.17                         3372                         2862
##                    TCGA-BH-A0E7-01A-11R-A034-07 TCGA-E9-A1NG-11A-52R-A14M-07
## ENSG00000135446.17                         3668                         1481
##                    TCGA-A0-A0JL-01A-11R-A056-07 TCGA-BH-A0DO-11A-22R-A12D-07
## ENSG00000135446.17                          444                         2372
##                    TCGA-AC-A8OS-01A-12R-A41B-07 TCGA-E2-A14Z-01A-11R-A115-07
## ENSG00000135446.17                         1845                         1910
##                    TCGA-HN-A2NL-01A-11R-A18M-07 TCGA-EW-A1PG-01A-11R-A144-07
## ENSG00000135446.17                        10466                         2379
##                    TCGA-D8-A3Z5-01A-41R-A24H-07 TCGA-AR-A1AH-01A-11R-A12D-07
## ENSG00000135446.17                         3061                         7729
##                    TCGA-A8-A075-01A-11R-A084-07 TCGA-BH-A0AV-01A-31R-A115-07
## ENSG00000135446.17                         3808                         3436
##                    TCGA-BH-A1FM-11B-23R-A13Q-07 TCGA-A2-A0YJ-01A-11R-A109-07
## ENSG00000135446.17                         2676                         4722
##                    TCGA-B6-A402-01A-11R-A239-07 TCGA-BH-A0DE-01A-11R-A115-07
## ENSG00000135446.17                         1822                         2567
##                    TCGA-A2-A1G1-01A-21R-A13Q-07 TCGA-OL-A5S0-01A-11R-A28M-07
## ENSG00000135446.17                         4381                         2704
##                    TCGA-BH-A0E9-01B-11R-A115-07 TCGA-BH-A0GY-01A-11R-A056-07
## ENSG00000135446.17                         2544                         2740
##                    TCGA-GM-A2DH-01A-11R-A180-07 TCGA-BH-A1EO-11A-31R-A137-07
## ENSG00000135446.17                         5508                         4565
##                    TCGA-A2-A0T7-01A-21R-A084-07 TCGA-B6-A0I1-01A-11R-A21T-07
## ENSG00000135446.17                         3539                         3330
##                    TCGA-BH-A0B9-01A-11R-A056-07 TCGA-A2-A0CK-01A-11R-A22K-07
## ENSG00000135446.17                        11790                         2252
##                    TCGA-A0-A12G-01A-11R-A10J-07 TCGA-E2-A1L9-01A-11R-A13Q-07
## ENSG00000135446.17                         3556                         3537
##                    TCGA-WT-AB44-01A-11R-A41B-07 TCGA-D8-A1XF-01A-11R-A14D-07
## ENSG00000135446.17                          359                         3723
##                    TCGA-BH-A8G0-01A-11R-A352-07 TCGA-AR-A251-01A-12R-A169-07
## ENSG00000135446.17                         2141                         4376
##                    TCGA-D8-A13Z-01A-11R-A115-07 TCGA-AR-A0TQ-01A-11R-A084-07
## ENSG00000135446.17                         7347                         4199
##                    TCGA-A2-A4RY-01A-31R-A266-07 TCGA-A7-A13E-01B-06R-A277-07
## ENSG00000135446.17                         2009                           93
##                    TCGA-A7-A13E-11A-61R-A12P-07 TCGA-D8-A1X7-01A-11R-A14M-07
## ENSG00000135446.17                         1546                         2496
##                    TCGA-E9-A1R6-01A-11R-A14D-07 TCGA-A2-A04P-01A-31R-A034-07
## ENSG00000135446.17                         3588                          588
##                    TCGA-E2-A570-01A-11R-A29R-07 TCGA-AC-A8OR-01A-21R-A41B-07
## ENSG00000135446.17                         1625                         2356
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##                    TCGA-A8-A09W-01A-11R-A00Z-07 TCGA-BH-A0DH-11A-31R-A089-07
## ENSG00000135446.17                         4224                         2666
##                    TCGA-A2-A1FV-01A-11R-A13Q-07 TCGA-S3-AA0Z-01A-11R-A41B-07
## ENSG00000135446.17                         2627                         4059
##                    TCGA-D8-A1JT-01A-31R-A13Q-07 TCGA-LL-A6FQ-01A-11R-A31O-07
## ENSG00000135446.17                         1727                         2719
##                    TCGA-BH-A0AY-11A-23R-A089-07 TCGA-E2-A1IU-01A-11R-A14D-07
## ENSG00000135446.17                         2175                         2286
##                    TCGA-E9-A3X8-01A-31R-A22U-07 TCGA-AC-A3QQ-01B-06R-A22O-07
## ENSG00000135446.17                         1029                          268
##                    TCGA-D8-A1XK-01A-21R-A14M-07 TCGA-BH-A0AU-01A-11R-A12P-07
## ENSG00000135446.17                         5861                         2738
##                    TCGA-BH-A0BD-01A-11R-A034-07 TCGA-AO-A0J5-01A-11R-A034-07
## ENSG00000135446.17                         3725                         4351
##                    TCGA-PL-A8LV-01A-21R-A41B-07 TCGA-EW-A2FW-01A-11R-A17B-07
## ENSG00000135446.17                         2067                         4271
##                    TCGA-C8-A1HI-01A-11R-A137-07 TCGA-E9-A247-01A-11R-A169-07
## ENSG00000135446.17                         4268                         5753
##                    TCGA-BH-A28O-01A-11R-A22K-07 TCGA-OL-A6VQ-01A-12R-A41B-07
## ENSG00000135446.17                         2941                         1017
##                    TCGA-B6-A0WV-01A-11R-A109-07 TCGA-AR-A2LM-01A-11R-A180-07
## ENSG00000135446.17                         2460                         3365
##                    TCGA-A2-A0EP-01A-52R-A22U-07 TCGA-AR-A0TW-01A-11R-A084-07
## ENSG00000135446.17                         2885                         3429
##                    TCGA-3C-AALJ-01A-31R-A41B-07 TCGA-B6-A0RM-01A-11R-A084-07
## ENSG00000135446.17                         1492                         2491
##                    TCGA-AO-A0JD-01A-11R-A056-07 TCGA-E2-A15L-01A-11R-A12D-07
## ENSG00000135446.17                         4736                         3282
##                    TCGA-A7-A6VV-01A-22R-A33J-07 TCGA-B6-A0IA-01A-11R-A034-07
## ENSG00000135446.17                         3558                         1154
##                    TCGA-BH-A1FE-06A-11R-A213-07 TCGA-A8-A07F-01A-11R-A00Z-07
## ENSG00000135446.17                         3171                         4331
##                    TCGA-E2-A15T-01A-11R-A115-07 TCGA-C8-A12Y-01A-11R-A12D-07
## ENSG00000135446.17                         2903                         2605
##                    TCGA-EW-A1PC-01B-11R-A21T-07 TCGA-A2-A3XT-01A-11R-A22U-07
## ENSG00000135446.17                         6885                         4394
##                    TCGA-C8-A138-01A-11R-A115-07 TCGA-BH-A0B6-01A-11R-A19W-07
## ENSG00000135446.17                         1005                         2785
##                    TCGA-BH-A1FB-01A-11R-A13Q-07 TCGA-C8-A1HN-01A-11R-A137-07
## ENSG00000135446.17                         3417                         3530
##                    TCGA-BH-A1FB-11A-33R-A13Q-07 TCGA-A8-A084-01A-21R-A00Z-07
## ENSG00000135446.17                         1945                         2474
##                    TCGA-D8-A1Y3-01A-11R-A157-07 TCGA-BH-A5J0-01A-11R-A27Q-07
## ENSG00000135446.17                        11397                         2196
##                    TCGA-D8-A1XD-01A-11R-A14D-07 TCGA-BH-A0HL-01A-11R-A10U-07
## ENSG00000135446.17                         2425                         3562
##                    TCGA-AR-A0U3-01A-11R-A109-07 TCGA-B6-A0IE-01A-11R-A034-07
## ENSG00000135446.17                         2438                         3267
##                    TCGA-AO-A0J8-01A-21R-A034-07 TCGA-A7-A6VX-01A-12R-A33J-07
## ENSG00000135446.17                         4607                         2502
##                    TCGA-AC-A23E-01A-11R-A157-07 TCGA-AC-A3W7-01A-11R-A22K-07
## ENSG00000135446.17                         4225                         3230
##                    TCGA-BH-A0BF-01A-21R-A12P-07 TCGA-BH-A0H9-11A-22R-A466-07
## ENSG00000135446.17                         2612                         1498
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##                    TCGA-A8-A09B-01A-11R-A00Z-07 TCGA-EW-A1P8-01A-11R-A144-07
## ENSG00000135446.17                         2592                         5508
##                    TCGA-BH-A0GZ-01A-11R-A056-07 TCGA-D8-A1XC-01A-11R-A14D-07
## ENSG00000135446.17                         1836                         1646
##                    TCGA-A2-A3KD-01A-12R-A213-07 TCGA-C8-A1HK-01A-21R-A13Q-07
## ENSG00000135446.17                         2704                         2790
##                    TCGA-A0-A12H-01A-11R-A115-07 TCGA-AC-A2FK-01A-12R-A180-07
## ENSG00000135446.17                         2468                         1826
##                    TCGA-A2-A0CV-01A-31R-A115-07 TCGA-EW-A1IX-01A-12R-A144-07
## ENSG00000135446.17                         5027                         2619
##                    TCGA-AN-A0XP-01A-11R-A109-07 TCGA-EW-A6S9-01A-22R-A33J-07
## ENSG00000135446.17                         2404                         3878
##                    TCGA-A7-A4SF-01A-11R-A266-07 TCGA-AR-A24H-01A-11R-A169-07
## ENSG00000135446.17                         2028                         3168
##                    TCGA-EW-A1J5-01A-11R-A13Q-07 TCGA-E2-A1LB-11A-22R-A144-07
## ENSG00000135446.17                         3797                         2845
##                    TCGA-E2-A1LB-01A-11R-A144-07 TCGA-A8-A07R-01A-21R-A034-07
## ENSG00000135446.17                         2179                        14415
##                    TCGA-EW-A6SB-01A-12R-A32P-07 TCGA-E2-A15I-11A-32R-A137-07
## ENSG00000135446.17                         3298                         1391
##                    TCGA-E2-A15I-01A-21R-A137-07 TCGA-EW-A3E8-01B-11R-A24H-07
## ENSG00000135446.17                         2331                         4120
##                    TCGA-BH-A0BV-01A-11R-A00Z-07 TCGA-BH-A0BV-11A-31R-A089-07
## ENSG00000135446.17                         2402                         2919
##                    TCGA-AR-A24S-01A-11R-A169-07 TCGA-C8-A12X-01A-11R-A115-07
## ENSG00000135446.17                         3594                         1606
##                    TCGA-A2-A0EV-01A-11R-A034-07 TCGA-A0-A12C-01A-11R-A10J-07
## ENSG00000135446.17                         3544                         2808
##                    TCGA-A2-A0YI-01A-31R-A10J-07 TCGA-A2-A04X-01A-21R-A034-07
## ENSG00000135446.17                         2898                         1766
##                    TCGA-BH-A0DT-11A-12R-A12D-07 TCGA-E2-A15A-01A-11R-A12D-07
## ENSG00000135446.17                         2425                         3777
##                    TCGA-B6-A408-01A-12R-A24H-07 TCGA-E2-A1B0-01A-11R-A12P-07
## ENSG00000135446.17                         3422                         4359
##                    TCGA-A2-A0D2-01A-21R-A034-07 TCGA-BH-A1EW-01A-11R-A137-07
## ENSG00000135446.17                         5551                         2895
##                    TCGA-AN-A04C-01A-21R-A034-07 TCGA-A2-A0EX-01A-21R-A034-07
## ENSG00000135446.17                         4251                         3112
##                    TCGA-4H-AAAK-01A-12R-A41B-07 TCGA-A7-A0CE-01A-11R-A00Z-07
## ENSG00000135446.17                         2983                         6117
##                    TCGA-AC-A3TM-01A-11R-A22K-07 TCGA-GM-A2DI-01A-31R-A18M-07
## ENSG00000135446.17                         2372                         1424
##                    TCGA-BH-A1F2-11A-32R-A13Q-07 TCGA-C8-A8HQ-01A-11R-A36F-07
## ENSG00000135446.17                         2765                         3112
##                    TCGA-A7-A3IY-01A-21R-A21T-07 TCGA-E2-A1II-01A-11R-A144-07
## ENSG00000135446.17                         2760                         5450
##                    TCGA-A7-A13G-11A-51R-A13Q-07 TCGA-D8-A27T-01A-11R-A16F-07
## ENSG00000135446.17                         1935                         4328
##                    TCGA-A2-A0YH-01A-11R-A109-07 TCGA-D8-A1JI-01A-11R-A13Q-07
## ENSG00000135446.17                         2870                         2892
##                    TCGA-AN-A0FT-01A-11R-A034-07 TCGA-A0-A03V-01A-11R-A115-07
## ENSG00000135446.17                         1108                         2862
##                    TCGA-GM-A3NY-01A-11R-A21T-07 TCGA-E2-A10A-01A-21R-A115-07
## ENSG00000135446.17                          657                         2513
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##                    TCGA-D8-A3Z6-01A-11R-A239-07 TCGA-A8-A08C-01A-11R-A00Z-07
## ENSG00000135446.17                         3016                         3183
##                    TCGA-BH-A1EX-01A-11R-A13Q-07 TCGA-BH-A0C3-01A-21R-A12P-07
## ENSG00000135446.17                         3618                         2574
##                    TCGA-AR-A1AN-01A-11R-A12P-07 TCGA-AO-A0JM-01A-21R-A056-07
## ENSG00000135446.17                         3638                         1793
##                    TCGA-UU-A93S-01A-21R-A41B-07 TCGA-OL-A5RZ-01A-11R-A28M-07
## ENSG00000135446.17                         1216                         3211
##                    TCGA-GM-A2DN-01A-11R-A180-07 TCGA-A2-A0EN-01A-13R-A084-07
## ENSG00000135446.17                         2485                         2379
##                    TCGA-E2-A15M-01A-11R-A12D-07 TCGA-E9-A1RC-11A-33R-A157-07
## ENSG00000135446.17                         4335                         2061
##                    TCGA-A8-A08O-01A-21R-A056-07 TCGA-AN-A0XL-01A-11R-A10J-07
## ENSG00000135446.17                         2394                         2146
##                    TCGA-AO-A12F-01A-11R-A115-07 TCGA-B6-A401-01A-11R-A239-07
## ENSG00000135446.17                         2277                         2576
##                    TCGA-BH-A0DH-01A-11R-A084-07 TCGA-AN-A0FW-01A-11R-A034-07
## ENSG00000135446.17                         3088                         6420
##                    TCGA-D8-A1XM-01A-21R-A14M-07 TCGA-E9-A1RB-01A-11R-A157-07
## ENSG00000135446.17                         4047                        12068
##                    TCGA-PL-A8LZ-01A-31R-A36F-07 TCGA-MS-A51U-01A-31R-A266-07
## ENSG00000135446.17                         3331                         1569
##                    TCGA-BH-A0C7-01B-11R-A115-07 TCGA-E2-A1L7-11A-33R-A144-07
## ENSG00000135446.17                         4138                         2452
##                    TCGA-A8-A07G-01A-11R-A034-07 TCGA-GM-A2DO-01A-11R-A18M-07
## ENSG00000135446.17                         2886                          526
##                    TCGA-BH-A0C0-11A-21R-A089-07 TCGA-BH-A0C0-01A-21R-A056-07
## ENSG00000135446.17                         3174                         3195
##                    TCGA-D8-A143-01A-11R-A115-07 TCGA-D8-A1XV-01A-11R-A14M-07
## ENSG00000135446.17                         3560                         1316
##                    TCGA-E2-A1LI-01A-12R-A157-07 TCGA-AO-A03N-01B-11R-A10J-07
## ENSG00000135446.17                         5194                          587
##                    TCGA-B6-A0X5-01A-21R-A109-07 TCGA-E2-A15R-01A-11R-A115-07
## ENSG00000135446.17                         2375                         4455
##                    TCGA-E9-A1NA-11A-33R-A144-07 TCGA-BH-A0BM-11A-12R-A089-07
## ENSG00000135446.17                         2605                         2453
##                    TCGA-AN-A0FZ-01A-11R-A034-07 TCGA-AC-A2FF-01A-11R-A17B-07
## ENSG00000135446.17                         4277                         3491
##                    TCGA-BH-A0BW-11A-12R-A115-07 TCGA-AN-A0FL-01A-11R-A034-07
## ENSG00000135446.17                         2146                         1962
##                    TCGA-B6-A0WT-01A-11R-A109-07 TCGA-C8-A12V-01A-11R-A115-07
## ENSG00000135446.17                         1290                         2494
##                    TCGA-A7-A26F-01B-04R-A22O-07 TCGA-A7-A26F-01A-21R-A169-07
## ENSG00000135446.17                          813                         3104
##                    TCGA-A8-A093-01A-11R-A00Z-07 TCGA-A2-A1G0-01A-11R-A13Q-07
## ENSG00000135446.17                         3267                         2290
##                    TCGA-BH-A0BQ-01A-21R-A115-07 TCGA-E2-A1IN-01A-11R-A13Q-07
## ENSG00000135446.17                         2633                         3630
##                    TCGA-AR-A24K-01A-11R-A169-07 TCGA-BH-A0H0-01A-11R-A056-07
## ENSG00000135446.17                         3736                         3461
##                    TCGA-EW-A1PH-01A-11R-A14M-07 TCGA-BH-A18T-01A-11R-A12D-07
## ENSG00000135446.17                         4313                         6277
##                    TCGA-AR-A0U2-01A-11R-A109-07 TCGA-BH-A209-01A-11R-A157-07
## ENSG00000135446.17                         5755                         3680
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##                    TCGA-AR-A2LN-01A-21R-A18M-07 TCGA-E2-A14S-01A-11R-A12D-07
## ENSG00000135446.17                         1620                         2286
##                    TCGA-AO-A1KS-01A-11R-A13Q-07 TCGA-AC-A62X-01A-11R-A29R-07
## ENSG00000135446.17                         3201                         5736
##                    TCGA-A8-A083-01A-21R-A00Z-07 TCGA-B6-A0IH-01A-11R-A115-07
## ENSG00000135446.17                         2765                         2196
##                    TCGA-E2-A1IJ-01A-11R-A144-07 TCGA-A1-A0SK-01A-12R-A084-07
## ENSG00000135446.17                         1486                         9014
##                    TCGA-A8-A081-01A-11R-A00Z-07 TCGA-AR-A2LJ-01A-12R-A19W-07
## ENSG00000135446.17                         5415                         3546
##                    TCGA-BH-A0B5-01A-11R-A12P-07 TCGA-BH-A1F6-11B-94R-A13Q-07
## ENSG00000135446.17                         4226                         1811
##                    TCGA-BH-A0B4-01A-11R-A00Z-07 TCGA-BH-A0BL-01A-11R-A115-07
## ENSG00000135446.17                         3595                         4452
##                    TCGA-A2-A0EY-01A-11R-A034-07 TCGA-GM-A2DK-01A-21R-A180-07
## ENSG00000135446.17                         3892                         1316
##                    TCGA-D8-A27I-01A-11R-A16F-07 TCGA-A8-A08G-01A-11R-A00Z-07
## ENSG00000135446.17                         4277                         3996
##                    TCGA-D8-A1XA-01A-11R-A14D-07 TCGA-E2-A1B5-01A-21R-A12P-07
## ENSG00000135446.17                         4111                         1970
##                    TCGA-E2-A15A-06A-11R-A12D-07 TCGA-E2-A14N-01A-31R-A137-07
## ENSG00000135446.17                         4015                         3360
##                    TCGA-BH-A18I-01A-11R-A12D-07 TCGA-AC-A6IX-06A-11R-A32P-07
## ENSG00000135446.17                         3876                         3051
##                    TCGA-AC-A6IX-01A-12R-A32P-07 TCGA-E2-A10C-01A-21R-A10J-07
## ENSG00000135446.17                         2762                         2109
##                    TCGA-A2-A04Y-01A-21R-A034-07 TCGA-D8-A1JA-01A-11R-A13Q-07
## ENSG00000135446.17                         2256                         2263
##                    TCGA-E9-A1N4-11A-33R-A14M-07 TCGA-A7-A0CE-11A-21R-A089-07
## ENSG00000135446.17                         2755                         3074
##                    TCGA-LL-A441-01A-11R-A24H-07 TCGA-BH-A18N-11A-43R-A12D-07
## ENSG00000135446.17                         6321                         2260
##                    TCGA-BH-A1F2-01A-31R-A13Q-07 TCGA-A8-A06O-01A-11R-A00Z-07
## ENSG00000135446.17                         2213                         3752
##                    TCGA-A7-A26J-01B-02R-A277-07 TCGA-B6-A0RU-01A-11R-A084-07
## ENSG00000135446.17                          303                         4327
##                    TCGA-A7-A26J-01A-11R-A277-07 TCGA-A7-A26J-01A-11R-A169-07
## ENSG00000135446.17                         2649                         4148
##                    TCGA-C8-A12K-01A-21R-A115-07 TCGA-BH-A18V-01A-11R-A12D-07
## ENSG00000135446.17                         6734                         3098
##                    TCGA-A7-A13G-01B-04R-A22O-07 TCGA-D8-A1XS-01A-11R-A14M-07
## ENSG00000135446.17                          711                         2808
##                    TCGA-A7-A3J0-01A-11R-A213-07 TCGA-E2-A15C-01A-31R-A12D-07
## ENSG00000135446.17                         3335                         2024
##                    TCGA-AR-A1AV-01A-21R-A12P-07 TCGA-A8-A08H-01A-21R-A00Z-07
## ENSG00000135446.17                         4005                         2532
##                    TCGA-OL-A6VR-01A-32R-A33J-07 TCGA-BH-A0DD-01A-31R-A12P-07
## ENSG00000135446.17                         3709                         3425
##                    TCGA-BH-A18Q-01A-12R-A12D-07 TCGA-A7-A4SE-01A-11R-A266-07
## ENSG00000135446.17                         3754                         2761
##                    TCGA-BH-A0HB-01A-11R-A056-07 TCGA-BH-A1FN-01A-11R-A13Q-07
## ENSG00000135446.17                         2087                         1551
##                    TCGA-BH-A0BO-01A-23R-A12D-07 TCGA-D8-A1JS-01A-11R-A13Q-07
## ENSG00000135446.17                         2601                         4467
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##                    TCGA-E9-A1R7-01A-11R-A14M-07 TCGA-D8-A27H-01A-11R-A16F-07
## ENSG00000135446.17                         2066                         3541
##                    TCGA-E9-A1R7-11A-42R-A14M-07 TCGA-B6-A400-01A-11R-A239-07
## ENSG00000135446.17                         1655                         5106
##                    TCGA-B6-A1KN-01A-11R-A13Q-07 TCGA-E2-A15G-01A-11R-A12D-07
## ENSG00000135446.17                         4149                         2385
##                    TCGA-A2-A1G6-01A-11R-A13Q-07 TCGA-A2-A25E-01A-11R-A169-07
## ENSG00000135446.17                         2571                         4078
##                    TCGA-A7-A26H-01A-11R-A169-07 TCGA-D8-A1JB-01A-11R-A13Q-07
## ENSG00000135446.17                         3775                         1658
##                    TCGA-BH-A208-01A-11R-A157-07 TCGA-A2-A4S0-01A-21R-A266-07
## ENSG00000135446.17                         7447                         1221
##                    TCGA-A8-A08Z-01A-21R-A00Z-07 TCGA-OL-A66H-01A-11R-A29R-07
## ENSG00000135446.17                         2609                         2269
##                    TCGA-EW-A1J3-01A-11R-A13Q-07 TCGA-AQ-A04L-01B-21R-A10J-07
## ENSG00000135446.17                         1610                        17531
##                    TCGA-EW-A423-01A-11R-A24H-07 TCGA-GM-A2DM-01A-11R-A180-07
## ENSG00000135446.17                         1486                         2331
##                    TCGA-E2-A153-01A-12R-A12D-07 TCGA-BH-A0WA-01A-11R-A109-07
## ENSG00000135446.17                         1472                         3362
##                    TCGA-E2-A1IK-01A-11R-A144-07 TCGA-BH-A0DL-01A-11R-A115-07
## ENSG00000135446.17                         1774                         3202
##                    TCGA-A2-A25C-01A-11R-A169-07 TCGA-A2-A0CY-01A-12R-A034-07
## ENSG00000135446.17                         4780                         2343
##                    TCGA-D8-A141-01A-11R-A115-07 TCGA-AN-A0AS-01A-11R-A00Z-07
## ENSG00000135446.17                         2927                         3127
##                    TCGA-BH-A18S-01A-11R-A12D-07 TCGA-BH-A18S-11A-43R-A12D-07
## ENSG00000135446.17                         2798                         1530
##                    TCGA-BH-A6R9-01A-21R-A32P-07 TCGA-BH-A1EO-01A-11R-A137-07
## ENSG00000135446.17                         4163                          962
##                    TCGA-BH-A42V-01A-11R-A24H-07 TCGA-AQ-A54N-01A-11R-A266-07
## ENSG00000135446.17                         3910                          879
##                    TCGA-E9-A1RC-01A-11R-A157-07 TCGA-E2-A9RU-01A-11R-A41B-07
## ENSG00000135446.17                         5480                        29400
##                    TCGA-AC-A3QP-01A-11R-A22U-07 TCGA-D8-A145-01A-11R-A115-07
## ENSG00000135446.17                         3468                         2646
##                    TCGA-A7-A13E-01A-11R-A12P-07 TCGA-OK-A5Q2-01A-11R-A27Q-07
## ENSG00000135446.17                         2283                         2764
##                    TCGA-A7-A13E-01A-11R-A277-07 TCGA-BH-A0DZ-01A-11R-A00Z-07
## ENSG00000135446.17                         1480                         1896
##                    TCGA-A8-A07P-01A-11R-A00Z-07 TCGA-C8-A275-01A-21R-A16F-07
## ENSG00000135446.17                         4862                         4212
##                    TCGA-AR-A24X-01A-11R-A169-07 TCGA-A8-A07O-01A-11R-A00Z-07
## ENSG00000135446.17                         2084                         8403
##                    TCGA-B6-A0RS-01A-11R-A084-07 TCGA-E2-A1LS-11A-32R-A157-07
## ENSG00000135446.17                         5233                         2029
##                    TCGA-E2-A10F-01A-11R-A10J-07 TCGA-A7-A2KD-01A-31R-A21T-07
## ENSG00000135446.17                         3232                         3258
##                    TCGA-BH-A0HI-01A-11R-A084-07 TCGA-EW-A1OZ-01A-11R-A144-07
## ENSG00000135446.17                         2332                         6090
##                    TCGA-E2-A14T-01A-11R-A115-07 TCGA-AR-A1AK-01A-21R-A12P-07
## ENSG00000135446.17                         2443                         2520
##                    TCGA-A7-A3J1-01A-11R-A213-07 TCGA-E2-A1B6-01A-31R-A12P-07
## ENSG00000135446.17                         3240                         2929
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##                    TCGA-GM-A3XL-01A-11R-A22U-07 TCGA-C8-A3M7-01A-12R-A21T-07
## ENSG00000135446.17                         9255                         2339
##                    TCGA-A2-A0CU-01A-12R-A034-07 TCGA-OL-A66I-01A-21R-A29R-07
## ENSG00000135446.17                         2331                         3641
##                    TCGA-AN-A0G0-01A-11R-A034-07 TCGA-LL-A440-01A-11R-A24H-07
## ENSG00000135446.17                         4835                         2926
##                    TCGA-A8-A09N-01A-11R-A00Z-07 TCGA-BH-A0DV-11A-22R-A12P-07
## ENSG00000135446.17                         4059                         2401
##                    TCGA-AC-A3BB-01A-21R-A19W-07 TCGA-B6-A0IO-01A-11R-A034-07
## ENSG00000135446.17                         3671                         2963
##                    TCGA-C8-A1HF-01A-11R-A137-07 TCGA-A8-A07Z-01A-11R-A00Z-07
## ENSG00000135446.17                         5374                         3204
##                    TCGA-A2-A0ES-01A-11R-A115-07 TCGA-BH-A0DQ-11A-12R-A089-07
## ENSG00000135446.17                         2190                         3313
##                    TCGA-BH-A0DQ-01A-11R-A084-07 TCGA-A7-A26E-01A-11R-A169-07
## ENSG00000135446.17                         3025                         3544
##                    TCGA-A7-A26E-01B-06R-A277-07 TCGA-A7-A26E-01A-11R-A277-07
## ENSG00000135446.17                          225                         1938
##                    TCGA-B6-A0IQ-01A-11R-A034-07 TCGA-LL-A73Z-01A-11R-A32P-07
## ENSG00000135446.17                         9005                         2054
##                    TCGA-AN-A0AL-01A-11R-A00Z-07 TCGA-GM-A2DB-01A-31R-A18M-07
## ENSG00000135446.17                         2113                          579
##                    TCGA-A2-A0T3-01A-21R-A115-07 TCGA-AO-A03U-01B-21R-A10J-07
## ENSG00000135446.17                         3017                         2054
##                    TCGA-BH-A0W4-01A-11R-A109-07 TCGA-AR-A5QN-01A-12R-A28M-07
## ENSG00000135446.17                         3564                         2980
##                    TCGA-A8-A0A4-01A-11R-A00Z-07 TCGA-AO-A12E-01A-11R-A10J-07
## ENSG00000135446.17                         2770                         3764
##                    TCGA-E2-A1LL-01A-11R-A144-07 TCGA-LL-A7SZ-01A-32R-A352-07
## ENSG00000135446.17                         3943                         2213
##                    TCGA-E2-A1IE-01A-11R-A13Q-07 TCGA-AR-A24L-01A-11R-A169-07
## ENSG00000135446.17                         4261                         3490
##                    TCGA-S3-AA10-01A-21R-A41B-07 TCGA-A2-A3XY-01A-11R-A239-07
## ENSG00000135446.17                         6116                         4342
##                    TCGA-B6-A0RN-01A-12R-A084-07 TCGA-BH-A1F0-01A-11R-A137-07
## ENSG00000135446.17                         3029                         4684
##                    TCGA-EW-A1P3-01A-11R-A144-07 TCGA-A1-A0SJ-01A-11R-A084-07
## ENSG00000135446.17                         1327                         5611
##                    TCGA-A7-A4SA-01A-11R-A266-07 TCGA-BH-A0EA-01A-11R-A115-07
## ENSG00000135446.17                         1940                         1820
##                    TCGA-AO-A1KT-01A-11R-A13Q-07 TCGA-BH-A0BG-01A-11R-A115-07
## ENSG00000135446.17                         3207                         2573
##                    TCGA-BH-A8FZ-01A-11R-A352-07 TCGA-E2-A1LK-01A-21R-A14D-07
## ENSG00000135446.17                         2617                         1686
##                    TCGA-OL-A5RV-01A-12R-A28M-07 TCGA-B6-A409-01A-11R-A24H-07
## ENSG00000135446.17                         2575                         3306
##                    TCGA-AC-A2FB-01A-11R-A17B-07 TCGA-B6-A0X0-01A-21R-A115-07
## ENSG00000135446.17                         2629                         1809
##                    TCGA-LL-A6FR-01A-12R-A31O-07 TCGA-D8-A1JC-01A-11R-A13Q-07
## ENSG00000135446.17                         1524                         4732
##                    TCGA-BH-A0DX-01A-11R-A115-07 TCGA-A1-A0SI-01A-11R-A144-07
## ENSG00000135446.17                         2635                         5330
##                    TCGA-A7-A0D9-01A-31R-A056-07 TCGA-E2-A15K-06A-11R-A12P-07
## ENSG00000135446.17                         1620                         2331
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##                    TCGA-AR-A24Z-01A-11R-A169-07 TCGA-EW-A1OY-01A-11R-A144-07
## ENSG00000135446.17                         5081                         2349
##                    TCGA-C8-A130-01A-31R-A115-07 TCGA-E2-A14Y-01A-21R-A12D-07
## ENSG00000135446.17                         2076                         1661
##                    TCGA-D8-A27N-01A-11R-A16F-07 TCGA-AR-A0TY-01A-12R-A115-07
## ENSG00000135446.17                         3441                         5594
##                    TCGA-E2-A1BC-01A-11R-A12P-07 TCGA-E9-A243-01A-21R-A169-07
## ENSG00000135446.17                         1140                         4336
##                    TCGA-BH-A0BJ-11A-23R-A089-07 TCGA-BH-A0BJ-01A-11R-A056-07
## ENSG00000135446.17                         2579                         3243
##                    TCGA-AR-A24O-01A-11R-A169-07 TCGA-AR-A24R-01A-11R-A169-07
## ENSG00000135446.17                         3088                         3743
##                    TCGA-LL-A7T0-01A-31R-A352-07 TCGA-AC-A5XS-01A-11R-A29R-07
## ENSG00000135446.17                         3341                         2562
##                    TCGA-BH-A0DD-11A-23R-A12P-07 TCGA-BH-A18Q-11A-34R-A12D-07
## ENSG00000135446.17                         1508                         2693
##                    TCGA-E2-A1B4-01A-11R-A12P-07 TCGA-C8-A26Z-01A-11R-A16F-07
## ENSG00000135446.17                         2416                         4174
##                    TCGA-E2-A10E-01A-21R-A10J-07 TCGA-BH-A0HU-01A-11R-A034-07
## ENSG00000135446.17                         3106                         3752
##                    TCGA-OL-A5D8-01A-11R-A27Q-07 TCGA-C8-A1HG-01A-11R-A137-07
## ENSG00000135446.17                         1958                         1214
##                    TCGA-BH-A1FN-11A-34R-A13Q-07 TCGA-GM-A3NW-01A-21R-A22K-07
## ENSG00000135446.17                         3194                         2147
##                    TCGA-A8-A09X-01A-11R-A00Z-07 TCGA-E2-A1BD-01A-11R-A12P-07
## ENSG00000135446.17                         6349                         2832
##                    TCGA-D8-A1JN-01A-11R-A13Q-07 TCGA-A7-A5ZV-01A-11R-A28M-07
## ENSG00000135446.17                         1583                         2407
##                    TCGA-BH-A0HW-01A-11R-A034-07 TCGA-BH-A2L8-01A-11R-A18M-07
## ENSG00000135446.17                         3095                         3305
##                    TCGA-BH-A208-11A-51R-A157-07 TCGA-C8-A26Y-01A-11R-A16F-07
## ENSG00000135446.17                         3139                         2123
##                    TCGA-A1-A0SM-01A-11R-A084-07 TCGA-BH-A0DL-11A-13R-A115-07
## ENSG00000135446.17                         2176                         2895
##                    TCGA-E2-A153-11A-31R-A12D-07 TCGA-A7-A5ZX-01A-12R-A29R-07
## ENSG00000135446.17                         1792                         2670
##                    TCGA-BH-A0C3-11A-23R-A12P-07 TCGA-C8-A12L-01A-11R-A115-07
## ENSG00000135446.17                         2475                         3873
##                    TCGA-B6-A0I9-01A-11R-A034-07 TCGA-JL-A3YX-01A-11R-A22U-07
## ENSG00000135446.17                         3675                         4128
##                    TCGA-AC-A23H-01A-11R-A157-07 TCGA-OL-A5RX-01A-11R-A28M-07
## ENSG00000135446.17                         5103                         2627
##                    TCGA-AC-A23H-11A-12R-A157-07 TCGA-BH-A1FM-01A-11R-A13Q-07
## ENSG00000135446.17                         2963                         1439
##                    TCGA-B6-A0RP-01A-21R-A084-07 TCGA-EW-A1IW-01A-11R-A13Q-07
## ENSG00000135446.17                         2997                         4998
##                    TCGA-A0-A0JG-01A-31R-A084-07 TCGA-C8-A273-01A-11R-A16F-07
## ENSG00000135446.17                         2080                         4350
##                    TCGA-A8-A082-01A-11R-A00Z-07 TCGA-B6-A0X7-01A-11R-A10J-07
## ENSG00000135446.17                         2243                         2935
##                    TCGA-E2-A15M-11A-22R-A12D-07 TCGA-BH-A8FY-01A-11R-A36F-07
## ENSG00000135446.17                         1519                          919
##                    TCGA-E2-A56Z-01A-12R-A29R-07 TCGA-D8-A1XJ-01A-11R-A14M-07
## ENSG00000135446.17                         2041                         5155
```

```
##                    TCGA-D8-A142-01A-11R-A115-07 TCGA-BH-A0DZ-11A-22R-A089-07
## ENSG00000135446.17                         3928                         3124
##                    TCGA-E2-A1LA-01A-11R-A144-07 TCGA-E9-A229-01A-31R-A157-07
## ENSG00000135446.17                         1605                         3913
##                    TCGA-E9-A22D-01A-11R-A157-07 TCGA-E9-A1R3-01A-31R-A14M-07
## ENSG00000135446.17                         6577                         3358
##                    TCGA-D8-A1Y0-01A-11R-A14M-07 TCGA-B6-A0RV-01A-11R-A084-07
## ENSG00000135446.17                         5997                         2867
##                    TCGA-E2-A1LS-01A-12R-A157-07 TCGA-LL-A50Y-01A-11R-A266-07
## ENSG00000135446.17                         1611                         1768
##                    TCGA-LL-A8F5-01A-11R-A36F-07 TCGA-A2-A0CS-01A-11R-A115-07
## ENSG00000135446.17                         3109                         2337
##                    TCGA-A2-A0YK-01A-22R-A109-07 TCGA-B6-A3ZX-01A-11R-A239-07
## ENSG00000135446.17                         2025                         3885
##                    TCGA-AR-A0TU-01A-31R-A109-07 TCGA-AR-A1AP-01A-11R-A12P-07
## ENSG00000135446.17                         4687                         3013
##                    TCGA-S3-AA14-01A-11R-A41B-07 TCGA-E9-A1RB-11A-33R-A157-07
## ENSG00000135446.17                         2030                         3089
##                    TCGA-A0-A0J7-01A-11R-A034-07 TCGA-AC-A5XU-01A-11R-A28M-07
## ENSG00000135446.17                         4294                         3583
##                    TCGA-EW-A6SC-01A-12R-A32P-07 TCGA-AN-A0AM-01A-11R-A034-07
## ENSG00000135446.17                         2538                        55739
##                    TCGA-AQ-A54O-01A-11R-A266-07 TCGA-AN-A0AJ-01A-11R-A00Z-07
## ENSG00000135446.17                          516                         6541
##                    TCGA-C8-A135-01A-11R-A115-07 TCGA-BH-A0AU-11A-11R-A12P-07
## ENSG00000135446.17                         5035                         2360
##                    TCGA-AR-A2LH-01A-31R-A18M-07 TCGA-OL-A5RW-01A-11R-A28M-07
## ENSG00000135446.17                         2171                         1921
##                    TCGA-C8-A1HJ-01A-11R-A13Q-07 TCGA-C8-A1HL-01A-11R-A137-07
## ENSG00000135446.17                         2525                         1516
##                    TCGA-A8-A07W-01A-11R-A00Z-07 TCGA-D8-A1XY-01A-11R-A14M-07
## ENSG00000135446.17                         6185                         2474
##                    TCGA-BH-A5IZ-01A-11R-A27Q-07 TCGA-A2-A3XU-01A-12R-A22U-07
## ENSG00000135446.17                         2774                         6080
##                    TCGA-EW-A1P7-01A-21R-A144-07 TCGA-E9-A54X-01A-11R-A266-07
## ENSG00000135446.17                         1919                         1559
##                    TCGA-AC-A2FB-11A-13R-A17B-07 TCGA-AN-A0XT-01A-11R-A109-07
## ENSG00000135446.17                         3225                         2170
##                    TCGA-A0-A0JB-01A-11R-A32Y-07 TCGA-D8-A1JM-01A-11R-A13Q-07
## ENSG00000135446.17                          336                         8438
##                    TCGA-AC-A2B8-01A-11R-A17B-07 TCGA-B6-A0I8-01A-11R-A034-07
## ENSG00000135446.17                         4175                         3298
##                    TCGA-AN-A0AT-01A-11R-A034-07 TCGA-A2-A0YC-01A-11R-A109-07
## ENSG00000135446.17                         4908                         3351
##                    TCGA-BH-A0H3-01A-11R-A12P-07 TCGA-BH-A204-11A-53R-A157-07
## ENSG00000135446.17                         3324                         2113
##                    TCGA-A8-A09Q-01A-11R-A00Z-07 TCGA-A2-A0T6-01A-11R-A084-07
## ENSG00000135446.17                         4863                         3179
##                    TCGA-A2-A0D1-01A-11R-A034-07 TCGA-B6-A0WZ-01A-11R-A109-07
## ENSG00000135446.17                         1813                         2629
##                    TCGA-B6-A0RE-01A-11R-A056-07 TCGA-A7-A13F-01A-11R-A12P-07
## ENSG00000135446.17                         4373                         3315
##                    TCGA-D8-A1XG-01A-11R-A14D-07 TCGA-A2-A0YD-01A-11R-A109-07
## ENSG00000135446.17                         2087                         3021
```

```
##                    TCGA-E2-A14U-01A-11R-A22K-07 TCGA-A8-A076-01A-21R-A00Z-07
## ENSG00000135446.17                         3306                         6076
##                    TCGA-BH-A0B7-11A-34R-A115-07 TCGA-BH-A18U-01A-21R-A12D-07
## ENSG00000135446.17                         2234                         2583
##                    TCGA-A8-A06Z-01A-11R-A00Z-07 TCGA-BH-A0BT-11A-21R-A12P-07
## ENSG00000135446.17                         4350                         2661
##                    TCGA-BH-A0H7-01A-13R-A056-07 TCGA-BH-A0E0-01A-11R-A056-07
## ENSG00000135446.17                         7124                         2084
##                    TCGA-A7-A5ZW-01A-12R-A29R-07 TCGA-E9-A244-01A-11R-A169-07
## ENSG00000135446.17                         2709                         2828
##                    TCGA-D8-A1JD-01A-11R-A13Q-07 TCGA-BH-A18K-11A-13R-A12D-07
## ENSG00000135446.17                         6469                         1782
##                    TCGA-E9-A248-01A-11R-A169-07 TCGA-A2-A0T1-01A-21R-A084-07
## ENSG00000135446.17                         3233                         3703
##                    TCGA-BH-A0AZ-11A-22R-A12P-07 TCGA-AN-A0FX-01A-11R-A034-07
## ENSG00000135446.17                         2672                         6863
##                    TCGA-BH-A1FH-01A-12R-A13Q-07 TCGA-A7-A3IZ-01A-11R-A213-07
## ENSG00000135446.17                         3396                         2149
##                    TCGA-D8-A13Y-01A-11R-A115-07 TCGA-A2-A0SX-01A-12R-A084-07
## ENSG00000135446.17                         7255                         4136
##                    TCGA-B6-A0IN-01A-11R-A034-07 TCGA-A2-A3Y0-01A-11R-A239-07
## ENSG00000135446.17                         1647                        10738
##                    TCGA-BH-A0B3-01A-11R-A056-07 TCGA-AN-A0XO-01A-11R-A109-07
## ENSG00000135446.17                         4182                         3944
##                    TCGA-E2-A15K-11A-13R-A12P-07 TCGA-BH-A0HF-01A-11R-A056-07
## ENSG00000135446.17                         1720                         3308
##                    TCGA-GM-A2D9-01A-11R-A18M-07 TCGA-A2-A0CZ-01A-11R-A034-07
## ENSG00000135446.17                         1616                         2159
##                    TCGA-D8-A1JE-01A-11R-A13Q-07 TCGA-D8-A1XT-01A-11R-A14M-07
## ENSG00000135446.17                         3593                         5204
##                    TCGA-EW-A2FS-01A-11R-A17B-07 TCGA-A7-A0CJ-01A-21R-A00Z-07
## ENSG00000135446.17                          880                         3580
##                    TCGA-BH-A1FG-01A-11R-A13Q-07 TCGA-E2-A159-01A-11R-A115-07
## ENSG00000135446.17                         2351                         3155
##                    TCGA-AC-A62Y-01A-11R-A29R-07 TCGA-A7-A13G-01A-11R-A13Q-07
## ENSG00000135446.17                         1949                         1683
##                    TCGA-AN-A0XU-01A-11R-A109-07 TCGA-JL-A3YW-01A-12R-A239-07
## ENSG00000135446.17                         5845                         4391
##                    TCGA-BH-A1EY-01A-11R-A13Q-07 TCGA-BH-A204-01A-11R-A157-07
## ENSG00000135446.17                         1930                         5022
##                    TCGA-LL-A5YP-01A-21R-A28M-07 TCGA-BH-AB28-01A-31R-A41B-07
## ENSG00000135446.17                         3327                         2979
##                    TCGA-B6-A0IJ-01A-11R-A034-07 TCGA-E2-A15E-01A-11R-A12D-07
## ENSG00000135446.17                         4696                         2331
##                    TCGA-D8-A1XZ-01A-11R-A14M-07 TCGA-E2-A15E-06A-11R-A12D-07
## ENSG00000135446.17                         3815                         4331
##                    TCGA-A1-A0SQ-01A-21R-A144-07 TCGA-A7-A13F-11A-42R-A12P-07
## ENSG00000135446.17                         2976                         1665
##                    TCGA-GM-A3XN-01A-12R-A22U-07 TCGA-AC-A7VC-01A-11R-A352-07
## ENSG00000135446.17                         2753                         4078
##                    TCGA-S3-AA12-01A-11R-A41B-07 TCGA-A2-A3KC-01A-11R-A213-07
## ENSG00000135446.17                         1899                         2881
##                    TCGA-AN-A0FY-01A-11R-A034-07 TCGA-BH-A0B7-01A-12R-A115-07
## ENSG00000135446.17                         5679                         3020
```

```
##                    TCGA-E2-A15J-01A-11R-A12P-07 TCGA-BH-A1F0-11B-23R-A137-07
## ENSG00000135446.17                         3316                         2554
##                    TCGA-AR-A1AJ-01A-21R-A12P-07 TCGA-A8-A0AB-01A-11R-A034-07
## ENSG00000135446.17                         2537                         5701
##                    TCGA-E9-A1RE-01A-11R-A157-07 TCGA-BH-A18U-11A-23R-A12D-07
## ENSG00000135446.17                         3181                         1789
##                    TCGA-BH-A0H7-11A-13R-A089-07 TCGA-BH-A0BT-01A-11R-A12P-07
## ENSG00000135446.17                         4269                         3528
##                    TCGA-OL-A66O-01A-11R-A31O-07 TCGA-AC-A3YI-01A-21R-A239-07
## ENSG00000135446.17                         5699                         2142
##                    TCGA-BH-A0E0-11A-13R-A089-07 TCGA-E2-A14R-01A-11R-A115-07
## ENSG00000135446.17                         2769                         2421
##                    TCGA-LL-A5YL-01A-12R-A29R-07 TCGA-BH-A18K-01A-11R-A12D-07
## ENSG00000135446.17                         1775                         2098
##                    TCGA-A2-A0ST-01A-12R-A084-07 TCGA-D8-A27F-01A-11R-A16F-07
## ENSG00000135446.17                         2789                         4821
##                    TCGA-EW-A1P4-01A-21R-A144-07 TCGA-B6-A0IB-01A-11R-A034-07
## ENSG00000135446.17                         3892                         5085
##                    TCGA-A0-A0JC-01A-11R-A056-07 TCGA-BH-A1FH-11B-42R-A13Q-07
## ENSG00000135446.17                         1301                         1866
##                    TCGA-B6-A0X1-01A-11R-A109-07 TCGA-E9-A1N3-01A-12R-A157-07
## ENSG00000135446.17                         1825                         3397
##                    TCGA-BH-A0B3-11B-21R-A089-07 TCGA-AN-A0FS-01A-11R-A034-07
## ENSG00000135446.17                         2767                         4472
##                    TCGA-E2-A15K-01A-11R-A12P-07 TCGA-AC-A7VB-01A-11R-A352-07
## ENSG00000135446.17                         2864                         2996
##                    TCGA-LD-A7W5-01A-22R-A352-07 TCGA-B6-A0RL-01A-11R-A084-07
## ENSG00000135446.17                         4873                         4617
##                    TCGA-BH-A18P-11A-43R-A12D-07 TCGA-C8-A8HP-01A-11R-A36F-07
## ENSG00000135446.17                         1815                         1910
##                    TCGA-EW-A3U0-01A-11R-A22K-07 TCGA-D8-A1J9-01A-11R-A13Q-07
## ENSG00000135446.17                         2486                         4298
##                    TCGA-V7-A7HQ-01A-11R-A33J-07
## ENSG00000135446.17                         1253
```

```
dim(CDK4_data)
```

```
## [1]    1 1231
```

```
# Transpose
CDK4_data <- t(CDK4_data) # transpose the data
head(CDK4_data)
```

```
##                              ENSG00000135446.17
## TCGA-GM-A2DL-01A-11R-A18M-07               2709
## TCGA-AC-A2QI-01A-12R-A19W-07               3738
## TCGA-A8-A06R-01A-11R-A00Z-07               1722
## TCGA-EW-A1PD-01A-11R-A144-07               3313
## TCGA-A0-A12D-01A-11R-A115-07               1459
## TCGA-AR-A24N-01A-11R-A169-07               3656
```

```r
dim(CDK4_data)
```

```
## [1] 1231    1
```

```r
# Rename the column
colnames(CDK4_data) <- "CDK4"
head(CDK4_data)
```

```
##                              CDK4
## TCGA-GM-A2DL-01A-11R-A18M-07 2709
## TCGA-AC-A2QI-01A-12R-A19W-07 3738
## TCGA-A8-A06R-01A-11R-A00Z-07 1722
## TCGA-EW-A1PD-01A-11R-A144-07 3313
## TCGA-AO-A12D-01A-11R-A115-07 1459
## TCGA-AR-A24N-01A-11R-A169-07 3656
```

```r
dim(CDK4_data)
```

```
## [1] 1231    1
```

```r
class(CDK4_data)
```

```
## [1] "matrix" "array"
```

```r
# Change matrix to data frame
CDK4_data_df <- as.data.frame(CDK4_data)
head(CDK4_data_df)
```

```
##                              CDK4
## TCGA-GM-A2DL-01A-11R-A18M-07 2709
## TCGA-AC-A2QI-01A-12R-A19W-07 3738
## TCGA-A8-A06R-01A-11R-A00Z-07 1722
## TCGA-EW-A1PD-01A-11R-A144-07 3313
## TCGA-AO-A12D-01A-11R-A115-07 1459
## TCGA-AR-A24N-01A-11R-A169-07 3656
```

```r
# Convert to tibble
CDK4_data_tbl <- as_tibble(CDK4_data_df)
```

```r
# Create a scatter plot

# Combine the data from both gene data frames

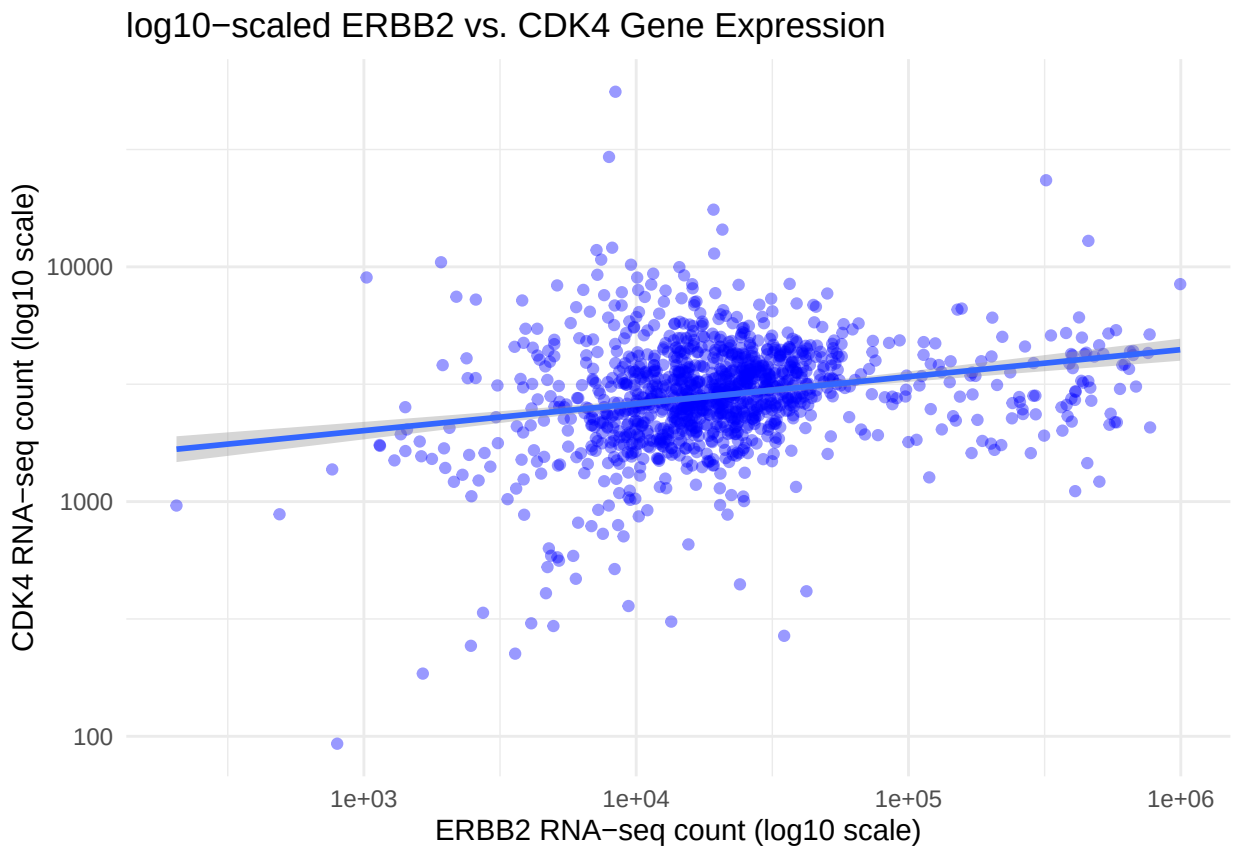
Two_gene_data_df <- cbind(ERBB2_data_df, CDK4_data_df)
head(Two_gene_data_df)
```

```
##                              ERBB2 CDK4
## TCGA-GM-A2DL-01A-11R-A18M-07  7613 2709
## TCGA-AC-A2QI-01A-12R-A19W-07 41006 3738
## TCGA-A8-A06R-01A-11R-A00Z-07  6858 1722
## TCGA-EW-A1PD-01A-11R-A144-07 52772 3313
## TCGA-AO-A12D-01A-11R-A115-07 453696 1459
## TCGA-AR-A24N-01A-11R-A169-07 24483 3656
```

```r
# Create scatterplot

scatterplot <- ggplot(Two_gene_data_df, # data to be used
       aes(x = ERBB2, y = CDK4)) + # map variables
  geom_point(alpha = 0.4, color = "blue") + # choose plot
  geom_smooth(method = "lm") + # add linear regression line based on peer feedback
  scale_x_log10() + # log10 scale is used to better visualize the distribution
  scale_y_log10() +
  labs(x = "ERBB2 RNA-seq count (log10 scale)", # add labels
       y = "CDK4 RNA-seq count (log10 scale)",
       title = "log10-scaled ERBB2 vs. CDK4 Gene Expression") +
  theme_minimal() # customize appearance
scatterplot
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# ggplotly(scatterplot)

# Most samples cluster within a relatively narrow range of ERBB2 and CDK4 expression
# The log10 scales allow the relationship to be visualized despite the wide range of RNA-seq counts
# The regression line suggests a modest positive relationship between ERBB2 and CDK4 expression
```

3c. Violin + box plot — ERBB2 expression across PAM50 subtypes

```
# Merge gene data to metadata

# row names were a bit different since read.csv() defaults to check.names = TRUE
# hyphens get converted to periods
# went back and switched to counts <- read.csv("counts.csv", row.names = 1, check.names = FALSE)
# Used ChatGPTs assistance to troubleshoot

ERBB2gene_metadata <- merge(ERBB2_data_df, metadata,
                        by = "row.names",
                        all = TRUE)

head(ERBB2gene_metadata)
```

```
##                      Row.names  ERBB2      patient           sample
## 1 TCGA-3C-AAAU-01A-11R-A41B-07  21360 TCGA-3C-AAAU TCGA-3C-AAAU-01A
## 2 TCGA-3C-AALI-01A-11R-A41B-07 367620 TCGA-3C-AALI TCGA-3C-AALI-01A
## 3 TCGA-3C-AALJ-01A-31R-A41B-07  12131 TCGA-3C-AALJ TCGA-3C-AALJ-01A
## 4 TCGA-3C-AALK-01A-11R-A41B-07  87332 TCGA-3C-AALK TCGA-3C-AALK-01A
## 5 TCGA-4H-AAAK-01A-12R-A41B-07  29303 TCGA-4H-AAAK TCGA-4H-AAAK-01A
## 6 TCGA-5L-AAT0-01A-12R-A41B-07  15966 TCGA-5L-AAT0 TCGA-5L-AAT0-01A
##   shortLetterCode          definition sample_submitter_id sample_type_id
## 1              TP Primary solid Tumor    TCGA-3C-AAAU-01A              1
## 2              TP Primary solid Tumor    TCGA-3C-AALI-01A              1
## 3              TP Primary solid Tumor    TCGA-3C-AALJ-01A              1
## 4              TP Primary solid Tumor    TCGA-3C-AALK-01A              1
## 5              TP Primary solid Tumor    TCGA-4H-AAAK-01A              1
## 6              TP Primary solid Tumor    TCGA-5L-AAT0-01A              1
##   tumor_descriptor                             sample_id   sample_type
## 1          Primary 197b76a1-a09a-4659-83aa-2c14fd1023a9 Primary Tumor
## 2          Primary 0050d7c9-ece9-4b6c-8023-1ff2efcb3c9c Primary Tumor
## 3          Primary e355913a-6e4c-4d63-8a53-8be4c5b003b8 Primary Tumor
## 4          Primary 2a532346-0008-4af8-99bc-d709585da1d6 Primary Tumor
## 5          Primary 685b5df4-0620-40e2-aa09-0dd736bb5e2b Primary Tumor
## 6          Primary 8434f4a2-0ead-40f6-9b63-9c958a3fb812 Primary Tumor
##    composition days_to_collection    state initial_weight preservation_method
## 1 Not Reported               3599 released             60                 OCT
## 2 Not Reported               3772 released            120                 OCT
## 3 Not Reported               1026 released            420                 OCT
## 4 Not Reported               1022 released            140                 OCT
## 5 Not Reported                 35 released            770                 OCT
## 6 Not Reported               1320 released             80             Unknown
##                  pathology_report_uuid submitter_id oct_embedded specimen_type
## 1 0CD23E1B-3FA3-4A43-AE6E-C8E7B51252F8 TCGA-3C-AAAU         true       Unknown
## 2 84E6A935-1A49-4BC1-9669-3DEA161CF6FC TCGA-3C-AALI         true       Unknown
## 3 265E5A9A-64FD-4B86-89BC-5E89F253C118 TCGA-3C-AALJ         true       Unknown
## 4 F43B01E6-E1DB-44B1-8003-93870606346A TCGA-3C-AALK         true       Unknown
## 5 8894688F-7167-48A1-BB1B-FC219B7675C2 TCGA-4H-AAAK         true       Unknown
## 6 F9B6971F-23C0-465F-BFEC-778BF228A1AE TCGA-5L-AAT0        false       Unknown
##   is_ffpe tissue_type synchronous_malignancy ajcc_pathologic_stage
## 1   FALSE       Tumor                  <NA>                  <NA>
## 2   FALSE       Tumor                    No             Stage IIB
```

```
## 3   FALSE       Tumor                    No             Stage IIB
## 4   FALSE       Tumor                    No              Stage IA
## 5   FALSE       Tumor                    No            Stage IIIA
## 6   FALSE       Tumor                  <NA>                  <NA>
##   days_to_diagnosis laterality
## 1              1808       <NA>
## 2                 0      Right
## 3                 0      Right
## 4                 0      Right
## 5                 0       Left
## 6                39       <NA>
##
## 1
## 2
## 3 c(490, NA, 126, 511, 511, NA, 126);c("No", NA, "No", "Yes", "No", NA, "No");c(154, NA, 63, 368, 15
## 4
## 5
## 6
##   tissue_or_organ_of_origin age_at_diagnosis               primary_diagnosis
## 1                 Lung, NOS            22019           Lobular carcinoma, NOS
## 2               Breast, NOS            18538 Infiltrating duct carcinoma, NOS
## 3               Breast, NOS            22848 Infiltrating duct carcinoma, NOS
## 4               Breast, NOS            19074 Infiltrating duct carcinoma, NOS
## 5               Breast, NOS            18371           Lobular carcinoma, NOS
## 6             Cervix uteri             15432              Adenocarcinoma, NOS
##   metastasis_at_diagnosis prior_malignancy year_of_diagnosis prior_treatment
## 1                    <NA>            <NA>                NA             Yes
## 2                    <NA>              no              2003              No
## 3                    <NA>              no              2011              No
## 4                    <NA>              no              2011              No
## 5                    <NA>              no              2013              No
## 6                    <NA>            <NA>                NA            <NA>
##   diagnosis_is_primary_disease method_of_diagnosis ajcc_staging_system_edition
## 1                        FALSE                <NA>                         <NA>
## 2                         TRUE         Core Biopsy                          6th
## 3                         TRUE         Core Biopsy                          7th
## 4                         TRUE         Core Biopsy                          7th
## 5                         TRUE         Core Biopsy                          7th
## 6                        FALSE                <NA>                         <NA>
##   ajcc_pathologic_t morphology ajcc_pathologic_n ajcc_pathologic_m
## 1              <NA>     8520/3              <NA>              <NA>
## 2                T2     8500/3               N1a                M0
## 3                T2     8500/3               N1a                M0
## 4               T1c     8500/3            N0 (i+)                M0
## 5                T2     8520/3               N2a                M0
## 6              <NA>     8140/3              <NA>              <NA>
##   classification_of_tumor                          diagnosis_id icd_10_code
## 1              metastasis 6e43f538-6f17-4eb9-82ea-d6028880660b        <NA>
## 2                 primary 8cafc022-585f-54a1-a7d4-cfa632b3991e       C50.9
## 3                 primary 63d85b81-8eba-5f17-8552-92babf137c00       C50.9
## 4                 primary 8c90b19d-54f7-5788-a5eb-49abe239ef0b       C50.9
## 5                 primary 81c406cd-ad2d-552f-b543-8b2b03044686       C50.9
## 6    Synchronous primary 6ac00b2d-7607-4c59-af06-47f425d98afc        <NA>
##   site_of_resection_or_biopsy                sites_of_involvement
```

```
## 1                Not Reported
## 2                 Breast, NOS            Breast, Right Upper Outer
## 3                 Breast, NOS                              Breast, NOS
## 4                 Breast, NOS                              Breast, NOS
## 5                 Breast, NOS Breast, NOS;Breast, Left Upper Outer
## 6                Not Reported
##   days_to_last_follow_up follow_ups_disease_response                      race
## 1                   4047              WT-With Tumor                     white
## 2                   4005              TF-Tumor Free black or african american
## 3                   1474              TF-Tumor Free black or african american
## 4                   1448              TF-Tumor Free black or african american
## 5                    348              TF-Tumor Free                     white
## 6                   1477              TF-Tumor Free                     white
##   gender              ethnicity vital_status age_at_index days_to_birth
## 1 female not hispanic or latino        Alive           55        -20211
## 2 female not hispanic or latino        Alive           50        -18538
## 3 female not hispanic or latino        Alive           62        -22848
## 4 female not hispanic or latino        Alive           52        -19074
## 5 female not hispanic or latino        Alive           50        -18371
## 6 female     hispanic or latino        Alive           42        -15393
##                         demographic_id age_is_obfuscated
## 1 cee0a94c-1d9e-5650-a500-a6b021fe138d             FALSE
## 2 583a1ee5-3175-523e-ba1f-75a30a3e1e41             FALSE
## 3 9619a908-1684-547f-a407-6adf93e15b8d             FALSE
## 4 e54b1469-fffc-5291-a8e3-df2092ab5f34             FALSE
## 5 64a204f0-b380-5962-a927-84af1841b6d6             FALSE
## 6 776bb6f9-f5c8-57b6-bf4d-4014b8025a06             FALSE
##   country_of_residence_at_enrollment bcr_patient_barcode primary_site
## 1                      United States    TCGA-3C-AAAU-01A       Breast
## 2                      United States    TCGA-3C-AALI-01A       Breast
## 3                      United States    TCGA-3C-AALJ-01A       Breast
## 4                      United States    TCGA-3C-AALK-01A       Breast
## 5                             Russia    TCGA-4H-AAAK-01A       Breast
## 6                             Brazil    TCGA-5L-AAT0-01A       Breast
##   project_id
## 1  TCGA-BRCA
## 2  TCGA-BRCA
## 3  TCGA-BRCA
## 4  TCGA-BRCA
## 5  TCGA-BRCA
## 6  TCGA-BRCA
##
## 1 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
## 2 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
## 3 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
## 4 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
## 5 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
## 6 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas and Adenocarcinomas;Cystic
##                         name releasable released figo_stage
## 1 Breast Invasive Carcinoma       TRUE     TRUE       <NA>
## 2 Breast Invasive Carcinoma       TRUE     TRUE       <NA>
## 3 Breast Invasive Carcinoma       TRUE     TRUE       <NA>
## 4 Breast Invasive Carcinoma       TRUE     TRUE       <NA>
## 5 Breast Invasive Carcinoma       TRUE     TRUE       <NA>
```

```
## 6 Breast Invasive Carcinoma       TRUE     TRUE  Stage IB2
##   figo_staging_edition_year days_to_death        tumor_of_origin
## 1                        NA            NA TCGA-3C-AAAU_diagnosis
## 2                        NA            NA                   <NA>
## 3                        NA            NA                   <NA>
## 4                        NA            NA                   <NA>
## 5                        NA            NA                   <NA>
## 6                      2009            NA                   <NA>
##   days_to_sample_procurement last_known_disease_status tumor_grade
## 1                         NA                      <NA>        <NA>
## 2                         NA                      <NA>        <NA>
## 3                         NA                      <NA>        <NA>
## 4                         NA                      <NA>        <NA>
## 5                         NA                      <NA>        <NA>
## 6                         NA                      <NA>        <NA>
##   progression_or_recurrence alcohol_history exposure_id year_of_birth
## 1                      <NA>            <NA>        <NA>            NA
## 2                      <NA>            <NA>        <NA>            NA
## 3                      <NA>            <NA>        <NA>            NA
## 4                      <NA>            <NA>        <NA>            NA
## 5                      <NA>            <NA>        <NA>            NA
## 6                      <NA>            <NA>        <NA>            NA
##   paper_patient paper_Tumor.Type paper_Included_in_previous_marker_papers
## 1  TCGA-3C-AAAU             BRCA                                       NO
## 2  TCGA-3C-AALI             BRCA                                       NO
## 3  TCGA-3C-AALJ             BRCA                                       NO
## 4  TCGA-3C-AALK             BRCA                                       NO
## 5  TCGA-4H-AAAK             BRCA                                       NO
## 6  TCGA-5L-AAT0             BRCA                                       NO
##   paper_vital_status paper_days_to_birth paper_days_to_death
## 1              Alive              -20211                  NA
## 2              Alive              -18538                  NA
## 3              Alive              -22848                  NA
## 4              Alive              -19074                  NA
## 5              Alive              -18371                  NA
## 6              Alive              -15393                  NA
##   paper_days_to_last_followup paper_age_at_initial_pathologic_diagnosis
## 1                        4047                                        55
## 2                        4005                                        50
## 3                        1474                                        62
## 4                        1448                                        52
## 5                         348                                        50
## 6                        1477                                        42
##   paper_pathologic_stage paper_Tumor_Grade paper_BRCA_Pathology
## 1                   <NA>               NA                 <NA>
## 2              Stage_II               NA                 <NA>
## 3              Stage_II               NA                 <NA>
## 4               Stage_I               NA                 <NA>
## 5             Stage_III               NA                 <NA>
## 6              Stage_II               NA                 <NA>
##   paper_BRCA_Subtype_PAM50 paper_MSI_status paper_HPV_Status
## 1                     LumA              NA              NA
## 2                     Her2              NA              NA
## 3                     LumB              NA              NA
```

```
## 4                      LumA              NA              NA
## 5                      LumA              NA              NA
## 6                      LumA              NA              NA
##   paper_tobacco_smoking_history paper_CNV.Clusters paper_Mutation.Clusters
## 1                            NA                C6                      C7
## 2                            NA                C6                      C9
## 3                            NA                C6                      C4
## 4                            NA                C1                      C5
## 5                            NA                C6                      C4
## 6                            NA                C1                      C9
##   paper_DNA.Methylation.Clusters paper_mRNA.Clusters paper_miRNA.Clusters
## 1                             C1                 C1                   C3
## 2                             C2                 C2                   C3
## 3                             C2                 C2                   C2
## 4                             C2                 C2                   C2
## 5                             C1                 C2                   C2
## 6                             C1                 C2                   C2
##   paper_lncRNA.Clusters paper_Protein.Clusters paper_PARADIGM.Clusters
## 1                  <NA>                  <NA>                      C5
## 2                  <NA>                    C2                      C4
## 3                  <NA>                  <NA>                      C4
## 4                  <NA>                    C2                      C6
## 5                  <NA>                    C2                      C6
## 6                  <NA>                  <NA>                      C6
##   paper_Pan.Gyn.Clusters
## 1                   <NA>
## 2                     C4
## 3                   <NA>
## 4                     C4
## 5                     C1
## 6                   <NA>
```

```
dim(ERBB2gene_metadata)
```

```
## [1] 1231   99
```

```
# Clean up merged data frame

rownames(ERBB2gene_metadata) <- ERBB2gene_metadata$Row.names
# put the sample IDs back into the row names

ERBB2gene_metadata <- subset(ERBB2gene_metadata,
                         select = -c(Row.names))
# remove the now-unnecessary Row.names column.

head(ERBB2gene_metadata)
```

```
##                              ERBB2      patient           sample
## TCGA-3C-AAAU-01A-11R-A41B-07  21360 TCGA-3C-AAAU TCGA-3C-AAAU-01A
## TCGA-3C-AALI-01A-11R-A41B-07 367620 TCGA-3C-AALI TCGA-3C-AALI-01A
## TCGA-3C-AALJ-01A-31R-A41B-07  12131 TCGA-3C-AALJ TCGA-3C-AALJ-01A
## TCGA-3C-AALK-01A-11R-A41B-07  87332 TCGA-3C-AALK TCGA-3C-AALK-01A
## TCGA-4H-AAAK-01A-12R-A41B-07  29303 TCGA-4H-AAAK TCGA-4H-AAAK-01A
```

```
## TCGA-5L-AAT0-01A-12R-A41B-07  15966 TCGA-5L-AAT0 TCGA-5L-AAT0-01A
##                               shortLetterCode          definition
## TCGA-3C-AAAU-01A-11R-A41B-07              TP Primary solid Tumor
## TCGA-3C-AALI-01A-11R-A41B-07              TP Primary solid Tumor
## TCGA-3C-AALJ-01A-31R-A41B-07              TP Primary solid Tumor
## TCGA-3C-AALK-01A-11R-A41B-07              TP Primary solid Tumor
## TCGA-4H-AAAK-01A-12R-A41B-07              TP Primary solid Tumor
## TCGA-5L-AAT0-01A-12R-A41B-07              TP Primary solid Tumor
##                               sample_submitter_id sample_type_id
## TCGA-3C-AAAU-01A-11R-A41B-07   TCGA-3C-AAAU-01A              1
## TCGA-3C-AALI-01A-11R-A41B-07   TCGA-3C-AALI-01A              1
## TCGA-3C-AALJ-01A-31R-A41B-07   TCGA-3C-AALJ-01A              1
## TCGA-3C-AALK-01A-11R-A41B-07   TCGA-3C-AALK-01A              1
## TCGA-4H-AAAK-01A-12R-A41B-07   TCGA-4H-AAAK-01A              1
## TCGA-5L-AAT0-01A-12R-A41B-07   TCGA-5L-AAT0-01A              1
##                               tumor_descriptor
## TCGA-3C-AAAU-01A-11R-A41B-07          Primary
## TCGA-3C-AALI-01A-11R-A41B-07          Primary
## TCGA-3C-AALJ-01A-31R-A41B-07          Primary
## TCGA-3C-AALK-01A-11R-A41B-07          Primary
## TCGA-4H-AAAK-01A-12R-A41B-07          Primary
## TCGA-5L-AAT0-01A-12R-A41B-07          Primary
##                                                              sample_id   sample_type
## TCGA-3C-AAAU-01A-11R-A41B-07 197b76a1-a09a-4659-83aa-2c14fd1023a9 Primary Tumor
## TCGA-3C-AALI-01A-11R-A41B-07 0050d7c9-ece9-4b6c-8023-1ff2efcb3c9c Primary Tumor
## TCGA-3C-AALJ-01A-31R-A41B-07 e355913a-6e4c-4d63-8a53-8be4c5b003b8 Primary Tumor
## TCGA-3C-AALK-01A-11R-A41B-07 2a532346-0008-4af8-99bc-d709585da1d6 Primary Tumor
## TCGA-4H-AAAK-01A-12R-A41B-07 685b5df4-0620-40e2-aa09-0dd736bb5e2b Primary Tumor
## TCGA-5L-AAT0-01A-12R-A41B-07 8434f4a2-0ead-40f6-9b63-9c958a3fb812 Primary Tumor
##                               composition days_to_collection    state
## TCGA-3C-AAAU-01A-11R-A41B-07 Not Reported               3599 released
## TCGA-3C-AALI-01A-11R-A41B-07 Not Reported               3772 released
## TCGA-3C-AALJ-01A-31R-A41B-07 Not Reported               1026 released
## TCGA-3C-AALK-01A-11R-A41B-07 Not Reported               1022 released
## TCGA-4H-AAAK-01A-12R-A41B-07 Not Reported                 35 released
## TCGA-5L-AAT0-01A-12R-A41B-07 Not Reported               1320 released
##                               initial_weight preservation_method
## TCGA-3C-AAAU-01A-11R-A41B-07             60                 OCT
## TCGA-3C-AALI-01A-11R-A41B-07            120                 OCT
## TCGA-3C-AALJ-01A-31R-A41B-07            420                 OCT
## TCGA-3C-AALK-01A-11R-A41B-07            140                 OCT
## TCGA-4H-AAAK-01A-12R-A41B-07            770                 OCT
## TCGA-5L-AAT0-01A-12R-A41B-07             80             Unknown
##                                                pathology_report_uuid submitter_id
## TCGA-3C-AAAU-01A-11R-A41B-07 0CD23E1B-3FA3-4A43-AE6E-C8E7B51252F8 TCGA-3C-AAAU
## TCGA-3C-AALI-01A-11R-A41B-07 84E6A935-1A49-4BC1-9669-3DEA161CF6FC TCGA-3C-AALI
## TCGA-3C-AALJ-01A-31R-A41B-07 265E5A9A-64FD-4B86-89BC-5E89F253C118 TCGA-3C-AALJ
## TCGA-3C-AALK-01A-11R-A41B-07 F43B01E6-E1DB-44B1-8003-93870606346A TCGA-3C-AALK
## TCGA-4H-AAAK-01A-12R-A41B-07 8894688F-7167-48A1-BB1B-FC219B7675C2 TCGA-4H-AAAK
## TCGA-5L-AAT0-01A-12R-A41B-07 F9B6971F-23C0-465F-BFEC-778BF228A1AE TCGA-5L-AAT0
##                               oct_embedded specimen_type is_ffpe tissue_type
## TCGA-3C-AAAU-01A-11R-A41B-07         true       Unknown   FALSE       Tumor
## TCGA-3C-AALI-01A-11R-A41B-07         true       Unknown   FALSE       Tumor
## TCGA-3C-AALJ-01A-31R-A41B-07         true       Unknown   FALSE       Tumor
```

```
## TCGA-3C-AALK-01A-11R-A41B-07         true       Unknown   FALSE       Tumor
## TCGA-4H-AAAK-01A-12R-A41B-07         true       Unknown   FALSE       Tumor
## TCGA-5L-AAT0-01A-12R-A41B-07        false       Unknown   FALSE       Tumor
##                              synchronous_malignancy ajcc_pathologic_stage
## TCGA-3C-AAAU-01A-11R-A41B-07                   <NA>                  <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                     No             Stage IIB
## TCGA-3C-AALJ-01A-31R-A41B-07                     No             Stage IIB
## TCGA-3C-AALK-01A-11R-A41B-07                     No              Stage IA
## TCGA-4H-AAAK-01A-12R-A41B-07                     No            Stage IIIA
## TCGA-5L-AAT0-01A-12R-A41B-07                   <NA>                  <NA>
##                              days_to_diagnosis laterality
## TCGA-3C-AAAU-01A-11R-A41B-07              1808       <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                 0      Right
## TCGA-3C-AALJ-01A-31R-A41B-07                 0      Right
## TCGA-3C-AALK-01A-11R-A41B-07                 0      Right
## TCGA-4H-AAAK-01A-12R-A41B-07                 0       Left
## TCGA-5L-AAT0-01A-12R-A41B-07                39       <NA>
##
## TCGA-3C-AAAU-01A-11R-A41B-07
## TCGA-3C-AALI-01A-11R-A41B-07
## TCGA-3C-AALJ-01A-31R-A41B-07 c(490, NA, 126, 511, 511, NA, 126);c("No", NA, "No", "Yes", "No", NA, "
## TCGA-3C-AALK-01A-11R-A41B-07
## TCGA-4H-AAAK-01A-12R-A41B-07
## TCGA-5L-AAT0-01A-12R-A41B-07
##                              tissue_or_organ_of_origin age_at_diagnosis
## TCGA-3C-AAAU-01A-11R-A41B-07                Lung, NOS           22019
## TCGA-3C-AALI-01A-11R-A41B-07             Breast, NOS            18538
## TCGA-3C-AALJ-01A-31R-A41B-07             Breast, NOS            22848
## TCGA-3C-AALK-01A-11R-A41B-07             Breast, NOS            19074
## TCGA-4H-AAAK-01A-12R-A41B-07             Breast, NOS            18371
## TCGA-5L-AAT0-01A-12R-A41B-07            Cervix uteri            15432
##                                        primary_diagnosis
## TCGA-3C-AAAU-01A-11R-A41B-07          Lobular carcinoma, NOS
## TCGA-3C-AALI-01A-11R-A41B-07 Infiltrating duct carcinoma, NOS
## TCGA-3C-AALJ-01A-31R-A41B-07 Infiltrating duct carcinoma, NOS
## TCGA-3C-AALK-01A-11R-A41B-07 Infiltrating duct carcinoma, NOS
## TCGA-4H-AAAK-01A-12R-A41B-07          Lobular carcinoma, NOS
## TCGA-5L-AAT0-01A-12R-A41B-07             Adenocarcinoma, NOS
##                              metastasis_at_diagnosis prior_malignancy
## TCGA-3C-AAAU-01A-11R-A41B-07                   <NA>             <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                   <NA>               no
## TCGA-3C-AALJ-01A-31R-A41B-07                   <NA>               no
## TCGA-3C-AALK-01A-11R-A41B-07                   <NA>               no
## TCGA-4H-AAAK-01A-12R-A41B-07                   <NA>               no
## TCGA-5L-AAT0-01A-12R-A41B-07                   <NA>             <NA>
##                              year_of_diagnosis prior_treatment
## TCGA-3C-AAAU-01A-11R-A41B-07                NA             Yes
## TCGA-3C-AALI-01A-11R-A41B-07              2003              No
## TCGA-3C-AALJ-01A-31R-A41B-07              2011              No
## TCGA-3C-AALK-01A-11R-A41B-07              2011              No
## TCGA-4H-AAAK-01A-12R-A41B-07              2013              No
## TCGA-5L-AAT0-01A-12R-A41B-07                NA            <NA>
##                              diagnosis_is_primary_disease method_of_diagnosis
## TCGA-3C-AAAU-01A-11R-A41B-07                        FALSE                <NA>
```

```
## TCGA-3C-AALI-01A-11R-A41B-07                        TRUE         Core Biopsy
## TCGA-3C-AALJ-01A-31R-A41B-07                        TRUE         Core Biopsy
## TCGA-3C-AALK-01A-11R-A41B-07                        TRUE         Core Biopsy
## TCGA-4H-AAAK-01A-12R-A41B-07                        TRUE         Core Biopsy
## TCGA-5L-AAT0-01A-12R-A41B-07                       FALSE               <NA>
##                              ajcc_staging_system_edition ajcc_pathologic_t
## TCGA-3C-AAAU-01A-11R-A41B-07                       <NA>              <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                        6th                T2
## TCGA-3C-AALJ-01A-31R-A41B-07                        7th                T2
## TCGA-3C-AALK-01A-11R-A41B-07                        7th               T1c
## TCGA-4H-AAAK-01A-12R-A41B-07                        7th                T2
## TCGA-5L-AAT0-01A-12R-A41B-07                       <NA>              <NA>
##                              morphology ajcc_pathologic_n ajcc_pathologic_m
## TCGA-3C-AAAU-01A-11R-A41B-07    8520/3              <NA>              <NA>
## TCGA-3C-AALI-01A-11R-A41B-07    8500/3               N1a                M0
## TCGA-3C-AALJ-01A-31R-A41B-07    8500/3               N1a                M0
## TCGA-3C-AALK-01A-11R-A41B-07    8500/3          N0 (i+)                 M0
## TCGA-4H-AAAK-01A-12R-A41B-07    8520/3               N2a                M0
## TCGA-5L-AAT0-01A-12R-A41B-07    8140/3              <NA>              <NA>
##                              classification_of_tumor
## TCGA-3C-AAAU-01A-11R-A41B-07             metastasis
## TCGA-3C-AALI-01A-11R-A41B-07                primary
## TCGA-3C-AALJ-01A-31R-A41B-07                primary
## TCGA-3C-AALK-01A-11R-A41B-07                primary
## TCGA-4H-AAAK-01A-12R-A41B-07                primary
## TCGA-5L-AAT0-01A-12R-A41B-07    Synchronous primary
##                                                    diagnosis_id icd_10_code
## TCGA-3C-AAAU-01A-11R-A41B-07 6e43f538-6f17-4eb9-82ea-d6028880660b        <NA>
## TCGA-3C-AALI-01A-11R-A41B-07 8cafc022-585f-54a1-a7d4-cfa632b3991e       C50.9
## TCGA-3C-AALJ-01A-31R-A41B-07 63d85b81-8eba-5f17-8552-92babf137c00       C50.9
## TCGA-3C-AALK-01A-11R-A41B-07 8c90b19d-54f7-5788-a5eb-49abe239ef0b       C50.9
## TCGA-4H-AAAK-01A-12R-A41B-07 81c406cd-ad2d-552f-b543-8b2b03044686       C50.9
## TCGA-5L-AAT0-01A-12R-A41B-07 6ac00b2d-7607-4c59-af06-47f425d98afc        <NA>
##                              site_of_resection_or_biopsy
## TCGA-3C-AAAU-01A-11R-A41B-07               Not Reported
## TCGA-3C-AALI-01A-11R-A41B-07                Breast, NOS
## TCGA-3C-AALJ-01A-31R-A41B-07                Breast, NOS
## TCGA-3C-AALK-01A-11R-A41B-07                Breast, NOS
## TCGA-4H-AAAK-01A-12R-A41B-07                Breast, NOS
## TCGA-5L-AAT0-01A-12R-A41B-07               Not Reported
##                                             sites_of_involvement
## TCGA-3C-AAAU-01A-11R-A41B-07
## TCGA-3C-AALI-01A-11R-A41B-07           Breast, Right Upper Outer
## TCGA-3C-AALJ-01A-31R-A41B-07                          Breast, NOS
## TCGA-3C-AALK-01A-11R-A41B-07                          Breast, NOS
## TCGA-4H-AAAK-01A-12R-A41B-07 Breast, NOS;Breast, Left Upper Outer
## TCGA-5L-AAT0-01A-12R-A41B-07
##                              days_to_last_follow_up follow_ups_disease_response
## TCGA-3C-AAAU-01A-11R-A41B-07                   4047                WT-With Tumor
## TCGA-3C-AALI-01A-11R-A41B-07                   4005                TF-Tumor Free
## TCGA-3C-AALJ-01A-31R-A41B-07                   1474                TF-Tumor Free
## TCGA-3C-AALK-01A-11R-A41B-07                   1448                TF-Tumor Free
## TCGA-4H-AAAK-01A-12R-A41B-07                    348                TF-Tumor Free
## TCGA-5L-AAT0-01A-12R-A41B-07                   1477                TF-Tumor Free
```

```
##                                                race gender
## TCGA-3C-AAAU-01A-11R-A41B-07                      white female
## TCGA-3C-AALI-01A-11R-A41B-07 black or african american female
## TCGA-3C-AALJ-01A-31R-A41B-07 black or african american female
## TCGA-3C-AALK-01A-11R-A41B-07 black or african american female
## TCGA-4H-AAAK-01A-12R-A41B-07                      white female
## TCGA-5L-AAT0-01A-12R-A41B-07                      white female
##                                          ethnicity vital_status age_at_index
## TCGA-3C-AAAU-01A-11R-A41B-07 not hispanic or latino        Alive           55
## TCGA-3C-AALI-01A-11R-A41B-07 not hispanic or latino        Alive           50
## TCGA-3C-AALJ-01A-31R-A41B-07 not hispanic or latino        Alive           62
## TCGA-3C-AALK-01A-11R-A41B-07 not hispanic or latino        Alive           52
## TCGA-4H-AAAK-01A-12R-A41B-07 not hispanic or latino        Alive           50
## TCGA-5L-AAT0-01A-12R-A41B-07     hispanic or latino        Alive           42
##                              days_to_birth                       demographic_id
## TCGA-3C-AAAU-01A-11R-A41B-07        -20211 cee0a94c-1d9e-5650-a500-a6b021fe138d
## TCGA-3C-AALI-01A-11R-A41B-07        -18538 583a1ee5-3175-523e-ba1f-75a30a3e1e41
## TCGA-3C-AALJ-01A-31R-A41B-07        -22848 9619a908-1684-547f-a407-6adf93e15b8d
## TCGA-3C-AALK-01A-11R-A41B-07        -19074 e54b1469-fffc-5291-a8e3-df2092ab5f34
## TCGA-4H-AAAK-01A-12R-A41B-07        -18371 64a204f0-b380-5962-a927-84af1841b6d6
## TCGA-5L-AAT0-01A-12R-A41B-07        -15393 776bb6f9-f5c8-57b6-bf4d-4014b8025a06
##                              age_is_obfuscated
## TCGA-3C-AAAU-01A-11R-A41B-07             FALSE
## TCGA-3C-AALI-01A-11R-A41B-07             FALSE
## TCGA-3C-AALJ-01A-31R-A41B-07             FALSE
## TCGA-3C-AALK-01A-11R-A41B-07             FALSE
## TCGA-4H-AAAK-01A-12R-A41B-07             FALSE
## TCGA-5L-AAT0-01A-12R-A41B-07             FALSE
##                              country_of_residence_at_enrollment
## TCGA-3C-AAAU-01A-11R-A41B-07                      United States
## TCGA-3C-AALI-01A-11R-A41B-07                      United States
## TCGA-3C-AALJ-01A-31R-A41B-07                      United States
## TCGA-3C-AALK-01A-11R-A41B-07                      United States
## TCGA-4H-AAAK-01A-12R-A41B-07                             Russia
## TCGA-5L-AAT0-01A-12R-A41B-07                             Brazil
##                              bcr_patient_barcode primary_site project_id
## TCGA-3C-AAAU-01A-11R-A41B-07   TCGA-3C-AAAU-01A       Breast  TCGA-BRCA
## TCGA-3C-AALI-01A-11R-A41B-07   TCGA-3C-AALI-01A       Breast  TCGA-BRCA
## TCGA-3C-AALJ-01A-31R-A41B-07   TCGA-3C-AALJ-01A       Breast  TCGA-BRCA
## TCGA-3C-AALK-01A-11R-A41B-07   TCGA-3C-AALK-01A       Breast  TCGA-BRCA
## TCGA-4H-AAAK-01A-12R-A41B-07   TCGA-4H-AAAK-01A       Breast  TCGA-BRCA
## TCGA-5L-AAT0-01A-12R-A41B-07   TCGA-5L-AAT0-01A       Breast  TCGA-BRCA
##
## TCGA-3C-AAAU-01A-11R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-3C-AALI-01A-11R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-3C-AALJ-01A-31R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-3C-AALK-01A-11R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-4H-AAAK-01A-12R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
## TCGA-5L-AAT0-01A-12R-A41B-07 Fibroepithelial Neoplasms;Adnexal and Skin Appendage Neoplasms;Adenomas
##                                                 name releasable released
## TCGA-3C-AAAU-01A-11R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-3C-AALI-01A-11R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-3C-AALJ-01A-31R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-3C-AALK-01A-11R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
```

```
## TCGA-4H-AAAK-01A-12R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
## TCGA-5L-AAT0-01A-12R-A41B-07 Breast Invasive Carcinoma       TRUE     TRUE
##                              figo_stage figo_staging_edition_year days_to_death
## TCGA-3C-AAAU-01A-11R-A41B-07      <NA>                      NA            NA
## TCGA-3C-AALI-01A-11R-A41B-07      <NA>                      NA            NA
## TCGA-3C-AALJ-01A-31R-A41B-07      <NA>                      NA            NA
## TCGA-3C-AALK-01A-11R-A41B-07      <NA>                      NA            NA
## TCGA-4H-AAAK-01A-12R-A41B-07      <NA>                      NA            NA
## TCGA-5L-AAT0-01A-12R-A41B-07  Stage IB2                    2009            NA
##                                    tumor_of_origin days_to_sample_procurement
## TCGA-3C-AAAU-01A-11R-A41B-07 TCGA-3C-AAAU_diagnosis                        NA
## TCGA-3C-AALI-01A-11R-A41B-07                  <NA>                         NA
## TCGA-3C-AALJ-01A-31R-A41B-07                  <NA>                         NA
## TCGA-3C-AALK-01A-11R-A41B-07                  <NA>                         NA
## TCGA-4H-AAAK-01A-12R-A41B-07                  <NA>                         NA
## TCGA-5L-AAT0-01A-12R-A41B-07                  <NA>                         NA
##                              last_known_disease_status tumor_grade
## TCGA-3C-AAAU-01A-11R-A41B-07                      <NA>        <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                      <NA>        <NA>
## TCGA-3C-AALJ-01A-31R-A41B-07                      <NA>        <NA>
## TCGA-3C-AALK-01A-11R-A41B-07                      <NA>        <NA>
## TCGA-4H-AAAK-01A-12R-A41B-07                      <NA>        <NA>
## TCGA-5L-AAT0-01A-12R-A41B-07                      <NA>        <NA>
##                              progression_or_recurrence alcohol_history
## TCGA-3C-AAAU-01A-11R-A41B-07                      <NA>            <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                      <NA>            <NA>
## TCGA-3C-AALJ-01A-31R-A41B-07                      <NA>            <NA>
## TCGA-3C-AALK-01A-11R-A41B-07                      <NA>            <NA>
## TCGA-4H-AAAK-01A-12R-A41B-07                      <NA>            <NA>
## TCGA-5L-AAT0-01A-12R-A41B-07                      <NA>            <NA>
##                              exposure_id year_of_birth paper_patient
## TCGA-3C-AAAU-01A-11R-A41B-07        <NA>            NA  TCGA-3C-AAAU
## TCGA-3C-AALI-01A-11R-A41B-07        <NA>            NA  TCGA-3C-AALI
## TCGA-3C-AALJ-01A-31R-A41B-07        <NA>            NA  TCGA-3C-AALJ
## TCGA-3C-AALK-01A-11R-A41B-07        <NA>            NA  TCGA-3C-AALK
## TCGA-4H-AAAK-01A-12R-A41B-07        <NA>            NA  TCGA-4H-AAAK
## TCGA-5L-AAT0-01A-12R-A41B-07        <NA>            NA  TCGA-5L-AAT0
##                              paper_Tumor.Type
## TCGA-3C-AAAU-01A-11R-A41B-07             BRCA
## TCGA-3C-AALI-01A-11R-A41B-07             BRCA
## TCGA-3C-AALJ-01A-31R-A41B-07             BRCA
## TCGA-3C-AALK-01A-11R-A41B-07             BRCA
## TCGA-4H-AAAK-01A-12R-A41B-07             BRCA
## TCGA-5L-AAT0-01A-12R-A41B-07             BRCA
##                              paper_Included_in_previous_marker_papers
## TCGA-3C-AAAU-01A-11R-A41B-07                                       NO
## TCGA-3C-AALI-01A-11R-A41B-07                                       NO
## TCGA-3C-AALJ-01A-31R-A41B-07                                       NO
## TCGA-3C-AALK-01A-11R-A41B-07                                       NO
## TCGA-4H-AAAK-01A-12R-A41B-07                                       NO
## TCGA-5L-AAT0-01A-12R-A41B-07                                       NO
##                              paper_vital_status paper_days_to_birth
## TCGA-3C-AAAU-01A-11R-A41B-07              Alive              -20211
## TCGA-3C-AALI-01A-11R-A41B-07              Alive              -18538
```

```
## TCGA-3C-AALJ-01A-31R-A41B-07              Alive              -22848
## TCGA-3C-AALK-01A-11R-A41B-07              Alive              -19074
## TCGA-4H-AAAK-01A-12R-A41B-07              Alive              -18371
## TCGA-5L-AAT0-01A-12R-A41B-07              Alive              -15393
##                               paper_days_to_death paper_days_to_last_followup
## TCGA-3C-AAAU-01A-11R-A41B-07                  NA                        4047
## TCGA-3C-AALI-01A-11R-A41B-07                  NA                        4005
## TCGA-3C-AALJ-01A-31R-A41B-07                  NA                        1474
## TCGA-3C-AALK-01A-11R-A41B-07                  NA                        1448
## TCGA-4H-AAAK-01A-12R-A41B-07                  NA                         348
## TCGA-5L-AAT0-01A-12R-A41B-07                  NA                        1477
##                               paper_age_at_initial_pathologic_diagnosis
## TCGA-3C-AAAU-01A-11R-A41B-07                                        55
## TCGA-3C-AALI-01A-11R-A41B-07                                        50
## TCGA-3C-AALJ-01A-31R-A41B-07                                        62
## TCGA-3C-AALK-01A-11R-A41B-07                                        52
## TCGA-4H-AAAK-01A-12R-A41B-07                                        50
## TCGA-5L-AAT0-01A-12R-A41B-07                                        42
##                               paper_pathologic_stage paper_Tumor_Grade
## TCGA-3C-AAAU-01A-11R-A41B-07                   <NA>                NA
## TCGA-3C-AALI-01A-11R-A41B-07               Stage_II                NA
## TCGA-3C-AALJ-01A-31R-A41B-07               Stage_II                NA
## TCGA-3C-AALK-01A-11R-A41B-07                Stage_I                NA
## TCGA-4H-AAAK-01A-12R-A41B-07              Stage_III                NA
## TCGA-5L-AAT0-01A-12R-A41B-07               Stage_II                NA
##                               paper_BRCA_Pathology paper_BRCA_Subtype_PAM50
## TCGA-3C-AAAU-01A-11R-A41B-07                 <NA>                      LumA
## TCGA-3C-AALI-01A-11R-A41B-07                 <NA>                      Her2
## TCGA-3C-AALJ-01A-31R-A41B-07                 <NA>                      LumB
## TCGA-3C-AALK-01A-11R-A41B-07                 <NA>                      LumA
## TCGA-4H-AAAK-01A-12R-A41B-07                 <NA>                      LumA
## TCGA-5L-AAT0-01A-12R-A41B-07                 <NA>                      LumA
##                               paper_MSI_status paper_HPV_Status
## TCGA-3C-AAAU-01A-11R-A41B-07               NA               NA
## TCGA-3C-AALI-01A-11R-A41B-07               NA               NA
## TCGA-3C-AALJ-01A-31R-A41B-07               NA               NA
## TCGA-3C-AALK-01A-11R-A41B-07               NA               NA
## TCGA-4H-AAAK-01A-12R-A41B-07               NA               NA
## TCGA-5L-AAT0-01A-12R-A41B-07               NA               NA
##                               paper_tobacco_smoking_history paper_CNV.Clusters
## TCGA-3C-AAAU-01A-11R-A41B-07                            NA                 C6
## TCGA-3C-AALI-01A-11R-A41B-07                            NA                 C6
## TCGA-3C-AALJ-01A-31R-A41B-07                            NA                 C6
## TCGA-3C-AALK-01A-11R-A41B-07                            NA                 C1
## TCGA-4H-AAAK-01A-12R-A41B-07                            NA                 C6
## TCGA-5L-AAT0-01A-12R-A41B-07                            NA                 C1
##                               paper_Mutation.Clusters
## TCGA-3C-AAAU-01A-11R-A41B-07                      C7
## TCGA-3C-AALI-01A-11R-A41B-07                      C9
## TCGA-3C-AALJ-01A-31R-A41B-07                      C4
## TCGA-3C-AALK-01A-11R-A41B-07                      C5
## TCGA-4H-AAAK-01A-12R-A41B-07                      C4
## TCGA-5L-AAT0-01A-12R-A41B-07                      C9
##                               paper_DNA.Methylation.Clusters paper_mRNA.Clusters
```

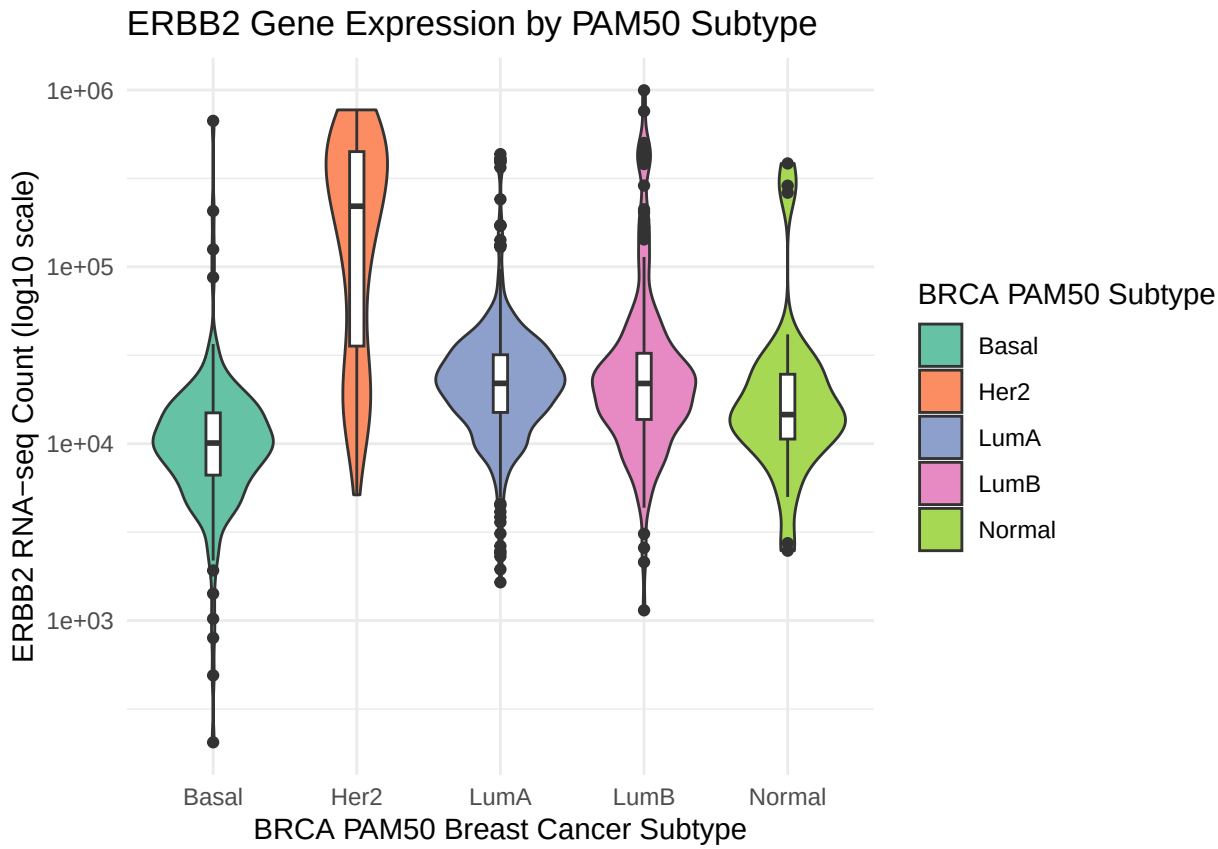
```
## TCGA-3C-AAAU-01A-11R-A41B-07                              C1                  C1
## TCGA-3C-AALI-01A-11R-A41B-07                              C2                  C2
## TCGA-3C-AALJ-01A-31R-A41B-07                              C2                  C2
## TCGA-3C-AALK-01A-11R-A41B-07                              C2                  C2
## TCGA-4H-AAAK-01A-12R-A41B-07                              C1                  C2
## TCGA-5L-AAT0-01A-12R-A41B-07                              C1                  C2
##                              paper_miRNA.Clusters paper_lncRNA.Clusters
## TCGA-3C-AAAU-01A-11R-A41B-07                   C3                  <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                   C3                  <NA>
## TCGA-3C-AALJ-01A-31R-A41B-07                   C2                  <NA>
## TCGA-3C-AALK-01A-11R-A41B-07                   C2                  <NA>
## TCGA-4H-AAAK-01A-12R-A41B-07                   C2                  <NA>
## TCGA-5L-AAT0-01A-12R-A41B-07                   C2                  <NA>
##                              paper_Protein.Clusters paper_PARADIGM.Clusters
## TCGA-3C-AAAU-01A-11R-A41B-07                   <NA>                      C5
## TCGA-3C-AALI-01A-11R-A41B-07                     C2                      C4
## TCGA-3C-AALJ-01A-31R-A41B-07                   <NA>                      C4
## TCGA-3C-AALK-01A-11R-A41B-07                     C2                      C6
## TCGA-4H-AAAK-01A-12R-A41B-07                     C2                      C6
## TCGA-5L-AAT0-01A-12R-A41B-07                   <NA>                      C6
##                              paper_Pan.Gyn.Clusters
## TCGA-3C-AAAU-01A-11R-A41B-07                   <NA>
## TCGA-3C-AALI-01A-11R-A41B-07                     C4
## TCGA-3C-AALJ-01A-31R-A41B-07                   <NA>
## TCGA-3C-AALK-01A-11R-A41B-07                     C4
## TCGA-4H-AAAK-01A-12R-A41B-07                     C1
## TCGA-5L-AAT0-01A-12R-A41B-07                   <NA>
```

```r
dim(ERBB2gene_metadata)
```

```
## [1] 1231   98
```

```r
# Create violin plot of ERBB2 expression by BRCA PAM50 subtype

ggplot(ERBB2gene_metadata %>%
    filter(!is.na(paper_BRCA_Subtype_PAM50)), # filtered out the NA values
       aes(x = paper_BRCA_Subtype_PAM50,
           y = ERBB2,
           fill = paper_BRCA_Subtype_PAM50)) +
  geom_violin() +
  geom_boxplot(width = 0.1, fill = "white") + # add box plot based on peer feedback
  labs(
    x = "BRCA PAM50 Breast Cancer Subtype",
    y = "ERBB2 RNA-seq Count (log10 scale)",
    fill = "BRCA PAM50 Subtype",
    title = "ERBB2 Gene Expression by PAM50 Subtype"
  ) +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2") +
  scale_y_log10() # Thanks Paul Skeffington for discussing log scale during the in-class exercises!!
```

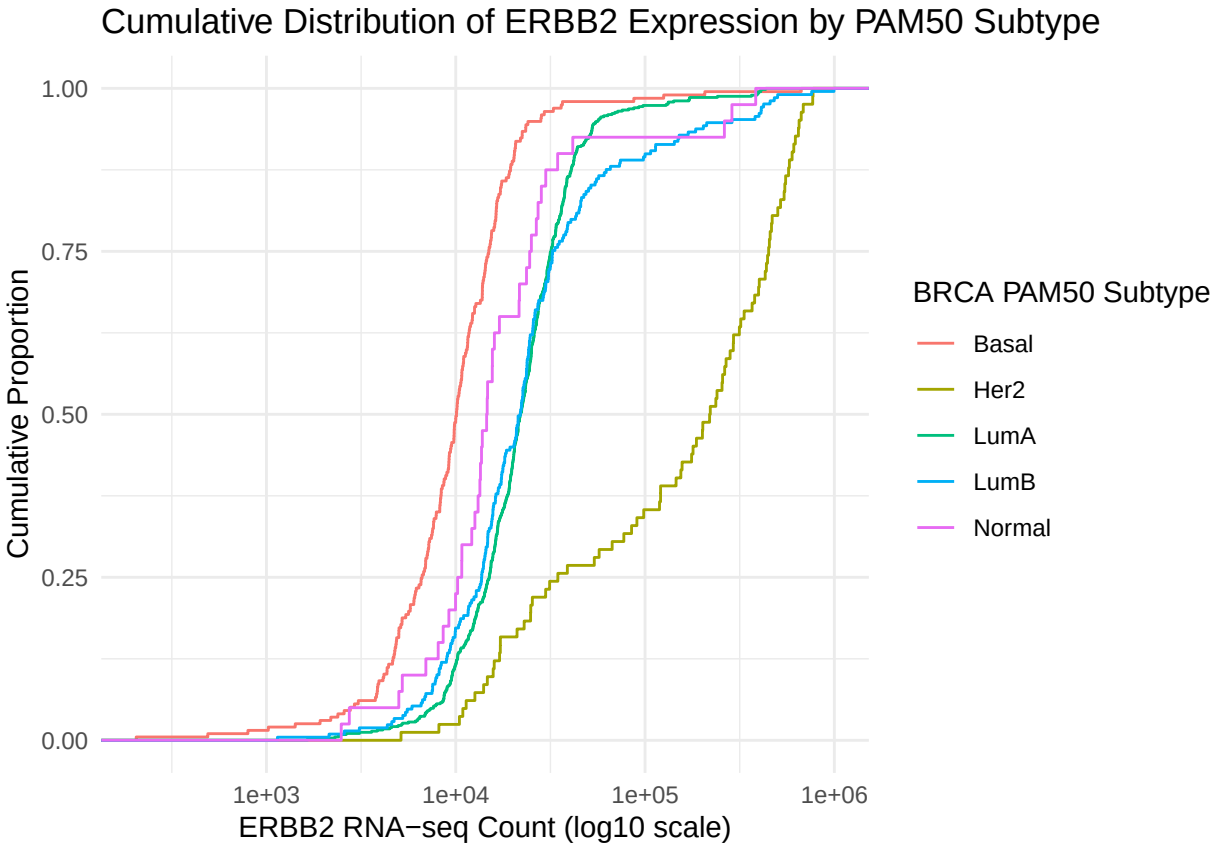



```
# without the log scale, the right-skewed distributions make
# the distributions near the bottom of violin plot difficult to see
# one option would be to de-lim the y-axis but then you hide the high-expression observations
```

3d. Cumulative distribution function — additional new plot type

I chose to create a cumulative distribution function to compare the distribution of ERBB2 expression across PAM50 molecular subtypes and to complement the histogram and violin plot by displaying the cumulative proportion of samples across the range of ERBB2 expression.

```
ggplot(
  ERBB2gene_metadata %>%
    filter(!is.na(paper_BRCA_Subtype_PAM50)), # filtered out the NA values
  aes(x = ERBB2, color = paper_BRCA_Subtype_PAM50)) + # map variables
  stat_ecdf() + # choose plot
  scale_x_log10() + # log10 scale is used to better visualize the distribution
  labs(
    x = "ERBB2 RNA-seq Count (log10 scale)", # add labels
    y = "Cumulative Proportion",
    color = "BRCA PAM50 Subtype",
    title = "Cumulative Distribution of ERBB2 Expression by PAM50 Subtype"
  ) +
  theme_minimal()
```




```
# The Her2 subtype curve is shifted to the right, indicating higher ERBB2 expression
# Basal samples generally show lower ERBB2 expression
# LumA, LumB, and Normal subtypes show intermediate distributions
```

3e. Heatmap — expression patterns across the 10 selected genes with PAM50 annotation

```
# load heatmap libraries
library(ComplexHeatmap)
```

```
## Loading required package: grid
```

```
## ========================================
## ComplexHeatmap version 2.28.0
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##     genomic data. Bioinformatics 2016.
##
##
```

```
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## ========================================
```

```
##
## Attaching package: 'ComplexHeatmap'
```

```
## The following object is masked from 'package:plotly':
##
##     add_heatmap
```

```
library(circlize)
```

```
## ========================================
## circlize version 0.4.18
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## ========================================
```

```
# Similar to above, in addition to ERBB2 and CDK4, I chose 8 more clinically relevant breast CA genes
# Use grep() to identify the exact Ensembl IDs for the genes in the dataset

grep("ENSG00000105810", rownames(counts), value = TRUE) # CDK6
```

```
## [1] "ENSG00000105810.10"
```

```
grep("ENSG00000091831", rownames(counts), value = TRUE) # ESR1
```

```
## [1] "ENSG00000091831.24"
```

```
grep("ENSG00000082175", rownames(counts), value = TRUE) # PGR
```

```
## [1] "ENSG00000082175.15"
```

```
grep("ENSG00000148773", rownames(counts), value = TRUE) # MKI67
```

```
## [1] "ENSG00000148773.14"
```

```r
grep("ENSG00000146648", rownames(counts), value = TRUE) # EGFR
```

```
## [1] "ENSG00000146648.19"
```

```r
grep("ENSG00000186081", rownames(counts), value = TRUE) # KRT5
```

```
## [1] "ENSG00000186081.12"
```

```r
grep("ENSG00000129514", rownames(counts), value = TRUE) # FOXA1
```

```
## [1] "ENSG00000129514.8"
```

```r
grep("ENSG00000107485", rownames(counts), value = TRUE) # GATA3
```

```
## [1] "ENSG00000107485.18"
```

```r
## Selecting final 10 genes

genes_for_heatmap <- c(
  ERBB2 = "ENSG00000141736.14",
  CDK4  = "ENSG00000135446.17",
  CDK6  = "ENSG00000105810.10",
  ESR1  = "ENSG00000091831.24",
  PGR   = "ENSG00000082175.15",
  MKI67 = "ENSG00000148773.14",
  EGFR  = "ENSG00000146648.19",
  KRT5  = "ENSG00000186081.12",
  FOXA1 = "ENSG00000129514.8",
  GATA3 = "ENSG00000107485.18"
)

genes_for_heatmap
```

```
##                ERBB2                 CDK4                 CDK6
## "ENSG00000141736.14" "ENSG00000135446.17" "ENSG00000105810.10"
##                 ESR1                  PGR                MKI67
## "ENSG00000091831.24" "ENSG00000082175.15" "ENSG00000148773.14"
##                 EGFR                 KRT5                FOXA1
## "ENSG00000146648.19" "ENSG00000186081.12"  "ENSG00000129514.8"
##                GATA3
## "ENSG00000107485.18"
```

```r
# Thanks to Steve Hutchins for feedback on this step!

## Keep only the 10 genes we want
counts_subset <- counts[genes_for_heatmap, ]
counts_subset <- as.matrix(counts_subset)

# Replace Ensembl IDs with gene symbols
```

```r
rownames(counts_subset) <- names(genes_for_heatmap)

# Initially I did not do the 2 steps - matrix and log and got both errors
# 1. Data is not a matrix
# 2. ComplexHeatmap told me that my raw counts contain some very large values -
# hence few of the huge counts could dominate the colors and hide patterns in the rest of the data
# Used ChatGPT to assist with these

# log2(counts_subset + 1) is a common transformation used with RNA-seq count data
# to reduce skewness and compress the large numerical range

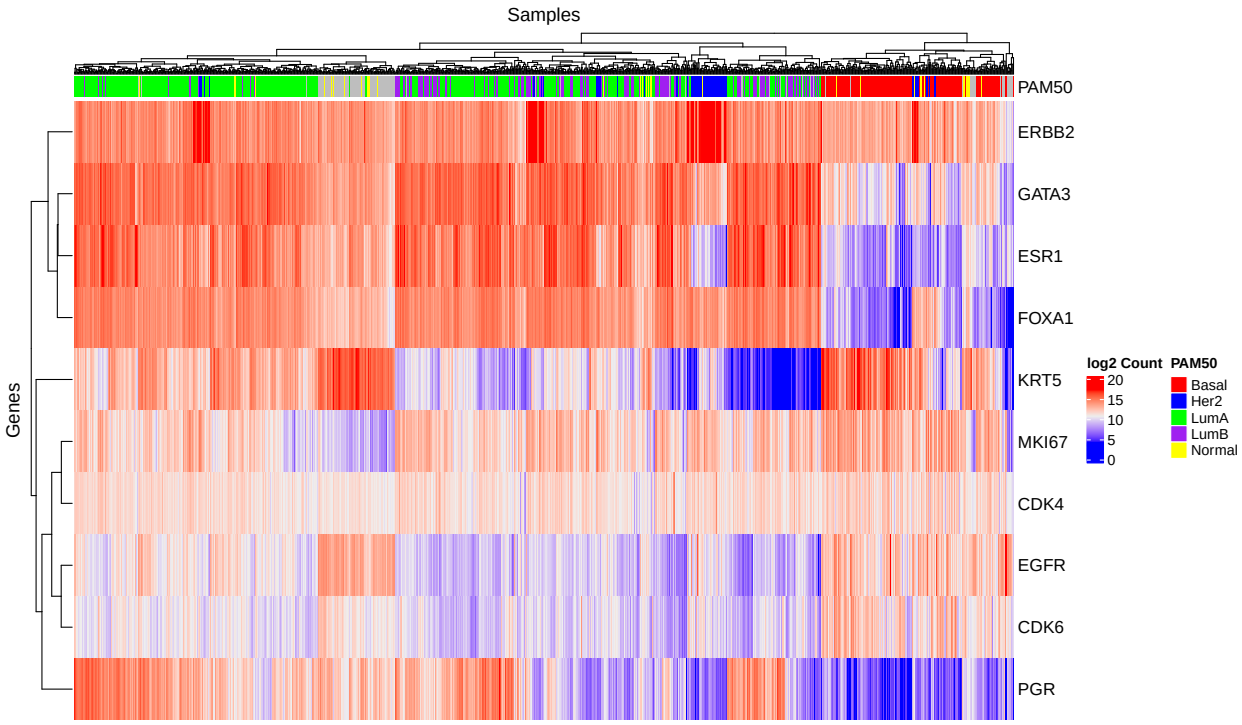
# Log transform counts
counts_subset_log <- log2(counts_subset + 1)

table(metadata$paper_BRCA_Subtype_PAM50)
```

```
##
##  Basal   Her2   LumA   LumB Normal
##    197     82    571    209     40
```

```r
## Heatmap annotations
column_ha <- HeatmapAnnotation(
  PAM50 = metadata$paper_BRCA_Subtype_PAM50,
  col = list(
    PAM50 = c(
      Basal = "red",
      Her2 = "blue",
      LumA = "green",
      LumB = "purple",
      Normal = "yellow"
    )
  )
)

# Create heatmap
Heatmap(
  counts_subset_log,
  name = "log2 Count",
  top_annotation = column_ha,
  show_row_names = TRUE,
  show_column_names = FALSE,
  cluster_columns = TRUE,
  cluster_rows = TRUE,
  column_title = "Samples",
  row_title = "Genes",
  heatmap_legend_param = list(title = "log2 Count")
)
```


Samples
PAM50
ERBB2
GATA3
ESR1
FOXA1
KRT5
MKI67
CDK4
EGFR
CDK6
PGR
Genes
log2 Count
20
15
10
5
0
PAM50
Basal
Her2
LumA
LumB
Normal